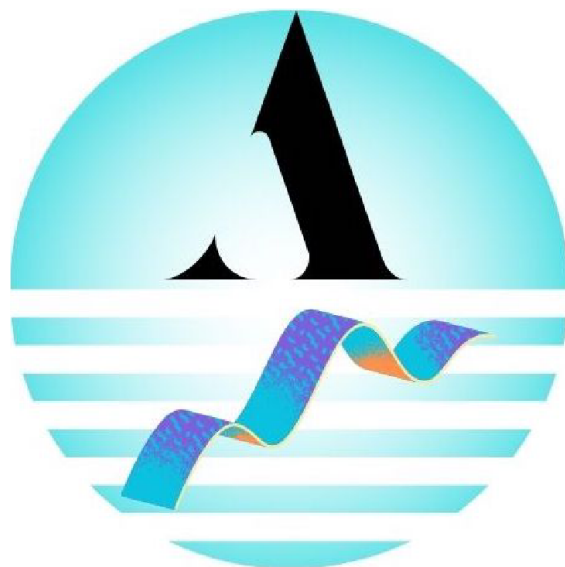




Number Theory (MMT422J3)

||||||| Written by: **DR. AIJAZ** |||||

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Bachelors with Mathematics as Major

4th Semester

MMT422J3: Mathematics/Applied Mathematics: Theory of Numbers

Credits: 4 THEORY + 2 TUTORIAL

Theory: 60 Hours & Tutorial: 30 Hours

Course Objectives: To enlighten students about the extensive study of integers, prime numbers and their properties (ii) to understand number theoretic functions and their applications.

Course Outcome: Upon successful completion of this course students will be able to

- (i) deal with the problems arising in cryptography and information theory especially in RSA encryption and decryption.
- (ii) solve congruences, linear diophantine equations, and other higher concepts of Discrete Mathematics.

Theory: 4 Credits

Unit –I

Divisibility of integers, prime numbers, fundamental theorem of arithmetic's, sequence of primes, Goldbach conjecture, Euclid's division algorithm, GCD and LCM of integers and their properties, Euclid's first theorem.

Unit –II

Linear Diophantine equations. Necessary and sufficient condition for solvability of linear Diophantine equations, Positive solutions. Linear Congruence's, solutions and applications, Euclid's Second theorem, Fermat Numbers and their properties.

Unit –III

Complete Residue System (CRS), Reduced Residue System (RRS) and their properties. Fermat and Euler's theorems with applications. Number theoretic functions, Euler's ϕ -function, $\phi(mn) = \phi(m)\phi(n)$ where $(m, n) = 1$, $\sum \phi(d) = n$, $\phi(m) = m \prod (1 - \frac{1}{p})$ for $m > 1$. Chinese Remainder theorem with applications.

Unit –IV

Order of an integer modulo n , primitive roots for primes, composite numbers having primitive roots, Euler's criterion, the Legendre symbol, law of reciprocity and its applications, Fermat's Last Theorem (statements only).

Tutorial: 2 Credits

Unit –V

Theory of polynomial equations, fundamental theorem of algebra, Remainder theorem and factor theorems, complex roots of a polynomial and their properties, synthetic division, relation between roots and coefficients.

Unit –VI

Solution of cubic and biquadratic polynomial equations. Problems on symmetric functions, Cardon's method and Descarte's rule of signs.

Recommended Books:

1. G.H Hardy and E.M. Wright, An introduction to the theory of Numbers, Fifth Edition, Oxford Science Publications, 1980.
2. E. Landau, Elementary Number Theory, American Mathematical Society, 2nd Edition, 1999.
3. W. J. Leveque, Topics in Number Theory, Vol(I & II), Dover Publishers, 2002
4. Ivan Niven and H.S Zuckerman, An introduction of the Theory of Numbers, John Wiley & Sons, 1991
5. Thomas Koshy, Elementary Number Theory with Applications (2nd Edition), Academic Press, 2007.
6. Aziz and Nisar, Theory of Equations, Kapoor and Son's, Srinagar, 2011
7. C.C.MacDuffee, Theory of Equations, John Wiley and Sons Inc., 1954.

Chapter 1

Fundamentals of Number Theory

This unit lays the foundation for the entire course by exploring the atomic structure of integers. It introduces the concept of divisibility, the Euclidean algorithm for finding the Greatest Common Divisor (GCD), and the Fundamental Theorem of Arithmetic, which asserts that prime numbers are the unique building blocks of all integers.

1.1 Divisibility of Integers

The concept of divisibility is central to number theory. It describes the relationship between two integers when one can be divided by the other without leaving a remainder.

Definition 1.1.1 (Divisibility). Let a and b be integers with $a \neq 0$. We say that a **divides** b , denoted by $a \mid b$, if there exists an integer c such that:

$$b = ac$$

If a divides b , we also say that a is a *divisor* or *factor* of b , and that b is a *multiple* of a . If a does not divide b , we write $a \nmid b$.

Example 1.1.2. • $3 \mid 12$ because $12 = 3 \times 4$.

- $-5 \mid 30$ because $30 = (-5) \times (-6)$.
- $4 \nmid 13$ because there is no integer c such that $13 = 4c$.

1.1.1 Basic Properties of Divisibility

The divisibility relation satisfies several elementary properties that are used frequently in mathematical proofs.

Theorem 1.1.3 (Properties of Divisibility). Let $a, b, c \in \mathbb{Z}$.

1. **Reflexivity:** $a \mid a$ for all $a \neq 0$.
2. **Transitivity:** If $a \mid b$ and $b \mid c$, then $a \mid c$.
3. **Linearity:** If $a \mid b$ and $a \mid c$, then $a \mid (bx + cy)$ for any integers x and y .
4. If $a \mid b$ and $a > 0, b > 0$, then $a \leq b$.
5. $1 \mid a$ for any a , and $a \mid 0$ for any $a \neq 0$.

Proof. We prove each property based on the definition of divisibility: $x \mid y$ implies $y = xk$ for some integer k .

1. **Reflexivity:** Since $a = a \cdot 1$ and $1 \in \mathbb{Z}$, it follows that $a \mid a$.

2. **Transitivity:** Since $a \mid b$ and $b \mid c$, there exist integers k and m such that $b = ak$ and $c = bm$. Substituting the first equation into the second yields $c = (ak)m = a(km)$. Since km is an integer, $a \mid c$.

3. **Linearity:** Since $a \mid b$ and $a \mid c$, there exist integers k and j such that $b = ak$ and $c = aj$. Substituting these into the linear combination:

$$bx + cy = (ak)x + (aj)y = a(kx + jy)$$

Since integers are closed under addition and multiplication, $(kx + jy) \in \mathbb{Z}$. Therefore, $a \mid (bx + cy)$.

4. Since $a \mid b$, we have $b = ak$ for some integer k . Because $a > 0$ and $b > 0$, k must be positive (i.e., $k \geq 1$). Multiplying $1 \leq k$ by the positive integer a gives $a \leq ak$, or $a \leq b$.

5. Note that $a = 1 \cdot a$, which implies $1 \mid a$. Similarly, $0 = a \cdot 0$, which implies $a \mid 0$.

■

Remark 1.1.4. The expression $bx + cy$ is called a *linear combination* of b and c . The theorem states that a common divisor of two integers divides any linear combination of them.

1.2 Prime Numbers

Integers greater than 1 can be categorized based on their divisibility properties.

Definition 1.2.1 (Prime and Composite Numbers). An integer $p > 1$ is called a **prime number** (or simply a prime) if its only positive divisors are 1 and p .

An integer $n > 1$ that is not prime is called a **composite number**.

Remark 1.2.2. The number 1 is neither prime nor composite. It is a special number often referred to as a *unit*.

Example 1.2.3. • The first few primes are: 2, 3, 5, 7, 11, 13, 17, 19, ...

- The number 2 is the only even prime number.
- 6 is composite because $2 \mid 6$ and $2 \neq 1, 2 \neq 6$.

To understand the structure of integers, we must first establish that primes act as the "building blocks" of numbers.

Theorem 1.2.4 (Existence of Prime Factors). *Every integer $n > 1$ has at least one prime divisor.*

Proof. We proceed by strong induction on n .

- **Base Case:** For $n = 2$, the number 2 is prime and divides itself. The statement holds.
- **Inductive Step:** Assume that every integer k satisfying $2 \leq k < n$ has a prime divisor. We consider the integer n :
 1. If n is prime, then $n \mid n$, and n itself is the prime divisor.
 2. If n is composite, then by definition, n has a divisor d such that $1 < d < n$. By the induction hypothesis, d must have a prime divisor p . Since $p \mid d$ and $d \mid n$, by transitivity, $p \mid n$.

In either case, n has a prime divisor.

■

1.3 The Fundamental Theorem of Arithmetic

The Fundamental Theorem of Arithmetic (FTA) is one of the most significant results in number theory. It asserts that prime factorization is the unique signature of any integer greater than 1.

Before stating the theorem, we require a critical lemma known as *Euclid's Lemma*.

Lemma 1.3.1 (Euclid's Lemma). *If p is a prime and $p \mid ab$ (where $a, b \in \mathbb{Z}$), then $p \mid a$ or $p \mid b$.*

Proof. (Note: This proof generally relies on Bézout's identity involving GCD, but we assume the standard properties here for the flow of the text). If $p \mid a$, the statement is true. Suppose $p \nmid a$. Since p is prime, the greatest common divisor $(p, a) = 1$. Therefore, there exist integers x and y such that $px + ay = 1$. Multiplying by b , we get:

$$pxb + aby = b$$

Since $p \mid pxb$ (obviously) and $p \mid ab$ (by hypothesis, so $p \mid aby$), p must divide the sum $pxb + aby$. Therefore, $p \mid b$. ■

Theorem 1.3.2 (The Fundamental Theorem of Arithmetic). *Every integer $n > 1$ can be expressed as a product of prime numbers. Furthermore, this factorization is unique, apart from the order in which the factors occur.*

Proof. Existence. We want to show that $n = p_1 p_2 \dots p_k$. We use strong induction on n .

Base Case: $n = 2$ is prime, so $2 = 2$ is the product of a single prime.

Inductive Hypothesis: Assume the statement is true for all integers k such that $2 \leq k < n$.

Inductive Step: Consider n .

- If n is prime, it is a product of a single prime (itself). Existence is established.
- If n is composite, then $n = ab$ where $1 < a < n$ and $1 < b < n$. By the inductive hypothesis, a and b can be written as products of primes.

$$a = p_1 \dots p_i, \quad b = q_1 \dots q_j$$

Thus, $n = ab = p_1 \dots p_i q_1 \dots q_j$, which is a product of primes.

Uniqueness: Suppose that n can be written as a product of primes in two ways:

$$n = p_1 p_2 \dots p_s = q_1 q_2 \dots q_t$$

where p_i and q_j are primes, and $p_1 \leq p_2 \leq \dots \leq p_s$ and $q_1 \leq q_2 \leq \dots \leq q_t$.

We must show that $s = t$ and $p_i = q_i$ for all i .

Consider p_1 . Since $p_1 \mid n$, it follows that $p_1 \mid q_1 q_2 \dots q_t$. By Euclid's Lemma extended to multiple factors, p_1 must divide at least one q_j . Since q_j is prime, its only divisors are 1 and q_j . Thus, $p_1 = q_j$ for some j .

Since the q 's are ordered (q_1 is the smallest), we must have $p_1 \geq q_1$. Similarly, considering q_1 , since $q_1 \mid n$, $q_1 \mid p_1 \dots p_s$, implying q_1 divides some p_i , so $q_1 \geq p_1$.

Since $p_1 \geq q_1$ and $q_1 \geq p_1$, we conclude $p_1 = q_1$.

We can now cancel p_1 and q_1 from both sides of the equation:

$$p_2 \dots p_s = q_2 \dots q_t$$

Repeating this process, we match each prime pair. If $s < t$, we would eventually arrive at $1 = q_{s+1} \dots q_t$, which is impossible since prime products are > 1 . Similarly, $s > t$ is impossible. Therefore, $s = t$ and each $p_i = q_i$. ■

1.3.1 Canonical Representation

As a consequence of the FTA, any integer $n > 1$ can be uniquely represented in the **canonical form**:

$$n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$$

where $p_1 < p_2 < \dots < p_k$ are distinct prime numbers and a_i are positive integers.

Example 1.3.3. The canonical factorization of 360:

$$360 = 2^3 \cdot 3^2 \cdot 5^1$$

The following theorem is footed here from Chapter II knwin as Euclid's Second Theorem.

Theorem 1.3.4 (Euclid's Theorem: Infinitude of Primes). *There are infinitely many prime numbers.*

Proof. Assume, for the sake of contradiction, that there is a finite number of primes. Let this complete list be $p_1, p_2, \dots, p_n$. Consider the integer

$$N = p_1 p_2 \dots p_n + 1$$

Since $N > 1$, by the Fundamental Theorem of Arithmetic, N must have at least one prime divisor, say q .

If q were one of the primes in our finite list ($q = p_i$ for some i), then q would divide the product $p_1 p_2 \dots p_n$. Since q divides N and q divides the product, q must divide the difference:

$$N - (p_1 p_2 \dots p_n) = 1$$

However, no prime divides 1. This is a contradiction. Therefore, q cannot be in the list $p_1, \dots, p_n$, implying there exists a prime not in our initial set. Thus, the number of primes is infinite. ■

Theorem 1.3.5 (Upper Bound for the n -th Prime). *If p_n denotes the n -th prime number (where $p_1 = 2$), then for $n \geq 1$:*

$$p_n \leq 2^{2^{n-1}}$$

Proof. We proceed by induction on n . **Base Case:** For $n = 1$, $p_1 = 2$ and $2^{2^{1-1}} = 2^0 = 2^1 = 2$. The inequality holds.

Inductive Step: Assume the inequality holds for all integers k such that $1 \leq k \leq n$. Consider Euclid's number N formed by the first n primes:

$$N = p_1 p_2 \dots p_n + 1$$

The prime factors of N are all distinct from $p_1, \dots, p_n$. Therefore, the next prime p_{n+1} must be less than or equal to N .

$$p_{n+1} \leq p_1 p_2 \dots p_n + 1$$

Using the inductive hypothesis $p_k \leq 2^{2^{k-1}}$ for each factor:

$$p_{n+1} \leq \left(\prod_{k=1}^n 2^{2^{k-1}} \right) + 1 = 2^{\sum_{k=1}^n 2^{k-1}} + 1$$

The exponent is a geometric series sum: $\sum_{k=0}^{n-1} 2^k = 2^n - 1$. Thus:

$$p_{n+1} \leq 2^{2^n - 1} + 1$$

For all $n \geq 1$, we have $1 < 2^{2^n - 1}$, so:

$$2^{2^n - 1} + 1 < 2^{2^n - 1} + 2^{2^n - 1} = 2 \cdot 2^{2^n - 1} = 2^{2^n}$$

Therefore, $p_{n+1} \leq 2^{2^n}$, satisfying the inductive step. ■

Theorem 1.3.6 (Primes of the form $4n + 3$). *There are infinitely many primes of the form $4n + 3$.*

Proof. Assume there are finitely many primes of the form $4n + 3$. Let this set be $\{q_1, q_2, \dots, q_k\}$, where $q_1 = 3$. Consider the integer:

$$N = 4(q_1 q_2 \dots q_k) - 1$$

Note that $N = 4(\dots) - 1 \equiv -1 \equiv 3 \pmod{4}$. Since N is odd, all its prime factors are odd. The odd primes are either of the form $4m + 1$ or $4m + 3$.

If all prime factors of N were of the form $4m + 1$, their product would also be of the form $4m + 1$ (since $(4a + 1)(4b + 1) = 4(4ab + a + b) + 1$). However, $N \equiv 3 \pmod{4}$. Therefore, N must have at least one prime factor, say p , of the form $4m + 3$.

Now, check if p is in our list $\{q_1, \dots, q_k\}$. Since $p \mid N$ and if $p = q_i$, then $p \mid 4(q_1 \dots q_k)$. This implies $p \mid (4q_1 \dots q_k - N)$, so $p \mid 1$, which is impossible. Thus, p is a new prime of the form $4n + 3$ not in the list. This contradicts the finiteness assumption. ■

Theorem 1.3.7 (Dirichlet's Theorem on Arithmetic Progressions). *If a and b are relatively prime positive integers (i.e., $\gcd(a, b) = 1$), then the arithmetic progression*

$$a, a + b, a + 2b, a + 3b, \dots$$

contains infinitely many prime numbers.

Note on Proof. The proof of this theorem requires advanced techniques from analytic number theory, specifically the use of Dirichlet L -functions and characters, which are beyond the scope of this text. The theorem generalizes the result regarding primes of the form $4n + 3$ (where $a = 3, b = 4$) to any general arithmetic progression where the starting term and the common difference are coprime. ■

Prime-Generating Polynomials

A famous problem in number theory is whether there exists a polynomial that generates only prime numbers.

Euler's Polynomial

Consider the quadratic polynomial discovered by Euler:

$$f(n) = n^2 + n + 41$$

This polynomial exhibits remarkable behavior for small integers.

- For $n = 0$, $f(0) = 41$ (Prime)
- For $n = 1$, $f(1) = 43$ (Prime)
- For $n = 2$, $f(2) = 47$ (Prime)
- ...
- For $n = 39$, $f(39) = 39^2 + 39 + 41 = 39(39 + 1) + 41 = 39(40) + 41 = 1560 + 41 = 1601$ (Prime)

The polynomial yields a prime number for every integer n from 0 to 39.

The Failure of the Pattern

However, the pattern breaks at $n = 40$:

$$f(40) = 40^2 + 40 + 41 = 40(40 + 1) + 41 = 40(41) + 41 = 41(40 + 1) = 41^2$$

Since $41^2 = 1681$ is clearly composite (divisible by 41), $f(40)$ is not prime. Similarly for $n = 41$:

$$f(41) = 41^2 + 41 + 41 = 41(41 + 1 + 1) = 41 \times 43$$

which is composite.

Remark 1.3.8. This example serves as a cautionary tale in mathematics: empirical evidence, no matter how consistent for initial values, does not constitute a proof. It has been proven (by Goldbach) that no non-constant polynomial with integer coefficients can evaluate to a prime number for *all* integer values of n .

The Goldbach conjectures remain some of the most famous unproven statements regarding the additive properties of primes.

Conjecture 1.3.9 (Strong (Binary) Goldbach Conjecture). Every even integer $n > 2$ can be expressed as the sum of two prime numbers.

Conjecture 1.3.10 (Weak (Ternary) Goldbach Conjecture). Every odd integer $n > 5$ can be expressed as the sum of three prime numbers.

Theorem 1.3.11 (Vinogradov's Theorem). Every sufficiently large odd integer can be represented as the sum of three primes.

1.4 The Division Algorithm

Despite its name, the Division Algorithm is not an algorithm in the computational sense (like a computer program), but a fundamental theorem ensuring the existence and uniqueness of the quotient and remainder when dividing integers.

Theorem 1.4.1 (The Division Algorithm). Let a and b be integers with $b > 0$. Then there exist unique integers q (the quotient) and r (the remainder) such that:

$$a = bq + r, \quad 0 \leq r < b$$

Proof. The proof consists of two parts: establishing the **existence** of q and r , and establishing their **uniqueness**.

Existence Consider the set S of all non-negative integers of the form $a - bx$:

$$S = \{a - bx \mid x \in \mathbb{Z} \text{ and } a - bx \geq 0\}$$

We must show S is non-empty.

- If $a \geq 0$, choose $x = 0$. Then $a - b(0) = a \geq 0$, so $a \in S$.
- If $a < 0$, choose $x = a$. Since $b \geq 1$ (as integers $b > 0$), we have $a - ba = a(1 - b)$. Since a is negative and $(1 - b) \leq 0$, the product is non-negative. Thus S is not empty.

By the **Well-Ordering Principle**, every non-empty set of non-negative integers contains a least element. Let r be the least element of S . Since $r \in S$, there exists an integer q corresponding to it such that:

$$r = a - bq \implies a = bq + r$$

By definition of S , $r \geq 0$. We must now show $r < b$. Assume, for contradiction, that $r \geq b$. Consider $r' = r - b$.

$$r' = (a - bq) - b = a - b(q + 1)$$

Since $r \geq b$, $r' = r - b \geq 0$. Thus $r' \in S$. However, $r' < r$ (since $b > 0$). This contradicts the fact that r is the *least* element of S . Therefore, $r < b$.

Uniqueness Suppose there are two pairs of integers (q, r) and (q', r') satisfying the conditions:

$$a = bq + r = bq' + r'$$

where $0 \leq r < b$ and $0 \leq r' < b$. Without loss of generality, assume $r' \geq r$. Subtracting the equations:

$$b(q - q') = r' - r$$

This implies $b \mid (r' - r)$. Since $0 \leq r \leq r' < b$, the difference satisfies $0 \leq r' - r < b$. The only multiple of b strictly less than b (and non-negative) is 0. Therefore, $r' - r = 0 \implies r' = r$. Since $b(q - q') = 0$ and $b \neq 0$, we must have $q - q' = 0 \implies q = q'$. Thus, the integers q and r are unique. ■

1.5 Greatest Common Divisor (GCD) and Least Common Multiple (LCM)

Definition 1.5.1. Let a and b be integers, not both zero. The **greatest common divisor** of a and b , denoted by $\gcd(a, b)$ or simply (a, b) , is the positive integer d satisfying:

1. $d \mid a$ and $d \mid b$.
2. If $c \mid a$ and $c \mid b$, then $c \leq d$.

1.5.1 The Euclidean Algorithm

The Division Algorithm provides an efficient method for calculating the GCD of two numbers, known as the Euclidean Algorithm. It relies on the following lemma.

Lemma 1.5.2. If $a = bq + r$, then $\gcd(a, b) = \gcd(b, r)$.

Proof. Let $d = \gcd(a, b)$. Since $d \mid a$ and $d \mid b$, it divides the linear combination $a - bq$, which is r . Thus d is a common divisor of b and r . Let c be any common divisor of b and r . Then c divides $bq + r$, which is a . Thus c is a common divisor of a and b , meaning $c \leq d$. Hence, $d = \gcd(b, r)$. ■

The Algorithm Process: To find $\gcd(a, b)$ where $a \geq b > 0$:

$$\begin{array}{ll}
 a = bq_1 + r_1 & 0 < r_1 < b \\
 b = r_1q_2 + r_2 & 0 < r_2 < r_1 \\
 r_1 = r_2q_3 + r_3 & 0 < r_3 < r_2 \\
 \vdots & \\
 r_{n-2} = r_{n-1}q_n + r_n & 0 < r_n < r_{n-1} \\
 r_{n-1} = r_nq_{n+1} + 0 &
 \end{array}$$

The last non-zero remainder r_n is the $\gcd(a, b)$.

Example 1.5.3 (Finding the GCD of Large Numbers). Find the greatest common divisor of 2740 and 1760 using the Euclidean Algorithm.

Solution: We apply the division algorithm repeatedly, starting with $a = 2740$ and $b = 1760$.

$$\begin{aligned}
 2740 &= 1760 \cdot 1 + 980 \\
 1760 &= 980 \cdot 1 + 780 \\
 980 &= 780 \cdot 1 + 200 \\
 780 &= 200 \cdot 3 + 180 \\
 200 &= 180 \cdot 1 + 20 \\
 180 &= 20 \cdot 9 + 0
 \end{aligned}$$

Since the last non-zero remainder is 20, we conclude that:

$$\gcd(2740, 1760) = 20$$

Example 1.5.4 (GCD and Linear Combinations). Find $\gcd(1071, 462)$ and express it as a linear combination of the two numbers.

Solution: First, we apply the Euclidean Algorithm:

$$\begin{aligned}
 1071 &= 462 \cdot 2 + 147 & (1) \\
 462 &= 147 \cdot 3 + 21 & (2) \\
 147 &= 21 \cdot 7 + 0 & (3)
 \end{aligned}$$

The last non-zero remainder is 21, so $\gcd(1071, 462) = 21$.

To express 21 as a linear combination, we work backward from equation (2):

$$\begin{aligned}
 21 &= 462 - 147 \cdot 3 \\
 \text{From (1), substitute } 147 &= 1071 - 462 \cdot 2 : \\
 21 &= 462 - (1071 - 462 \cdot 2) \cdot 3 \\
 21 &= 462 - 1071 \cdot 3 + 462 \cdot 6 \\
 21 &= 462 \cdot 7 + 1071 \cdot (-3)
 \end{aligned}$$

Thus, $21 = 1071(-3) + 462(7)$.

1.5.2 Properties of GCD

The utility of the GCD is reinforced by Bézout's Identity, a crucial theorem for solving Diophantine equations.

Theorem 1.5.5 (Bézout's Identity). *Let a and b be integers, not both zero. Then there exist integers x and y such that:*

$$\gcd(a, b) = ax + by$$

Moreover, $\gcd(a, b)$ is the smallest positive integer that can be written in this form.

Proof. Let $S = \{ax + by \mid x, y \in \mathbb{Z}, ax + by > 0\}$. Since not both a, b are zero, S is non-empty. By the Well-Ordering Principle, S contains a least element, say $d = ax_0 + by_0$. We claim $d = \gcd(a, b)$.

1. **d divides a and b :** By the Division Algorithm, $a = dq + r$ with $0 \leq r < d$.

$$r = a - dq = a - q(ax_0 + by_0) = a(1 - qx_0) + b(-qy_0)$$

If $r > 0$, then $r \in S$. But $r < d$, which contradicts the minimality of d . Thus $r = 0$, so $d \mid a$. Similarly, $d \mid b$.

2. **Greatest property:** Let c be any common divisor of a and b . Then $c \mid (ax_0 + by_0)$ by linearity, so $c \mid d$. Therefore $c \leq d$. ■

Corollary 1.5.6 (Relatively Prime Integers). *Two integers a and b are relatively prime (i.e., $\gcd(a, b) = 1$) if and only if there exist integers x and y such that $ax + by = 1$.*

The GCD function exhibits a scaling property known as homogeneity.

Theorem 1.5.7. *For any non-zero integers a, b and any positive integer k :*

$$\gcd(ka, kb) = k \cdot \gcd(a, b)$$

Proof. We utilize the property derived from Bézout's Identity: $\gcd(x, y)$ is the smallest positive integer that can be expressed in the form $xu + yv$ for integers u, v .

Let $S_{a,b}$ be the set of all positive linear combinations of a and b :

$$S_{a,b} = \{ax + by \mid x, y \in \mathbb{Z}, ax + by > 0\}$$

By definition, $\gcd(a, b) = \min(S_{a,b})$.

Now consider the set $S_{ka,kb}$ for the integers ka and kb :

$$S_{ka,kb} = \{(ka)x + (kb)y \mid x, y \in \mathbb{Z}, (ka)x + (kb)y > 0\}$$

We can factor out k from the expression in the set:

$$(ka)x + (kb)y = k(ax + by)$$

Thus, the set $S_{ka,kb}$ consists of elements of $S_{a,b}$ multiplied by k :

$$S_{ka,kb} = \{k \cdot s \mid s \in S_{a,b}\}$$

Since $k > 0$, multiplying a set of positive numbers by k scales the minimum element by exactly k . Therefore:

$$\min(S_{ka,kb}) = k \cdot \min(S_{a,b})$$

Substituting the definition of GCD back into this equality:

$$\gcd(ka, kb) = k \cdot \gcd(a, b)$$

■

2nd Proof. Let $d = \gcd(a, b)$ and $D = \gcd(ka, kb)$. Since d divides both a and b , kd divides both ka and kb . Therefore, $kd \leq D$.

Conversely, since k divides both ka and kb , it must divide their GCD, so we can write $D = kQ$ for some integer Q . Since kQ divides ka and kb , Q must divide a and b . Thus Q is a common divisor of a and b , implying $Q \leq d$. Multiplying by k , we get $D = kQ \leq kd$.

Since $kd \leq D$ and $D \leq kd$, it follows that $D = kd$.

■

3rd Proof. We apply the Euclidean Algorithm to the integers a and b (assuming $a \geq b > 0$) to find their GCD. Let r_n be the last non-zero remainder.

$$\begin{aligned} a &= bq_1 + r_1, & 0 < r_1 < b \\ b &= r_1q_2 + r_2, & 0 < r_2 < r_1 \\ &\vdots \\ r_{n-2} &= r_{n-1}q_n + r_n, & 0 < r_n < r_{n-1} \\ r_{n-1} &= r_nq_{n+1} + 0 \end{aligned}$$

By the definition of the algorithm, $\gcd(a, b) = r_n$.

Now, we multiply every equation in this system by the positive integer k :

$$\begin{aligned} (ka) &= (kb)q_1 + (kr_1) \\ (kb) &= (kr_1)q_2 + (kr_2) \\ &\vdots \\ (kr_{n-2}) &= (kr_{n-1})q_n + (kr_n) \\ (kr_{n-1}) &= (kr_n)q_{n+1} + 0 \end{aligned}$$

Since $k > 0$, the inequality condition for the remainders is preserved at every step (e.g., $0 < r_1 < b \implies 0 < kr_1 < kb$). Thus, this new sequence of equations constitutes the valid Euclidean Algorithm for finding the GCD of ka and kb .

The last non-zero remainder in this scaled sequence is kr_n . Therefore:

$$\gcd(ka, kb) = kr_n = k \cdot \gcd(a, b)$$

■

Euclid's First Theorem

In modern mathematics, "Euclid's First Theorem" often refers to the fundamental property of prime numbers regarding divisibility, formally known as Euclid's Lemma. This is Book VII, Proposition 30 of Euclid's *Elements*.

Theorem 1.5.8 (Euclid's First Theorem / Euclid's Lemma). *If p is a prime number and $p \mid ab$, where a and b are integers, then $p \mid a$ or $p \mid b$.*

Proof. Suppose $p \mid ab$. If $p \mid a$, the theorem holds. Suppose $p \nmid a$. Since p is prime, its only divisors are 1 and p . Since p does not divide a , the only common divisor of p and a is 1. Thus, $\gcd(p, a) = 1$. By Bézout's Identity, there exist integers x and y such that:

$$px + ay = 1$$

Multiply the entire equation by b :

$$\begin{aligned} pxb + aby &= b \\ p(xb) + (ab)y &= b \end{aligned}$$

We observe that:

- $p \mid p(xb)$ (obviously).
- $p \mid (ab)y$ (since $p \mid ab$ by hypothesis).

Since p divides both terms on the left-hand side, p divides their sum.

$$p \mid b$$

Therefore, if p does not divide a , it must divide b . ■

Theorem 1.5.9 (Generalized Euclid's Lemma). *If $a \mid bc$ and $\gcd(a, b) = 1$, then $a \mid c$.*

Proof. Since $\gcd(a, b) = 1$, there exist x, y such that $ax + by = 1$. Multiplying by c : $acx + bcy = c$. Clearly $a \mid acx$. Also $a \mid bcy$ because $a \mid bc$. Thus $a \mid (acx + bcy) \implies a \mid c$. ■

Least Common Multiple (LCM)

Definition 1.5.10. The **least common multiple** of two non-zero integers a and b , denoted by $\text{lcm}(a, b)$ or $[a, b]$, is the positive integer m satisfying:

1. $a \mid m$ and $b \mid m$.
2. If $a \mid c$ and $b \mid c$ for some $c > 0$, then $m \leq c$.

1.5.3 Relationship between GCD and LCM

Theorem 1.5.11. *For any positive integers a and b :*

$$\gcd(a, b) \cdot \text{lcm}(a, b) = ab$$

Proof. Let $d = \gcd(a, b)$. Since $d \mid ab$, the integer $m = \frac{ab}{d}$ exists. We claim $m = \text{lcm}(a, b)$.
 m is a common multiple:

$$m = a \left(\frac{b}{d} \right) = b \left(\frac{a}{d} \right)$$

Since $d \mid a$ and $d \mid b$, both factors in parentheses are integers. Thus $a \mid m$ and $b \mid m$.

m is the least common multiple: Let c be any positive common multiple of a and b . Thus $c = ax = by$ for some integers x, y . By Bézout's Identity, there exist u, v such that $d = au + bv$. We can express $\frac{c}{m}$ as:

$$\frac{c}{m} = \frac{cd}{ab} = \frac{c(au + bv)}{ab} = \frac{cau}{ab} + \frac{cbv}{ab} = \frac{c}{b}u + \frac{c}{a}v$$

Since $b \mid c$, $\frac{c}{b}$ is an integer. Since $a \mid c$, $\frac{c}{a}$ is an integer. Thus $\frac{c}{m}$ is an integer, implying $m \mid c$. Since m and c are positive, $m \leq c$. ■

Exercise 1.5.12. Assuming that $\gcd(a, b) = 1$, prove the following:

- (a) $\gcd(a + b, a - b) = 1$ or 2 .

(b) $\gcd(2a + b, a + 2b) = 1$ or 3 .

(c) $\gcd(a + b, a^2 + b^2) = 1$ or 2 .

(d) $\gcd(a + b, a^2 - ab + b^2) = 1$ or 3 .

Solution. **(a)** Let $d = \gcd(a + b, a - b)$. By definition, $d \mid (a + b)$ and $d \mid (a - b)$. Therefore, d must divide their sum and their difference:

$$d \mid ((a + b) + (a - b)) \implies d \mid 2a$$

$$d \mid ((a + b) - (a - b)) \implies d \mid 2b$$

Thus, d is a common divisor of $2a$ and $2b$, so $d \mid \gcd(2a, 2b)$. Using the homogeneity property of GCD:

$$\gcd(2a, 2b) = 2 \gcd(a, b)$$

Since we are given $\gcd(a, b) = 1$, it follows that $d \mid 2$. The positive divisors of 2 are 1 and 2. Hence, the GCD is either 1 or 2.

(b) Let $d = \gcd(2a + b, a + 2b)$. We perform linear combinations to isolate terms involving a and b .

$$d \mid 2(a + 2b) - (2a + b) \implies d \mid (2a + 4b - 2a - b) \implies d \mid 3b$$

$$d \mid 2(2a + b) - (a + 2b) \implies d \mid (4a + 2b - a - 2b) \implies d \mid 3a$$

Thus, d divides both $3a$ and $3b$, so $d \mid \gcd(3a, 3b)$.

$$d \mid 3 \gcd(a, b) \implies d \mid 3(1) \implies d \mid 3$$

The positive divisors of 3 are 1 and 3. Hence, the GCD is either 1 or 3.

(c) Let $d = \gcd(a + b, a^2 + b^2)$. We observe the algebraic identity: $a^2 + b^2 = (a + b)(a - b) + 2b^2$. Since $d \mid (a + b)$, d divides the first term $(a + b)(a - b)$. Since d divides the sum $a^2 + b^2$ by definition, it must divide the remaining term:

$$d \mid 2b^2$$

Now, consider the relationship between d and b . Let $k = \gcd(d, b)$. If a prime p divides k , then $p \mid b$ and $p \mid d$. Since $d \mid (a + b)$, then $p \mid (a + b)$, which implies $p \mid a$. This contradicts $\gcd(a, b) = 1$. Therefore, $\gcd(d, b) = 1$, which implies $\gcd(d, b^2) = 1$. Since $d \mid 2b^2$ and is coprime to b^2 , it must be that $d \mid 2$. Thus, d is 1 or 2.

(d) Let $d = \gcd(a + b, a^2 - ab + b^2)$. We observe the identity: $a^2 - ab + b^2 = (a + b)^2 - 3ab$. Since $d \mid (a + b)$, clearly $d \mid (a + b)^2$. It follows that d must divide the remaining term:

$$d \mid -3ab \implies d \mid 3ab$$

Similar to part (c), if any prime factor shared by d and ab existed, it would have to divide a (implies dividing b) or b (implies dividing a). Since $\gcd(a, b) = 1$, d is coprime to both a and b , and thus coprime to ab . Therefore, $d \mid 3$. Thus, d is 1 or 3. ■

Exercise 1.5.13. For $n \geq 1$, and positive integers a, b , show the following:

(a) If $\gcd(a, b) = 1$, then $\gcd(a^n, b^n) = 1$.

(b) The relation $a^n \mid b^n$ implies that $a \mid b$.

Solution. **(a)** We proceed by contradiction. Assume $\gcd(a^n, b^n) = d > 1$. Then there exists a prime number p such that $p \mid d$. Consequently, $p \mid a^n$ and $p \mid b^n$. If a prime divides a product of integers, it must divide at least one of the factors. Since $p \mid \underbrace{a \cdot a \cdots a}_{n \text{ times}}$, it follows that $p \mid a$. Similarly, $p \mid b$. This

implies that p is a common divisor of a and b , so $\gcd(a, b) \geq p > 1$. This contradicts the hypothesis that $\gcd(a, b) = 1$. Therefore, $\gcd(a^n, b^n) = 1$.

(b) Let $g = \gcd(a, b)$. We can write $a = rg$ and $b = sg$ where $\gcd(r, s) = 1$. Substitute these expressions into the divisibility relation $a^n \mid b^n$:

$$(rg)^n \mid (sg)^n \implies r^n g^n \mid s^n g^n$$

Since $g \geq 1$, we can divide by g^n :

$$r^n \mid s^n$$

From part (a), since $\gcd(r, s) = 1$, we know that $\gcd(r^n, s^n) = 1$. The only positive integer that divides s^n while being relatively prime to it is 1. Therefore, $r^n = 1$, which implies $r = 1$ (as r is a positive integer). Since $r = 1$, we have $a = 1 \cdot g = g$. By definition, $g = \gcd(a, b)$ divides b . Since $a = g$, we conclude $a \mid b$. ■

Exercise 1.5.14. Prove that if $\gcd(a, b) = 1$, then $\gcd(a + b, ab) = 1$.

Solution. Let $d = \gcd(a + b, ab)$. Assume for the sake of contradiction that $d > 1$. Then there exists a prime p such that $p \mid d$. This implies $p \mid (a + b)$ and $p \mid ab$. By Euclid's Lemma, since $p \mid ab$, then $p \mid a$ or $p \mid b$.

Case 1: Suppose $p \mid a$. Since $p \mid (a + b)$ and $p \mid a$, then p must divide the difference:

$$p \mid ((a + b) - a) \implies p \mid b$$

If $p \mid a$ and $p \mid b$, then $\gcd(a, b) \geq p > 1$, which contradicts the hypothesis.

Case 2: Suppose $p \mid b$. Similarly, $p \mid ((a + b) - b) \implies p \mid a$. This also implies $\gcd(a, b) > 1$, a contradiction.

Since assuming $d > 1$ leads to a contradiction in all cases, we must have $d = 1$. ■

Exercise 1.5.15. 1. Use Euclid's Division Algorithm to find $\gcd(12345, 54321)$ and then find integers x and y such that $12345x + 54321y = \gcd(12345, 54321)$.

2. Prove that for any integer n , the product of three consecutive integers $n(n + 1)(n + 2)$ is always divisible by 6.
3. Let a, b , and c be integers. If $\gcd(a, b) = 1$ and $a \mid bc$, use Euclid's First Theorem (Euclid's Lemma) to provide a rigorous proof that $a \mid c$.
4. Determine the prime factorization of $n = 20!$ and find the largest power of 7 that divides 20! using Legendre's Formula.
5. Following Euclid's proof for the infinitude of primes, prove that there are infinitely many prime numbers of the form $4n + 3$.
6. Verify the Strong (Even) Goldbach Conjecture for every even integer in the range $90 \leq n \leq 110$ by expressing each as the sum of two primes.
7. Given that $\gcd(a, b) = d$, prove that the quotient of the integers and their greatest common divisor are coprime, i.e., $\gcd\left(\frac{a}{d}, \frac{b}{d}\right) = 1$.
8. Prove that $\sqrt{p}$ is an irrational number for any prime p by using the uniqueness of the Fundamental Theorem of Arithmetic.
9. Show that if a and b are positive integers such that $a^2 \mid b^2$, then it must follow that $a \mid b$.
10. Calculate $\text{lcm}(2^3 \cdot 3^2 \cdot 5 \cdot 7, 2^2 \cdot 3^4 \cdot 7^2)$ and verify the identity $\gcd(a, b) \cdot \text{lcm}(a, b) = ab$ for these two values.
11. Use algebraic identities to show that for any integer $n > 1$, the expression $n^4 + 4$ is always a composite number.
12. Determine all possible prime numbers p such that $p, p + 2$, and $p + 4$ are all prime (this is known as the prime triplet).

13. Let a and b be coprime integers ($a > b$). Prove that the $\gcd(a + b, a - b)$ can only be equal to 1 or 2.
14. Prove that for any positive integer k , there exist k consecutive composite integers, thereby demonstrating that there are arbitrarily large gaps between successive primes.
15. Apply Euclid's Division Algorithm to find the unique quotient q and remainder r when $a = -100$ is divided by $b = 7$, ensuring that $0 \leq r < |b|$.
16. Use the Fundamental Theorem of Arithmetic to prove that if n is a perfect square, then every exponent in its canonical prime factorization must be even.
17. Prove the distributive property of the greatest common divisor over the least common multiple: $\gcd(a, \text{lcm}(b, c)) = \text{lcm}(\gcd(a, b), \gcd(a, c))$.
18. If p is a prime and $p \mid a^n$, use Euclid's First Theorem and induction to show that p^n must divide a^n .
19. Demonstrate that the Weak Goldbach Conjecture (every odd integer $n > 5$ is the sum of three primes) is a necessary consequence of the Strong Goldbach Conjecture.
20. Prove that $n = 5$ is the only integer for which the set $\{n, n + 2, n + 6, n + 8, n + 12, n + 14\}$ consists entirely of prime numbers.
21. **Lamé's Theorem Preparation:** Prove that if Euclid's Algorithm requires n steps to find $\gcd(a, b)$ with $a > b > 0$, then $a \geq F_{n+2}$ and $b \geq F_{n+1}$, where F_k denotes the k -th Fibonacci number.
22. Let a, m, n be positive integers with $a > 1$. Prove that $\gcd(a^m - 1, a^n - 1) = a^{\gcd(m, n)} - 1$.
23. **Legendre's Formula Derivative:** Let $v_p(n!)$ denote the exponent of the highest power of a prime p dividing $n!$. Prove that $v_p(n!) = \frac{n - s_p(n)}{p - 1}$, where $s_p(n)$ is the sum of the digits of n when written in base p .
24. Prove that for any $n > 1$, the sum $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$ is never an integer. (Hint: Consider the highest power of 2 less than or equal to n and use the Fundamental Theorem of Arithmetic).
25. **Mersenne and Fermat Numbers:**
 - (a) Prove that if $a^n - 1$ is prime ($a > 1, n > 1$), then a must be 2 and n must be prime.
 - (b) Prove that if $a^n + 1$ is prime ($a > 1, n > 1$), then a must be even and n must be a power of 2.
26. Let p_n be the n -th prime number ($p_1 = 2, p_2 = 3, \dots$). Prove Bonse's Inequality: for $n \geq 4$, $p_{n+1}^2 < p_1 p_2 \cdots p_n$.
27. Show that for any composite integer $n > 4$, n divides $(n - 1)!$. How does this result differ when n is prime?
28. Prove that there are no polynomials $P(n) = a_k n^k + \cdots + a_1 n + a_0$ with integer coefficients (where $k \geq 1$) that produce a prime number for every integer n .
29. Suppose that a and b are positive integers such that $\gcd(a, b) = 1$. Prove that $\gcd(a + b, a^2 - ab + b^2)$ is either 1 or 3.
30. **Euclidean Algorithm Efficiency:** Prove that the number of divisions performed in the Euclidean algorithm to find $\gcd(a, b)$ is at most 5 times the number of digits in the smaller of the two integers (base 10).
31. Use the Fundamental Theorem of Arithmetic to prove that if n is a positive integer such that n is not a perfect k -th power, then $\sqrt[k]{n}$ is irrational.
32. Prove that there are infinitely many primes of the form $6n + 5$.

33. Let a, b, c be integers such that $a^2 + b^2 = c^2$. Using Euclid's First Theorem and the properties of divisibility, prove that 3 must divide ab and 4 must divide ab .
34. Show that the set of all integers that can be expressed in the form $ax + by$ for $x, y \in \mathbb{Z}$ is exactly the set of all multiples of $\gcd(a, b)$.
35. **Sylvester's Frobenius Problem (Special Case):** If $\gcd(a, b) = 1$ and $a, b > 1$, prove that the largest integer that cannot be written in the form $ax + by$ for non-negative integers x, y is $ab - a - b$.

Challenges:

36. Let a, m , and n be positive integers with $a > 1$. Prove that $\gcd(a^m - 1, a^n - 1) = a^{\gcd(m, n)} - 1$.
37. Let $v_p(n!)$ denote the exponent of the highest power of a prime p dividing $n!$.
- (a) Prove that $v_p(n!) = \sum_{k=1}^{\infty} \lfloor \frac{n}{p^k} \rfloor$.
- (b) Use this to show that for any $n > 0$, the binomial coefficient $\binom{2n}{n}$ is always divisible by every prime p such that $n < p \leq 2n$.
38. Prove that for any $n > 1$, the n -th harmonic number $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$ is never an integer. (Hint: Apply the Fundamental Theorem of Arithmetic to the denominators and consider the parity of the numerator).
39. Prove Lamé's Theorem: The number of divisions required to find the $\gcd(a, b)$ using Euclid's Algorithm is at most 5 times the number of digits in $\min(a, b)$ in base 10. (You must first prove that if the algorithm takes n steps, then $a \geq F_{n+2}$ and $b \geq F_{n+1}$).
40. Prove that for any positive integer k , there exists a sequence of k consecutive composite integers. Then, provide a constructive proof for $k = 10$.
41. Prove the distributive law for $\gcd$ over lcm :

$$\gcd(a, \text{lcm}(b, c)) = \text{lcm}(\gcd(a, b), \gcd(a, c))$$

Explain how the Fundamental Theorem of Arithmetic (prime factorization) makes this proof more elegant than using the Division Algorithm alone.

42. If p is a prime and $p \mid a^n$, use Euclid's First Theorem and mathematical induction to prove that $p^n \mid a^n$. Is the statement still true if p is a composite number? Provide a counter-example if it is false.
43. Using a method analogous to Euclid's proof for the infinitude of primes, prove that there are infinitely many prime numbers of the form $4n + 3$. Explain why this specific proof cannot be easily modified to show the infinitude of primes of the form $4n + 1$.
44. Prove that the product of k consecutive integers is always divisible by $k!$. (Note: While this is a property of binomial coefficients, you must prove it using the properties of primes and the Fundamental Theorem of Arithmetic).
45. The Strong Goldbach Conjecture implies the Weak Goldbach Conjecture (every odd integer $n > 5$ is the sum of three primes).
- (a) Formally prove this implication.
- (b) Show that if the Strong Goldbach Conjecture is true, every integer $n > 1$ is the sum of at most 3 primes.

Chapter 2

Linear Diophantine Equations and Congruences

In this unit, the focus shifts from static properties of numbers to solving equations in integers. We study Linear Diophantine equations and the necessary conditions for their solvability. The unit also introduces the powerful language of modular arithmetic (congruences), providing tools to solve linear congruences and explore properties of special numbers like Fermat numbers.

2.1 Linear Diophantine Equations

A linear Diophantine equation is an equation of the form $ax + by = c$, where a, b , and c are integers, and we seek integer solutions for x and y .

2.1.1 Solvability Condition

Theorem 2.1.1 (Necessary and Sufficient Condition). *The linear Diophantine equation $ax + by = c$ has an integer solution if and only if $d \mid c$, where $d = \gcd(a, b)$.*

Proof. Necessity: Let (x, y) be an integer solution. Since $d = \gcd(a, b)$, d divides both a and b . Consequently, d must divide the linear combination $ax + by$. Thus, $d \mid c$.

Sufficiency: Assume $d \mid c$, so $c = kd$ for some integer k . By Bézout's Identity, there exist integers u, v such that $au + bv = d$. Multiplying this equation by k yields $a(uk) + b(vk) = kd = c$. Hence, (uk, vk) is a solution. ■

2.1.2 The General Solution

If one solution is known, all other solutions can be determined systematically.

Theorem 2.1.2. *Let $d = \gcd(a, b)$. If (x_0, y_0) is a particular solution to $ax + by = c$, then the general solution consists of the set of integers:*

$$x = x_0 + \frac{b}{d}t, \quad y = y_0 - \frac{a}{d}t$$

where t is an arbitrary integer.

Proof. First, we verify these are solutions by substitution:

$$a(x_0 + \frac{b}{d}t) + b(y_0 - \frac{a}{d}t) = (ax_0 + by_0) + t\left(\frac{ab}{d} - \frac{ab}{d}\right) = c$$

To show these are the *only* solutions, let (x, y) be any solution. Then $ax + by = c = ax_0 + by_0$, which implies $a(x - x_0) = b(y_0 - y)$. Dividing by d , we get $\frac{a}{d}(x - x_0) = \frac{b}{d}(y_0 - y)$. Since $\gcd(\frac{a}{d}, \frac{b}{d}) = 1$, by Euclid's Lemma, $\frac{b}{d}$ must divide $(x - x_0)$. Thus, $x - x_0 = \frac{b}{d}t$, leading to the stated formulas. ■

Example 2.1.3. Find the general solution of the equation $172x + 20y = 1000$.

Solution: We first determine the greatest common divisor of 172 and 20 using the Euclidean Algorithm:

$$\begin{aligned} 172 &= 20(8) + 12 \\ 20 &= 12(1) + 8 \\ 12 &= 8(1) + 4 \\ 8 &= 4(2) + 0 \end{aligned}$$

The GCD is 4. Since $4 \mid 1000$, the equation is solvable. To find a particular solution, we express 4 as a linear combination of 172 and 20 by working backward:

$$\begin{aligned} 4 &= 12 - 8(1) \\ 4 &= 12 - (20 - 12(1)) = 12(2) - 20 \\ 4 &= (172 - 20(8))(2) - 20 \\ 4 &= 172(2) + 20(-17) \end{aligned}$$

This gives the relation $172(2) + 20(-17) = 4$. To satisfy the original equation equals to 1000, we multiply the entire relation by 250 (since $1000/4 = 250$):

$$172(500) + 20(-4250) = 1000$$

Thus, a particular solution is $x_0 = 500$ and $y_0 = -4250$. The general solution is given by:

$$\begin{aligned} x &= 500 + \frac{20}{4}t = 500 + 5t \\ y &= -4250 - \frac{172}{4}t = -4250 - 43t \end{aligned}$$

2.1.3 Positive Solutions

To find solutions where both x and y are positive, we solve the inequalities $x > 0$ and $y > 0$ for the integer parameter t .

Example 2.1.4. Determine all positive solutions of $5x + 3y = 52$.

Solution: We begin by finding a particular solution. By inspection (or algorithm), we know that $5(2) + 3(-3) = 1$. Multiplying by 52 yields:

$$5(104) + 3(-156) = 52$$

So, $x_0 = 104$ and $y_0 = -156$ is a particular solution. Since $\gcd(5, 3) = 1$, the general solution is:

$$x = 104 + 3t, \quad y = -156 - 5t$$

We require $x > 0$ and $y > 0$:

$$\begin{aligned} 104 + 3t > 0 &\implies 3t > -104 \implies t > -34.66 \\ -156 - 5t > 0 &\implies -5t > 156 \implies t < -31.2 \end{aligned}$$

Combining these inequalities, we find $-34.66 < t < -31.2$. The possible integer values for t are -34 , -33 , and -32 . Substituting these values into the general solution:

- $t = -34 \implies x = 104 + 3(-34) = 2, \quad y = -156 - 5(-34) = 14$
- $t = -33 \implies x = 104 + 3(-33) = 5, \quad y = -156 - 5(-33) = 9$
- $t = -32 \implies x = 104 + 3(-32) = 8, \quad y = -156 - 5(-32) = 4$

Therefore, the positive solutions are $(2, 14)$, $(5, 9)$, and $(8, 4)$.

2.2 Congruences

Definition 2.2.1. Let n be a fixed positive integer. Two integers a and b are said to be congruent modulo n , symbolized by

$$a \equiv b \pmod{n}$$

if n divides the difference $a - b$; that is, provided that $a - b = kn$ for some integer k .

Theorem 2.2.2. Let $n > 1$ be fixed and a, b, c, d be arbitrary integers. Then the following properties hold:

- (a) $a \equiv a \pmod{n}$.
- (b) If $a \equiv b \pmod{n}$, then $b \equiv a \pmod{n}$.
- (c) If $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then $a \equiv c \pmod{n}$.
- (d) If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, then $a + c \equiv b + d \pmod{n}$ and $ac \equiv bd \pmod{n}$.
- (e) If $a \equiv b \pmod{n}$, then $a + c \equiv b + c \pmod{n}$ and $ac \equiv bc \pmod{n}$.
- (f) If $a \equiv b \pmod{n}$, then $a^k \equiv b^k \pmod{n}$ for any positive integer k .

Proof. Recall the definition of congruence: $x \equiv y \pmod{n}$ means that $n \mid (x - y)$, or equivalently, $x - y = nk$ for some integer k .

- (a) **Reflexivity:** Consider $a - a = 0$. Since $0 = n \cdot 0$, it follows that $n \mid (a - a)$. Thus, $a \equiv a \pmod{n}$.
- (b) **Symmetry:** Given $a \equiv b \pmod{n}$, we have $n \mid (a - b)$, so $a - b = nk$ for some integer k . Multiplying by -1 , we get $b - a = -(a - b) = -nk = n(-k)$. Since $-k$ is an integer, $n \mid (b - a)$. Thus, $b \equiv a \pmod{n}$.
- (c) **Transitivity:** Given $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, there exist integers k and l such that:

$$a - b = nk \quad \text{and} \quad b - c = nl$$

Adding these two equations:

$$\begin{aligned} (a - b) + (b - c) &= nk + nl \\ a - c &= n(k + l) \end{aligned}$$

Since $k + l$ is an integer, $n \mid (a - c)$. Thus, $a \equiv c \pmod{n}$.

- (d) **Addition and Multiplication:** Given $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, we have $a - b = nk$ and $c - d = nl$. This implies $a = b + nk$ and $c = d + nl$.

For Addition:

$$(a + c) - (b + d) = (a - b) + (c - d) = nk + nl = n(k + l)$$

Since n divides the difference, $a + c \equiv b + d \pmod{n}$.

For Multiplication:

$$\begin{aligned} ac - bd &= (b + nk)(d + nl) - bd \\ &= bd + bnl + dnk + n^2kl - bd \\ &= n(bn + dk + nkl) \end{aligned}$$

Since $bn + dk + nkl$ is an integer, $n \mid (ac - bd)$. Thus, $ac \equiv bd \pmod{n}$.

- (e) **Scalar Properties:** This is a special case of property (d). Since c is an integer, by reflexivity (a), $c \equiv c \pmod{n}$. Using (d) with $a \equiv b$ and $c \equiv c$:

$$a + c \equiv b + c \pmod{n} \quad \text{and} \quad ac \equiv bc \pmod{n}$$

(f) **Exponentiation:** Given $a \equiv b \pmod{n}$, then $n \mid (a-b)$. Recall the algebraic factorization identity for $a^k - b^k$:

$$a^k - b^k = (a - b)(a^{k-1} + a^{k-2}b + \cdots + ab^{k-2} + b^{k-1})$$

Since n divides the factor $(a - b)$, n must divide the entire product. Therefore, $n \mid (a^k - b^k)$, which implies $a^k \equiv b^k \pmod{n}$. ■

Theorem 2.2.3. Let $n > 1$ be a fixed integer. Two integers a and b are congruent modulo n if and only if they leave the same non-negative remainder when divided by n .

Proof. By the Division Algorithm, we can express a and b as:

$$a = nq_1 + r_1, \quad 0 \leq r_1 < n$$

$$b = nq_2 + r_2, \quad 0 \leq r_2 < n$$

where r_1 and r_2 are the remainders of a and b upon division by n .

We prove the biconditional statement in two parts.

($\Leftarrow$) Assume that a and b leave the same remainder, i.e., $r_1 = r_2 = r$. Then:

$$a = nq_1 + r \quad \text{and} \quad b = nq_2 + r$$

Subtracting the two equations:

$$a - b = (nq_1 + r) - (nq_2 + r) = nq_1 - nq_2 = n(q_1 - q_2)$$

Since $q_1 - q_2$ is an integer, $n \mid (a - b)$. Therefore, $a \equiv b \pmod{n}$.

($\Rightarrow$) Assume that $a \equiv b \pmod{n}$. By definition, this means $n \mid (a - b)$, so $a - b = nk$ for some integer k . Substituting the expressions from the Division Algorithm:

$$(nq_1 + r_1) - (nq_2 + r_2) = nk$$

Rearranging terms to isolate the remainders:

$$n(q_1 - q_2) + (r_1 - r_2) = nk$$

$$r_1 - r_2 = nk - n(q_1 - q_2)$$

$$r_1 - r_2 = n(k - q_1 + q_2)$$

This equation implies that n divides the difference of the remainders, $n \mid (r_1 - r_2)$. However, we know the bounds of the remainders:

$$0 \leq r_1 < n \quad \text{and} \quad 0 \leq r_2 < n$$

From these inequalities, the maximum possible difference is strictly less than n , and the minimum is strictly greater than $-n$:

$$-n < r_1 - r_2 < n$$

The only integer strictly between $-n$ and n that is divisible by n is 0. Therefore, $r_1 - r_2 = 0$, which implies $r_1 = r_2$. ■

Theorem 2.2.4. If $ca \equiv cb \pmod{n}$, then $a \equiv b \pmod{n/d}$, where $d = \gcd(c, n)$.

Proof. By hypothesis, we can write

$$c(a - b) = ca - cb = kn$$

for some integer k . Knowing that $\gcd(c, n) = d$, there exist relatively prime integers r and s satisfying $c = dr$, $n = ds$. When these values are substituted in the displayed equation and the common factor d canceled, the net result is

$$r(a - b) = ks$$

Hence, $s \mid r(a - b)$ and $\gcd(r, s) = 1$. Euclid's lemma yields $s \mid a - b$, which may be recast as $a \equiv b \pmod{s}$; in other words, $a \equiv b \pmod{n/d}$. ■

Exercise 2.2.5. Prove each of the following assertions:

- (a) If $a \equiv b \pmod{n}$ and $m \mid n$, then $a \equiv b \pmod{m}$.
- (b) If $a \equiv b \pmod{n}$ and $c > 0$, then $ca \equiv cb \pmod{cn}$.
- (c) If $a \equiv b \pmod{n}$ and the integers a, b, n are all divisible by $d > 0$, then $a/d \equiv b/d \pmod{n/d}$.

Solution. **(a)** Since $a \equiv b \pmod{n}$, by definition $n \mid (a - b)$. We are given that $m \mid n$. By the transitivity of divisibility (if $x \mid y$ and $y \mid z$, then $x \mid z$), since $m \mid n$ and $n \mid (a - b)$, it follows that $m \mid (a - b)$. Thus, $a \equiv b \pmod{m}$.

(b) Since $a \equiv b \pmod{n}$, there exists an integer k such that $a - b = nk$. Multiplying both sides by c , we get $c(a - b) = c(nk)$, which simplifies to $ca - cb = k(cn)$. Since k is an integer, $cn \mid (ca - cb)$. Thus, $ca \equiv cb \pmod{cn}$.

(c) Let $a = dA$, $b = dB$, and $n = dN$ for integers A, B, N . We wish to prove $A \equiv B \pmod{N}$. From $a \equiv b \pmod{n}$, we have $n \mid (a - b)$, so $a - b = kn$ for some integer k . Substituting the expressions involving d :

$$dA - dB = k(dN)$$

Dividing the entire equation by d (since $d > 0$):

$$A - B = kN$$

This implies $N \mid (A - B)$. In terms of the original variables, $n/d \mid (a/d - b/d)$. Thus, $a/d \equiv b/d \pmod{n/d}$. ■

Exercise 2.2.6. Give an example to show that $a^2 \equiv b^2 \pmod{n}$ need not imply that $a \equiv b \pmod{n}$.

Solution. We need a case where n divides $a^2 - b^2 = (a - b)(a + b)$, but n does not divide $a - b$. Consider $n = 8$, $a = 1$, and $b = 3$. Calculate squares modulo 8:

$$a^2 = 1^2 = 1$$

$$b^2 = 3^2 = 9 \equiv 1 \pmod{8}$$

So $1^2 \equiv 3^2 \pmod{8}$. However, $1 \not\equiv 3 \pmod{8}$. Therefore, the implication holds false. ■

Exercise 2.2.7. If $a \equiv b \pmod{n}$, prove that $\gcd(a, n) = \gcd(b, n)$.

Solution. Since $a \equiv b \pmod{n}$, we can write $a = b + kn$ for some integer k . Let $d = \gcd(b, n)$. By definition, $d \mid b$ and $d \mid n$. Since d divides both terms on the right side of $a = b + kn$, d must divide a . Thus, d is a common divisor of a and n . Consequently, $d \leq \gcd(a, n)$.

Now let $g = \gcd(a, n)$. By definition, $g \mid a$ and $g \mid n$. Rearranging the congruence relation gives $b = a - kn$. Since g divides both terms on the right side, $g \mid b$. Thus, g is a common divisor of b and n . Consequently, $g \leq \gcd(b, n)$.

Since $\gcd(b, n) \leq \gcd(a, n)$ and $\gcd(a, n) \leq \gcd(b, n)$, they must be equal. ■

Exercise 2.2.8. (a) Find the remainders when 2^{50} and 41^{65} are divided by 7.

(b) What is the remainder when the following sum is divided by 4?

$$1^5 + 2^5 + 3^5 + \cdots + 99^5 + 100^5$$

Solution. **(a)** For $2^{50} \pmod{7}$: We look for a power of 2 that is close to a multiple of 7.

$$2^1 = 2, \quad 2^2 = 4, \quad 2^3 = 8 \equiv 1 \pmod{7}$$

Since $2^3 \equiv 1 \pmod{7}$, we can decompose the exponent 50 using division by 3: $50 = 3(16) + 2$.

$$2^{50} = 2^{3(16)+2} = (2^3)^{16} \cdot 2^2 \equiv (1)^{16} \cdot 4 \equiv 4 \pmod{7}$$

The remainder is 4.

For $41^{65} \pmod{7}$: First, reduce the base modulo 7: $41 = 5(7) + 6 \equiv 6 \equiv -1 \pmod{7}$.

$$41^{65} \equiv (-1)^{65} \pmod{7}$$

Since 65 is an odd number, $(-1)^{65} = -1$.

$$-1 \equiv 6 \pmod{7}$$

The remainder is 6.

(b) We calculate $k^5 \pmod{4}$ for the residue classes $k = 0, 1, 2, 3$.

- $0^5 \equiv 0 \pmod{4}$
- $1^5 \equiv 1 \pmod{4}$
- $2^5 = 32 \equiv 0 \pmod{4}$
- $3^5 \equiv (-1)^5 \equiv -1 \equiv 3 \pmod{4}$

The sum of any four consecutive integers modulo 4 is:

$$0 + 1 + 0 + 3 = 4 \equiv 0 \pmod{4}$$

The series $1^5 + \dots + 100^5$ contains 100 terms. This can be grouped into exactly $100/4 = 25$ sets of consecutive residues $\{1, 2, 3, 0\} \pmod{4}$.

$$\sum_{k=1}^{100} k^5 \equiv 25 \times (1 + 0 + 3 + 0) \equiv 25 \times 0 \equiv 0 \pmod{4}$$

The remainder is 0. ■

Exercise 2.2.9. Prove that the integer $53^{103} + 103^{53}$ is divisible by 39, and that $111^{333} + 333^{111}$ is divisible by 7.

Solution. Part 1: Divisibility by 39. Since $39 = 3 \times 13$ and $\gcd(3, 13) = 1$, we show divisibility by 3 and 13 separately. Let $S = 53^{103} + 103^{53}$.

Modulo 3: $53 = 3(17) + 2 \equiv -1 \pmod{3}$. $103 = 3(34) + 1 \equiv 1 \pmod{3}$.

$$S \equiv (-1)^{103} + (1)^{53} \equiv -1 + 1 \equiv 0 \pmod{3}$$

Modulo 13: $53 = 13(4) + 1 \equiv 1 \pmod{13}$. $103 = 13(8) - 1 \equiv -1 \pmod{13}$.

$$S \equiv (1)^{103} + (-1)^{53} \equiv 1 + (-1) \equiv 0 \pmod{13}$$

Since $3 \mid S$ and $13 \mid S$, it follows that $39 \mid S$.

Part 2: Divisibility by 7. Let $T = 111^{333} + 333^{111}$. We reduce the bases modulo 7. $111 = 7(15) + 6 \equiv -1 \pmod{7}$. $333 = 7(47) + 4 \equiv 4 \pmod{7}$.

$$T \equiv (-1)^{333} + 4^{111} \equiv -1 + 4^{111} \pmod{7}$$

We need to evaluate $4^{111} \pmod{7}$. We examine powers of 4 modulo 7:

$$4^1 \equiv 4 \pmod{7}$$

$$4^2 = 16 \equiv 2 \pmod{7}$$

$$4^3 = 4 \cdot 2 = 8 \equiv 1 \pmod{7}$$

Since $4^3 \equiv 1 \pmod{7}$, we can reduce the exponent 111. Note that $111 = 3 \times 37$.

$$4^{111} = (4^3)^{37} \equiv (1)^{37} \equiv 1 \pmod{7}$$

Substituting this back into the expression for T :

$$T \equiv -1 + 1 \equiv 0 \pmod{7}$$

Thus, T is divisible by 7. ■

2.3 Linear Congruences

A linear congruence is the modular arithmetic equivalent of a linear equation. It plays a vital role in cryptography and algorithmic number theory.

Definition 2.3.1. A linear congruence in one variable is an expression of the form:

$$ax \equiv b \pmod{n}$$

where a, b, n are integers, $n > 0$, and x is the unknown integer.

2.3.1 Solvability and Solutions

Finding a solution means finding an integer x_0 that satisfies the congruence. If x_0 is a solution, then any integer x_1 such that $x_1 \equiv x_0 \pmod{n}$ is also a solution. We generally look for the number of mutually incongruent solutions modulo n (i.e., distinct solutions in the set $\{0, 1, \dots, n-1\}$).

The theory of linear congruences is directly linked to Linear Diophantine Equations.

$$ax \equiv b \pmod{n} \iff ax - ny = b \text{ for some } y \in \mathbb{Z}$$

Theorem 2.3.2 (Existence and Number of Solutions). *The linear congruence $ax \equiv b \pmod{n}$ has a solution if and only if $d \mid b$, where $d = \gcd(a, n)$. If $d \mid b$, then there are exactly d mutually incongruent solutions modulo n .*

Proof. The congruence is equivalent to the Diophantine equation $ax + n(-y) = b$. From the theory of Linear Diophantine Equations, this equation has a solution if and only if $\gcd(a, n)$ divides b . Let $d = \gcd(a, n)$.

If $d \mid b$, the Diophantine equation has a general solution:

$$x = x_0 + \frac{n}{d}t$$

where x_0 is a particular solution and $t \in \mathbb{Z}$. We seek values of t that produce distinct values of x modulo n . Consider two solutions corresponding to t_1 and t_2 :

$$x_0 + \frac{n}{d}t_1 \equiv x_0 + \frac{n}{d}t_2 \pmod{n}$$

$$\frac{n}{d}t_1 \equiv \frac{n}{d}t_2 \pmod{n}$$

Since $\gcd(\frac{n}{d}, n) = \frac{n}{d}$, we can divide by $\frac{n}{d}$ (carefully, effectively modulo d):

$$t_1 \equiv t_2 \pmod{d}$$

Thus, a complete set of incongruent solutions is obtained by choosing t from the set $\{0, 1, 2, \dots, d-1\}$. This yields exactly d distinct solutions modulo n . ■

Example 2.3.3. Solve the congruence $18x \equiv 30 \pmod{42}$.

Solution: 1. Find $d = \gcd(18, 42)$. $18 = 2 \cdot 3^2$, $42 = 2 \cdot 3 \cdot 7$. Thus $d = 6$. 2. Check condition: Does $6 \mid 30$? Yes. Therefore, there are exactly 6 incongruent solutions. 3. Simplify: Divide the entire congruence by $d = 6$:

$$3x \equiv 5 \pmod{7}$$

4. Solve for x : Multiply by 5 (since $3 \cdot 5 = 15 \equiv 1 \pmod{7}$):

$$15x \equiv 25 \pmod{7} \implies x \equiv 4 \pmod{7}$$

5. Lift back to modulo 42: The solutions are of the form $x = 4 + 7t$ for $t = 0, 1, \dots, 5$.

$$x \in \{4, 11, 18, 25, 32, 39\}$$

2.3.2 Applications: Modular Inverse

An important application of linear congruences is finding the modular multiplicative inverse.

Definition 2.3.4. An integer $\bar{a}$ is the inverse of a modulo n if $a\bar{a} \equiv 1 \pmod{n}$.

This is a specific case of linear congruence $ax \equiv 1 \pmod{n}$. By the theorem above, an inverse exists if and only if $\gcd(a, n) \mid 1$, which implies $\gcd(a, n) = 1$.

2.4 Euclid's Second Theorem

While nomenclature varies slightly in different mathematical traditions, "Euclid's First Theorem" usually refers to Euclid's Lemma, and "Euclid's Second Theorem" refers to the infinitude of prime numbers.

Theorem 2.4.1 (Euclid's Second Theorem). *The number of prime numbers is infinite.*

Proof. See Theorem 1.3.4. ■

2.5 Fermat Numbers

Pierre de Fermat studied numbers of the form $2^m + 1$. He observed that if $2^m + 1$ is prime, then m must be a power of 2.

Definition 2.5.1. The integers of the form $F_n = 2^{2^n} + 1$ for non-negative integers n are called **Fermat Numbers**.

The first few Fermat numbers are:

- $F_0 = 2^{2^0} + 1 = 2^1 + 1 = 3$ (Prime)
- $F_1 = 2^{2^1} + 1 = 4 + 1 = 5$ (Prime)
- $F_2 = 2^{2^2} + 1 = 16 + 1 = 17$ (Prime)
- $F_3 = 2^{2^3} + 1 = 256 + 1 = 257$ (Prime)
- $F_4 = 2^{2^4} + 1 = 65536 + 1 = 65537$ (Prime)

Fermat conjectured that all such numbers are prime. However, Euler showed in 1732 that $F_5 = 2^{32} + 1$ is divisible by 641, disproving the conjecture.

Properties of Fermat Numbers

Lemma 2.5.2 (Recurrence Relation). *For any $n \geq 1$:*

$$F_n = F_0 F_1 F_2 \dots F_{n-1} + 2$$

Proof. We use mathematical induction. **Base Case** ($n = 1$):

$$F_1 = 5 \quad \text{and} \quad F_0 + 2 = 3 + 2 = 5$$

The relation holds.

Inductive Step: Assume the relation holds for $n = k$:

$$\prod_{i=0}^{k-1} F_i = F_k - 2$$

We examine the product for $n = k + 1$:

$$\prod_{i=0}^k F_i = \left(\prod_{i=0}^{k-1} F_i \right) F_k = (F_k - 2) F_k$$

Since $F_k = 2^{2^k} + 1$:

$$(2^{2^k} + 1 - 2)(2^{2^k} + 1) = (2^{2^k} - 1)(2^{2^k} + 1)$$

Using difference of squares:

$$(2^{2^k})^2 - 1 = 2^{2 \cdot 2^k} - 1 = 2^{2^{k+1}} - 1$$

We want to show this equals $F_{k+1} - 2$:

$$F_{k+1} - 2 = (2^{2^{k+1}} + 1) - 2 = 2^{2^{k+1}} - 1$$

The sides match. Thus, the property holds for all n . ■

Theorem 2.5.3 (Goldbach's Theorem on Fermat Numbers). *Any two distinct Fermat numbers are relatively prime. That is, if $m \neq n$, then $\gcd(F_m, F_n) = 1$.*

Proof. Let $m < n$. By the recurrence lemma derived above:

$$F_n - 2 = F_0 F_1 \dots F_m \dots F_{n-1}$$

Thus, F_m is a factor of the product $F_0 \dots F_{n-1}$. This implies that F_m divides $(F_n - 2)$. Let $d = \gcd(F_m, F_n)$. Since $d \mid F_m$ and $F_m \mid (F_n - 2)$, it follows that $d \mid (F_n - 2)$. We also know $d \mid F_n$. Therefore, d must divide the difference:

$$d \mid (F_n - (F_n - 2)) \implies d \mid 2$$

The divisors of 2 are 1 and 2. Since Fermat numbers are of the form $2^k + 1$, they are all odd integers. Thus, d cannot be 2. Therefore, $d = 1$. ■

Corollary 2.5.4. *This theorem provides an alternative proof for Euclid's Second Theorem (Infinitude of Primes). Since there are infinitely many Fermat numbers, and they are pairwise coprime, each must define at least one unique prime factor. Hence, there are infinitely many primes.*

Exercise 2.5.5. 1. Determine whether the linear Diophantine equation $172x + 20y = 1000$ is solvable. If so, find the general solution in integers.

2. Find all positive integer solutions to the equation $18x + 7y = 520$.

3. A man has \$100 and wants to buy 100 animals. Cows cost \$5, sheep cost \$1, and rabbits cost \$0.50. If he must buy at least one of each, find all possible combinations of animals he can purchase.

4. Solve the linear congruence $15x \equiv 9 \pmod{21}$. How many mutually incongruent solutions exist modulo 21?

5. Find the general solution to the congruence $12x \equiv 48 \pmod{18}$.

6. Use the theory of linear congruences to find the remainder when 3^{100} is divided by 7.

7. Calculate the values of the first five Fermat numbers: F_0, F_1, F_2, F_3 , and F_4 . Verify that they are all prime.

8. Prove that every Fermat number $F_n = 2^{2^n} + 1$ (for $n > 0$) ends in the digit 7 when written in base 10.

9. Prove that for any $n > 0$, $F_n = F_0 F_1 F_2 \dots F_{n-1} + 2$.

10. Use Euclid's Second Theorem (the infinitude of primes) to show that the n -th prime p_n satisfies the inequality $p_n \leq 2^{2^{n-1}}$.

11. Find all integers x such that $x \equiv 1 \pmod{3}$, $x \equiv 2 \pmod{4}$, and $x \equiv 3 \pmod{5}$.

12. Determine the necessary and sufficient condition on c for the equation $ax + by + cz = d$ to have integer solutions.

13. Prove that if $\gcd(a, n) = 1$, then the linear congruence $ax \equiv b \pmod{n}$ has exactly one solution modulo n .
14. Verify that $F_5 = 2^{2^5} + 1$ is divisible by 641, thus showing F_5 is composite.
15. Show that any two distinct Fermat numbers are relatively prime, i.e., $\gcd(F_m, F_n) = 1$ for $m \neq n$.
16. Use the result of the previous problem to provide an alternative proof for Euclid's Second Theorem.
17. Solve the simultaneous linear Diophantine equations:

$$\begin{aligned}x + y + z &= 100 \\x + 8y + 50z &= 156\end{aligned}$$

for positive integers x, y, z .

18. Prove that if x_0, y_0 is a particular solution to $ax + by = c$, then all solutions are given by $x = x_0 + (b/d)t$ and $y = y_0 - (a/d)t$, where $d = \gcd(a, b)$.
19. Find the inverse of 17 modulo 101 using the Euclidean Algorithm.
20. Show that for $n > 1$, any prime factor p of the Fermat number F_n must be of the form $p = k \cdot 2^{n+1} + 1$.

Challenges:

21. Prove that for $n \geq 2$, the last two digits of F_n are always 17.
22. Let a and b be positive integers such that $ax + by = ab - a - b$ has no non-negative integer solutions. Prove that this is the largest such integer (The Frobenius Coin Problem for $n = 2$).
23. Prove that the sum of the reciprocals of the primes, $\sum_{p \in \text{Primes}} \frac{1}{p}$, diverges. How does this strengthen Euclid's Second Theorem?
24. Prove that for $n > 0$, $F_n \mid 2^{F_n} - 2$.
25. Show that no Fermat number F_n (for $n > 1$) can be a perfect square or a perfect cube.
26. If p is an odd prime, prove that all prime factors of $2^p - 1$ are of the form $2kp + 1$.
27. Prove that there are infinitely many primes of the form $4n + 3$ using a modification of Euclid's proof for the infinitude of primes.
28. Show that the recursive relation $F_n = (F_{n-1} - 1)^2 + 1$ holds for all $n \geq 1$.
29. Let x, y, z be a solution to $x^2 + y^2 = z^2$. Prove that at least one of x, y is divisible by 3 and at least one of x, y is divisible by 4.
30. Prove the Pepin Test: F_n is prime if and only if $3^{(F_n-1)/2} \equiv -1 \pmod{F_n}$ for $n > 0$.

Chapter 3

Residue Systems and Number Theoretic Functions

This unit delves deeper into the structural properties of modular arithmetic by defining Complete and Reduced Residue Systems. These systems form the basis for pivotal theorems by Fermat and Euler. Additionally, the unit examines arithmetic functions, such as Euler's ϕ -function, and techniques for solving simultaneous congruences using the Chinese Remainder Theorem.

3.1 Complete Residue System (CRS)

The concept of a Complete Residue System is fundamental to modular arithmetic. It represents a set of numbers that covers every possible remainder modulo n .

Definition 3.1.1 (Complete Residue System). A set of integers $S = \{x_1, x_2, \dots, x_k\}$ is called a **Complete Residue System (CRS)** modulo n if:

1. The set contains exactly n elements (i.e., $k = n$).
2. No two elements in S are congruent modulo n (i.e., $x_i \not\equiv x_j \pmod{n}$ for $i \neq j$).
3. Every integer y is congruent to exactly one element of S modulo n .

Remark 3.1.2. The most fundamental example of a CRS modulo n is the set of least non-negative residues:

$$\{0, 1, 2, \dots, n-1\}$$

However, a CRS can consist of any n integers, including negative numbers, provided they are mutually incongruent.

Example 3.1.3. Consider modulo $n = 5$.

- The set $\{0, 1, 2, 3, 4\}$ is the standard CRS.
- The set $\{5, 6, 7, 8, 9\}$ is also a CRS (since $5 \equiv 0, 6 \equiv 1$, etc.).
- The set $\{-2, -1, 0, 1, 2\}$ is a CRS (often useful for symmetry).
- The set $\{0, 1, 2, 3, 8\}$ is **not** a CRS because $8 \equiv 3 \pmod{5}$, so the elements are not distinct modulo 5.

3.1.1 Properties of CRS

Theorem 3.1.4. A set of n integers $\{x_1, x_2, \dots, x_n\}$ is a Complete Residue System modulo n if and only if no two elements are congruent modulo n .

Proof. If the set is a CRS, then by definition, elements are pairwise incongruent. Conversely, suppose we have n integers that are mutually incongruent modulo n . Since there are exactly n distinct equivalence classes modulo n (represented by $0, 1, \dots, n-1$), and we have n distinct representatives, by the Pigeonhole Principle, each equivalence class must be represented exactly once. Thus, every integer is congruent to exactly one element in the set. ■

Theorem 3.1.5 (Linear Transformation of a CRS). *If $\{x_1, x_2, \dots, x_n\}$ is a complete residue system modulo n , and a is an integer such that $\gcd(a, n) = 1$, then the set*

$$\{ax_1 + b, ax_2 + b, \dots, ax_n + b\}$$

is also a complete residue system modulo n for any integer b .

Proof. Let $S' = \{ax_1 + b, \dots, ax_n + b\}$. Since the original set has n elements, S' also has n elements. To prove it is a CRS, we only need to show that no two elements are congruent modulo n .

Suppose, for the sake of contradiction, that two distinct elements are congruent:

$$ax_i + b \equiv ax_j + b \pmod{n}$$

for some $i \neq j$. Subtracting b from both sides:

$$ax_i \equiv ax_j \pmod{n}$$

Since $\gcd(a, n) = 1$, we can cancel a from the congruence (by the cancellation law of modular arithmetic):

$$x_i \equiv x_j \pmod{n}$$

This contradicts the hypothesis that $\{x_1, \dots, x_n\}$ is a CRS (where all elements are distinct modulo n). Therefore, the elements of S' are mutually incongruent, and S' is a CRS. ■

3.2 Reduced Residue System (RRS)

While a CRS considers all integers, a Reduced Residue System focuses only on integers that are relatively prime to the modulus. This structure is crucial for the study of multiplicative groups modulo n .

Definition 3.2.1 (Euler's ϕ -function). For $n \geq 1$, Euler's totient function, denoted by $\phi(n)$, is the number of positive integers not exceeding n that are relatively prime to n , that is, for any positive integer n , the function is defined as:

$$\phi(n) = |\{m \in \mathbb{Z} : 1 \leq m \leq n, \gcd(m, n) = 1\}|$$

Key Properties and Formulas (Proofs shall be given later in this chapter.)

1. **Prime Numbers:** If p is a prime number, then every number less than p is coprime to it.

$$\phi(p) = p - 1$$

2. **Prime Powers:** If $n = p^k$ for some prime p and integer $k \geq 1$:

$$\phi(p^k) = p^k - p^{k-1} = p^k \left(1 - \frac{1}{p}\right)$$

3. **Multiplicative Property:** If $\gcd(m, n) = 1$, then:

$$\phi(mn) = \phi(m)\phi(n)$$

4. General Formula: For an integer n with prime factorization $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$:

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

Example 3.2.2. The prime factorization of 10 is 2×5 . Using the general formula:

$$\phi(10) = 10 \left(1 - \frac{1}{2}\right) \left(1 - \frac{1}{5}\right) = 10 \left(\frac{1}{2}\right) \left(\frac{4}{5}\right) = 4$$

The numbers coprime to 10 are $\{1, 3, 7, 9\}$.

The prime factorization is 3×7 . Since $\gcd(3, 7) = 1$, we can use the multiplicative property:

$$\phi(21) = \phi(3) \times \phi(7) = (3 - 1) \times (7 - 1) = 2 \times 6 = 12$$

Definition 3.2.3 (Reduced Residue System). A set of integers $R = \{r_1, r_2, \dots, r_k\}$ is called a **Reduced Residue System (RRS)** modulo n if:

1. The set contains exactly $\phi(n)$ elements (i.e., $k = \phi(n)$).
2. $\gcd(r_i, n) = 1$ for all i (every element is coprime to n).
3. No two elements in R are congruent modulo n .

Remark 3.2.4. A standard RRS is obtained by taking a CRS $\{0, 1, \dots, n-1\}$ and removing all numbers that share a common factor with n greater than 1.

Example 3.2.5. Consider modulo $n = 12$. The standard CRS is $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11\}$. We remove numbers not coprime to 12:

- Divisible by 2: 0, 2, 4, 6, 8, 10
- Divisible by 3: 3, 9

Remaining numbers: $\{1, 5, 7, 11\}$. Thus, $\phi(12) = 4$, and $\{1, 5, 7, 11\}$ is an RRS modulo 12.

Example 3.2.6. Consider modulo p where p is a prime. Since every positive integer less than p is coprime to p , the set $\{1, 2, \dots, p-1\}$ is an RRS. Thus, $\phi(p) = p-1$.

3.2.1 Properties of RRS

Theorem 3.2.7. If $\{r_1, r_2, \dots, r_{\phi(n)}\}$ is a reduced residue system modulo n , and a is an integer such that $\gcd(a, n) = 1$, then the set

$$\{ar_1, ar_2, \dots, ar_{\phi(n)}\}$$

is also a reduced residue system modulo n .

Proof. Let $R' = \{ar_1, ar_2, \dots, ar_{\phi(n)}\}$. The set R' has $\phi(n)$ elements. To prove it is an RRS, we must satisfy two conditions:

1. **Coprimality:** We must show $\gcd(ar_i, n) = 1$ for all i . We are given that $\gcd(r_i, n) = 1$ (definition of RRS) and $\gcd(a, n) = 1$ (hypothesis). Since the product of two numbers coprime to n is also coprime to n , it follows that $\gcd(ar_i, n) = 1$.
2. **Incongruence:** We must show elements are distinct modulo n . Suppose $ar_i \equiv ar_j \pmod{n}$. Since $\gcd(a, n) = 1$, we can cancel a , yielding $r_i \equiv r_j \pmod{n}$. Since the original set was an RRS, r_i and r_j are distinct for $i \neq j$. Thus, the elements of R' are distinct.

Since both conditions are met, R' is an RRS modulo n . ■

Corollary 3.2.8 (Basis for Euler's Theorem). *Let $\{r_1, \dots, r_{\phi(n)}\}$ be an RRS modulo n . If $\gcd(a, n) = 1$, then the product of elements in the original set is congruent to the product of elements in the scaled set:*

$$\prod_{i=1}^{\phi(n)} (ar_i) \equiv \prod_{i=1}^{\phi(n)} r_i \pmod{n}$$

Proof. Since $\{ar_1, \dots, ar_{\phi(n)}\}$ is also an RRS, its elements are simply the elements of the original set $\{r_1, \dots, r_{\phi(n)}\}$ in a different order (modulo n). Therefore, the product of the elements must be congruent.

$$a^{\phi(n)}(r_1 r_2 \dots r_{\phi(n)}) \equiv (r_1 r_2 \dots r_{\phi(n)}) \pmod{n}$$

Since each r_i is coprime to n , their product $P = \prod r_i$ is also coprime to n . We can cancel P from both sides, yielding Euler's Theorem:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

■

Theorem 3.2.9. *If $n = p^k$ for some prime p and integer $k \geq 1$, then:*

$$\phi(p^k) = p^k - p^{k-1} = p^k \left(1 - \frac{1}{p}\right)$$

where $\phi(n)$ is the Euler's totient function, which counts the number of positive integers less than or equal to n that are relatively prime to n .

By definition, $\phi(p^k)$ counts the number of integers in the set $S = \{1, 2, 3, \dots, p^k\}$ that are relatively prime to p^k . Two integers are relatively prime if their greatest common divisor is 1.

Since p is the only prime factor of p^k , an integer $m \in S$ shares a common factor with p^k if and only if m is a multiple of p . Therefore, to find $\phi(p^k)$, we must count how many multiples of p exist in the set S and subtract that count from the total number of elements.

The multiples of p in the set $\{1, 2, \dots, p^k\}$ are given by the sequence:

$$p, 2p, 3p, \dots, (p^{k-1})p$$

The largest multiple is $(p^{k-1})p = p^k$. By observing the coefficients $1, 2, 3, \dots, p^{k-1}$, we see that there are exactly p^{k-1} such multiples.

To calculate the number of integers relatively prime to p^k , we subtract the number of multiples of p from the total number of elements in S :

$$\begin{aligned} \phi(p^k) &= p^k - p^{k-1} \\ &= p^k \left(1 - \frac{p^{k-1}}{p^k}\right) \\ &= p^k \left(1 - \frac{1}{p}\right) \end{aligned}$$

This completes the proof.

3.3 The Chinese Remainder Theorem

Theorem 3.3.1 (Chinese Remainder Theorem). *Let $m_1, m_2, \dots, m_k$ be positive integers such that $\gcd(m_i, m_j) = 1$ for $i \neq j$. Then for any integers $a_1, a_2, \dots, a_k$, the system of simultaneous congruences:*

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ &\vdots \\ x &\equiv a_k \pmod{m_k} \end{aligned}$$

has a solution, and the solution is unique modulo $M = m_1 m_2 \dots m_k$.

Proof. We first prove the existence of a solution. Let $M = m_1 m_2 \cdots m_k$. For each $i \in \{1, 2, \dots, k\}$, define:

$$M_i = \frac{M}{m_i} = m_1 m_2 \cdots m_{i-1} m_{i+1} \cdots m_k$$

Since $\gcd(m_i, m_j) = 1$ for all $i \neq j$, it follows that $\gcd(M_i, m_i) = 1$. Therefore, there exists an integer y_i such that:

$$M_i y_i \equiv 1 \pmod{m_i}$$

Now, consider the sum:

$$x = a_1 M_1 y_1 + a_2 M_2 y_2 + \cdots + a_k M_k y_k = \sum_{i=1}^k a_i M_i y_i$$

For any specific j , consider $x \pmod{m_j}$. Note that for all $i \neq j$, M_i contains m_j as a factor, so $M_i \equiv 0 \pmod{m_j}$. Thus:

$$x \equiv a_j M_j y_j \pmod{m_j}$$

Since we chose y_j such that $M_j y_j \equiv 1 \pmod{m_j}$, we have:

$$x \equiv a_j(1) \equiv a_j \pmod{m_j}$$

This holds for all $j = 1, \dots, k$, so x is a valid solution.

To prove uniqueness, suppose x_1 and x_2 are both solutions to the system. Then for each i :

$$x_1 \equiv a_i \pmod{m_i} \quad \text{and} \quad x_2 \equiv a_i \pmod{m_i}$$

This implies $x_1 - x_2 \equiv 0 \pmod{m_i}$ for all i . Thus, each m_i divides $x_1 - x_2$. Since the m_i are pairwise relatively prime, their product M must also divide $x_1 - x_2$. Hence:

$$x_1 \equiv x_2 \pmod{M}$$

The solution is therefore unique modulo M . ■

Example 3.3.2. Find the smallest positive integer x such that:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

Solution. We have $m_1 = 3, m_2 = 5, m_3 = 7$ and $a_1 = 2, a_2 = 3, a_3 = 2$. First, calculate $M = 3 \times 5 \times 7 = 105$. Then calculate M_i and their inverses $y_i \pmod{m_i}$:

- $M_1 = 105/3 = 35$. We solve $35y_1 \equiv 1 \pmod{3} \implies 2y_1 \equiv 1 \pmod{3}$. Thus $y_1 = 2$.
- $M_2 = 105/5 = 21$. We solve $21y_2 \equiv 1 \pmod{5} \implies 1y_2 \equiv 1 \pmod{5}$. Thus $y_2 = 1$.
- $M_3 = 105/7 = 15$. We solve $15y_3 \equiv 1 \pmod{7} \implies 1y_3 \equiv 1 \pmod{7}$. Thus $y_3 = 1$.

The solution is:

$$\begin{aligned} x &= a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 \\ x &= (2)(35)(2) + (3)(21)(1) + (2)(15)(1) = 140 + 63 + 30 = 233 \end{aligned}$$

To find the smallest positive integer, we take $233 \pmod{105}$:

$$233 = 2(105) + 23 \implies x = 23$$

Verification: $23 \equiv 2 \pmod{3}$, $23 \equiv 3 \pmod{5}$, and $23 \equiv 2 \pmod{7}$. ■

Example 3.3.3. A classic puzzle: A woman is carrying a basket of eggs. If she takes them out 2 at a time, 1 is left over. If 3 at a time, 2 are left over. If 5 at a time, 4 are left over. What is the smallest number of eggs she could have?

Solution. The problem can be written as a system of congruences:

$$\begin{aligned}x &\equiv 1 \pmod{2} \\x &\equiv 2 \pmod{3} \\x &\equiv 4 \pmod{5}\end{aligned}$$

Note that these can be rewritten as $x \equiv -1 \pmod{2}$, $x \equiv -1 \pmod{3}$, and $x \equiv -1 \pmod{5}$. Since $x \equiv -1$ satisfies all three conditions, by the uniqueness part of the CRT, the general solution is:

$$x \equiv -1 \pmod{2 \times 3 \times 5} \implies x \equiv -1 \pmod{30}$$

The smallest positive integer is $x = 30 - 1 = 29$. ■

Generalization of the Chinese Remainder Theorem for Non-Relatively Prime Moduli

Theorem 3.3.4. *The system of linear congruences*

$$\begin{aligned}x &\equiv a_1 \pmod{m_1} \\x &\equiv a_2 \pmod{m_2}\end{aligned}$$

has a solution if and only if $a_1 \equiv a_2 \pmod{\gcd(m_1, m_2)}$. If a solution exists, it is unique modulo $[m_1, m_2]$ (the least common multiple of m_1 and m_2).

Proof. Let $g = \gcd(m_1, m_2)$.

Necessity: Suppose a solution x exists. Then there exist integers k_1 and k_2 such that:

$$x = a_1 + k_1 m_1 \quad \text{and} \quad x = a_2 + k_2 m_2$$

Equating the two expressions for x , we have:

$$a_1 + k_1 m_1 = a_2 + k_2 m_2 \implies a_1 - a_2 = k_2 m_2 - k_1 m_1$$

Since g divides both m_1 and m_2 , it must divide the linear combination $k_2 m_2 - k_1 m_1$. Thus, g divides $a_1 - a_2$, which implies $a_1 \equiv a_2 \pmod{g}$.

Sufficiency: Suppose $a_1 \equiv a_2 \pmod{g}$. Then $a_2 - a_1 = qg$ for some integer q . By Bézout's Identity, there exist integers s and t such that $sm_1 + tm_2 = g$. Multiplying by q gives:

$$qsm_1 + qtm_2 = qg = a_2 - a_1$$

Rearranging the terms, we get:

$$a_1 + (qs)m_1 = a_2 - (qt)m_2$$

Let $x = a_1 + (qs)m_1$. By construction, $x \equiv a_1 \pmod{m_1}$. Also, $x = a_2 - (qt)m_2$, which implies $x \equiv a_2 \pmod{m_2}$. Thus, x is a solution.

Uniqueness: If x and y are two solutions, then $x \equiv y \pmod{m_1}$ and $x \equiv y \pmod{m_2}$. This means $x - y$ is a multiple of both m_1 and m_2 . By definition, $x - y$ must be a multiple of the least common multiple, $[m_1, m_2]$. Therefore, $x \equiv y \pmod{[m_1, m_2]}$. ■

Method of Solution: The Method of Successive Substitution

For a general system of k congruences where moduli are not necessarily coprime, we solve them pairs at a time.

1. Start with the first two congruences: $x \equiv a_1 \pmod{m_1}$ and $x \equiv a_2 \pmod{m_2}$.
2. Check if $a_1 \equiv a_2 \pmod{\gcd(m_1, m_2)}$. If not, there is no solution.
3. Write the first congruence as $x = m_1 k + a_1$.

4. Substitute this into the second congruence: $m_1k + a_1 \equiv a_2 \pmod{m_2} \implies m_1k \equiv a_2 - a_1 \pmod{m_2}$.
5. Solve this linear congruence for k . Since the consistency condition is met, a solution $k \equiv k_0 \pmod{m_2/g}$ exists.
6. Substitute $k = (m_2/g)j + k_0$ back into the expression for x to get a single congruence: $x \equiv a_{1,2} \pmod{[m_1, m_2]}$.
7. Repeat the process using this new congruence and the next one in the system.

Example 3.3.5. Find the solution to the system:

$$\begin{aligned} x &\equiv 2 \pmod{6} \\ x &\equiv 8 \pmod{10} \end{aligned}$$

Solution. First, we check the consistency condition. Here $a_1 = 2, m_1 = 6$ and $a_2 = 8, m_2 = 10$. The greatest common divisor is $g = \gcd(6, 10) = 2$. Checking the condition: $a_2 - a_1 = 8 - 2 = 6$. Since $2 \mid 6$, a solution exists.

We write the first congruence as $x = 6k + 2$. Substituting into the second:

$$\begin{aligned} 6k + 2 &\equiv 8 \pmod{10} \\ 6k &\equiv 6 \pmod{10} \end{aligned}$$

We can divide the congruence by $\gcd(6, 10) = 2$:

$$3k \equiv 3 \pmod{5}$$

Multiplying by the inverse of 3 modulo 5 (which is 2):

$$6k \equiv 6 \pmod{5} \implies k \equiv 1 \pmod{5}$$

Thus $k = 5j + 1$. Substituting this back into x :

$$x = 6(5j + 1) + 2 = 30j + 6 + 2 = 30j + 8$$

The solution is $x \equiv 8 \pmod{30}$. ■

Another Method of Solution

Example 3.3.6. Find the solution to the system of congruences:

$$\begin{aligned} x &\equiv 2 \pmod{6} \\ x &\equiv 8 \pmod{10} \end{aligned}$$

Solve this by breaking each congruence into its equivalent prime power components.

Solution. By 3.3.4, the given problem has a solution. Since the moduli 6 and 10 are not relatively prime, we begin by decomposing each congruence into a system of congruences with prime power moduli.

Decomposition of Congruences The first congruence, $x \equiv 2 \pmod{6}$, can be split based on the factors of $6 = 2 \times 3$:

$$\begin{aligned} x &\equiv 2 \pmod{2} \implies x \equiv 0 \pmod{2} \\ x &\equiv 2 \pmod{3} \end{aligned}$$

The second congruence, $x \equiv 8 \pmod{10}$, can be split based on the factors of $10 = 2 \times 5$:

$$\begin{aligned} x &\equiv 8 \pmod{2} \implies x \equiv 0 \pmod{2} \\ x &\equiv 8 \pmod{5} \implies x \equiv 3 \pmod{5} \end{aligned}$$

Collecting the distinct congruences, the problem is now reduced to a system with pairwise relatively prime moduli:

$$\begin{aligned}x &\equiv 0 \pmod{2} \\x &\equiv 2 \pmod{3} \\x &\equiv 3 \pmod{5}\end{aligned}$$

Apply the Chinese Remainder Theorem: Let $m_1 = 2, m_2 = 3, m_3 = 5$ and $a_1 = 0, a_2 = 2, a_3 = 3$. The product of the moduli is $M = 2 \times 3 \times 5 = 30$. We calculate $M_i = M/m_i$ and their respective modular inverses $y_i \pmod{m_i}$:

- $M_1 = 15$: We have $15 \equiv 1 \pmod{2}$, so the inverse is $y_1 = 1$.
- $M_2 = 10$: We have $10 \equiv 1 \pmod{3}$, so the inverse is $y_2 = 1$.
- $M_3 = 6$: We have $6 \equiv 1 \pmod{5}$, so the inverse is $y_3 = 1$.

The simultaneous solution is:

$$\begin{aligned}x &= a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 \\x &= (0)(15)(1) + (2)(10)(1) + (3)(6)(1) \\x &= 0 + 20 + 18 = 38\end{aligned}$$

Reduce the result modulo $M = 30$ to find the smallest positive solution:

$$x \equiv 38 \equiv 8 \pmod{30}$$

Thus, the solution to the system is $x \equiv 8 \pmod{30}$. ■

Theorem 3.3.7. Euler's totient function $\phi(n)$ is a multiplicative function. That is, if m and n are positive integers such that $\gcd(m, n) = 1$, then:

$$\phi(mn) = \phi(m)\phi(n)$$

Proof. Let m and n be coprime positive integers. By definition, $\phi(mn)$ is the number of integers x such that $1 \leq x \leq mn$ and $\gcd(x, mn) = 1$. Consider the mapping f from the set of residue classes modulo mn to the Cartesian product of the residue classes modulo m and modulo n :

$$f : \mathbb{Z}/mn\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$$

defined by

$$f(x \pmod{mn}) = (x \pmod{m}, x \pmod{n})$$

According to the Chinese Remainder Theorem, since $\gcd(m, n) = 1$, this mapping is a bijection. An integer x is relatively prime to mn if and only if it is relatively prime to both m and n . This can be expressed as:

$$\gcd(x, mn) = 1 \iff \gcd(x, m) = 1 \text{ and } \gcd(x, n) = 1$$

Under the bijection f , the element $x \pmod{mn}$ is an element of the multiplicative group of units $(\mathbb{Z}/mn\mathbb{Z})^\times$ if and only if its image $(x \pmod{m}, x \pmod{n})$ is an element of the product set $(\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$.

The size of the set $(\mathbb{Z}/mn\mathbb{Z})^\times$ is, by definition, $\phi(mn)$. The size of the set $(\mathbb{Z}/m\mathbb{Z})^\times \times (\mathbb{Z}/n\mathbb{Z})^\times$ is the product of the sizes of the individual sets, which is $\phi(m)\phi(n)$.

Since f provides a one-to-one correspondence between these two sets, their cardinalities must be equal. Therefore:

$$\phi(mn) = \phi(m)\phi(n)$$

This completes the proof. ■

2nd Proof. To prove the theorem, we arrange the mn integers from 1 to mn in a rectangular array with m columns and n rows as follows:

$$\begin{array}{ccccccc}
 1 & 2 & \dots & r & \dots & m \\
 m+1 & m+2 & \dots & m+r & \dots & 2m \\
 2m+1 & 2m+2 & \dots & 2m+r & \dots & 3m \\
 \vdots & \vdots & & \vdots & & \vdots \\
 (n-1)m+1 & (n-1)m+2 & \dots & (n-1)m+r & \dots & nm
 \end{array}$$

Recall that $\phi(mn)$ is the number of integers in this array that are relatively prime to mn . An integer x is relatively prime to mn if and only if $\gcd(x, m) = 1$ and $\gcd(x, n) = 1$.

First, we consider the condition $\gcd(x, m) = 1$. For any element in the r -th column, the elements are of the form $qm + r$ for $0 \leq q < n$. Since $qm + r \equiv r \pmod{m}$, it follows that $\gcd(qm + r, m) = \gcd(r, m)$. Therefore, an entire column consists of integers relatively prime to m if and only if the column header r is relatively prime to m . By the definition of the totient function, there are exactly $\phi(m)$ such columns.

Next, we examine how many integers in each of these $\phi(m)$ columns are also relatively prime to n . Consider one such column where $\gcd(r, m) = 1$. The elements in this column are:

$$\{r, m+r, 2m+r, \dots, (n-1)m+r\}$$

We claim that these n integers form a complete residue system modulo n . To see this, suppose that two elements in the column are congruent modulo n :

$$im + r \equiv jm + r \pmod{n}$$

where $0 \leq i, j < n$. Subtracting r from both sides gives $im \equiv jm \pmod{n}$. Since we are given that $\gcd(m, n) = 1$, we can divide by m to obtain $i \equiv j \pmod{n}$. Because both i and j are less than n , it must be that $i = j$.

Since the n elements in the column form a complete residue system modulo n , the set of remainders when these elements are divided by n is simply $\{0, 1, 2, \dots, n-1\}$. In any complete residue system modulo n , the number of elements relatively prime to n is exactly $\phi(n)$.

Thus, we have $\phi(m)$ columns containing integers relatively prime to m , and each of these columns contains exactly $\phi(n)$ integers relatively prime to n . The total number of integers $x \in \{1, \dots, mn\}$ such that $\gcd(x, m) = 1$ and $\gcd(x, n) = 1$ is therefore the product of these two counts:

$$\phi(mn) = \phi(m)\phi(n)$$

This completes the proof. ■

Theorem 3.3.8. For an integer n with prime factorization $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$, the Euler's totient function satisfies:

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

Proof. Since the prime powers $p_i^{a_i}$ are pairwise relatively prime, we apply the multiplicative property of the totient function:

$$\phi(n) = \phi(p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}) = \phi(p_1^{a_1}) \phi(p_2^{a_2}) \cdots \phi(p_k^{a_k})$$

Using the formula for prime powers, $\phi(p^k) = p^k(1 - \frac{1}{p})$, we substitute for each term:

$$\phi(n) = p_1^{a_1} \left(1 - \frac{1}{p_1}\right) p_2^{a_2} \left(1 - \frac{1}{p_2}\right) \cdots p_k^{a_k} \left(1 - \frac{1}{p_k}\right)$$

Rearranging the factors to group the prime powers together, we get:

$$\phi(n) = (p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}) \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right)$$

Substituting n for the product of its prime power factors yields:

$$\phi(n) = n \prod_{i=1}^k \left(1 - \frac{1}{p_i}\right) = n \prod_{p|n} \left(1 - \frac{1}{p}\right)$$

This completes the proof. ■

Theorem 3.3.9. For each positive integer $n \geq 1$:

$$n = \sum_{d|n} \phi(d)$$

where the sum is taken over all positive divisors d of n .

Proof. Consider the set of integers $S = \{1, 2, \dots, n\}$. We can partition this set into disjoint subsets S_d based on the greatest common divisor of each element with n . For each divisor d of n , let:

$$S_d = \{x \in \mathbb{Z} \mid 1 \leq x \leq n \text{ and } \gcd(x, n) = d\}$$

Since every $x \in S$ has a unique greatest common divisor with n , and that divisor must be a divisor of n , the collection of sets $\{S_d : d \mid n\}$ forms a partition of S . Therefore:

$$n = |S| = \sum_{d|n} |S_d|$$

Now we determine the size of each set S_d . The condition $\gcd(x, n) = d$ is equivalent to:

$$\gcd\left(\frac{x}{d}, \frac{n}{d}\right) = 1$$

where $1 \leq \frac{x}{d} \leq \frac{n}{d}$. Let $k = x/d$ and $m = n/d$. The number of such integers x is equal to the number of integers k such that $1 \leq k \leq m$ and $\gcd(k, m) = 1$. By definition, this is exactly $\phi(m) = \phi(n/d)$. Substituting this back into our sum:

$$n = \sum_{d|n} \phi\left(\frac{n}{d}\right)$$

As d ranges over all divisors of n , the value n/d also ranges over all divisors of n in reverse order. Thus, we can rewrite the summation index to obtain:

$$n = \sum_{d|n} \phi(d)$$

This completes the proof. ■

Example 3.3.10. Verify the theorem for $n = 6$.

Solution. The divisors of 6 are $\{1, 2, 3, 6\}$. We compute the totient values for each:

- $\phi(1) = 1$
- $\phi(2) = 1$
- $\phi(3) = 2$
- $\phi(6) = 2$

Summing these values: $1 + 1 + 2 + 2 = 6$. The theorem holds. ■

Example 3.3.11. Verify the theorem for $n = 10$.

Solution. The divisors of 10 are $\{1, 2, 5, 10\}$. We compute:

- $\phi(1) = 1$

- $\phi(2) = 1$
- $\phi(5) = 4$
- $\phi(10) = 4$

Summing these values: $1 + 1 + 4 + 4 = 10$. The theorem holds. ■

Theorem 3.3.12. For $n > 1$, the sum of the positive integers less than n and relatively prime to n is:

$$\sum_{\substack{1 \leq x < n \\ \gcd(x, n) = 1}} x = \frac{n\phi(n)}{2}$$

Proof. Let $A = \{x_1, x_2, \dots, x_{\phi(n)}\}$ be the set of integers less than n and relatively prime to n . A key property of the greatest common divisor is that $\gcd(x, n) = 1$ if and only if $\gcd(n - x, n) = 1$.

If x is an element of A , then $n - x$ is also an element of A . Furthermore, for $n > 2$, x cannot equal $n - x$ because $x = n/2$ would imply $\gcd(n/2, n) = n/2$, which is greater than 1. For the case $n = 2$, the only coprime is $x = 1$, and $n - x = 1$, so the set is $\{1\}$.

Let S be the sum of the elements in A . We can write S in two ways:

$$S = x_1 + x_2 + \dots + x_{\phi(n)}$$

$$S = (n - x_1) + (n - x_2) + \dots + (n - x_{\phi(n)})$$

Adding these two equations together:

$$2S = [x_1 + (n - x_1)] + [x_2 + (n - x_2)] + \dots + [x_{\phi(n)} + (n - x_{\phi(n)})]$$

Each term in the brackets simplifies to n . Since there are $\phi(n)$ such terms:

$$2S = n \cdot \phi(n)$$

Dividing by 2, we obtain the desired result:

$$S = \frac{n\phi(n)}{2}$$

Example 3.3.13. Find the sum of integers less than 12 that are relatively prime to 12. ■

Solution. The integers less than 12 and relatively prime to 12 are $\{1, 5, 7, 11\}$. The actual sum is $1 + 5 + 7 + 11 = 24$. Using the formula, we have $n = 12$ and $\phi(12) = 12(1 - 1/2)(1 - 1/3) = 4$:

$$\text{Sum} = \frac{12 \cdot 4}{2} = \frac{48}{2} = 24$$

The result matches. ■

Example 3.3.14. Find the sum of integers less than 8 that are relatively prime to 8.

Solution. The integers less than 8 and relatively prime to 8 are $\{1, 3, 5, 7\}$. The actual sum is $1 + 3 + 5 + 7 = 16$. Using the formula, we have $n = 8$ and $\phi(8) = 8(1 - 1/2) = 4$:

$$\text{Sum} = \frac{8 \cdot 4}{2} = \frac{32}{2} = 16$$

The result matches. ■

3.4 Fermat's Little Theorem and Euler's Theorem

Theorem 3.4.1 (Fermat's Little Theorem). *If p is a prime number and a is an integer such that $p \nmid a$, then:*

$$a^{p-1} \equiv 1 \pmod{p}$$

More generally, for any integer a , $a^p \equiv a \pmod{p}$.

Proof. Consider the set of the first $p - 1$ positive integers: $S = \{1, 2, 3, \dots, p - 1\}$. Since $p \nmid a$, we can multiply each element of S by a to form a new set:

$$S' = \{a, 2a, 3a, \dots, (p - 1)a\}$$

We claim that the elements of S' , when reduced modulo p , are simply a permutation of the elements of S . First, note that no element in S' is congruent to 0 (mod p) because p is prime and $p \nmid a$ and p does not divide any $k \in \{1, \dots, p - 1\}$. Second, no two elements in S' are congruent to each other modulo p . If $ia \equiv ja \pmod{p}$ for $1 \leq i < j \leq p - 1$, then $a(j - i) \equiv 0 \pmod{p}$. Since $\gcd(a, p) = 1$, we must have $j - i \equiv 0 \pmod{p}$, which is impossible as $0 < j - i < p$.

Since S' modulo p contains $p - 1$ distinct non-zero residues, it must be the set $\{1, 2, \dots, p - 1\}$ in some order. Taking the product of all elements in both sets, we have:

$$a \cdot 2a \cdot 3a \cdots (p - 1)a \equiv 1 \cdot 2 \cdot 3 \cdots (p - 1) \pmod{p}$$

$$a^{p-1}(p - 1)! \equiv (p - 1)! \pmod{p}$$

Since p is prime, $(p - 1)!$ is relatively prime to p . We can therefore divide both sides by $(p - 1)!$ to obtain:

$$a^{p-1} \equiv 1 \pmod{p}$$

Multiplying both sides by a gives the more general form $a^p \equiv a \pmod{p}$, which holds even if $p \mid a$. ■

Alternative Proof of Fermat's Little Theorem. We will prove the statement $a^p \equiv a \pmod{p}$ for all non-negative integers a using induction.

Base Case: For $a = 0$, $0^p = 0 \equiv 0 \pmod{p}$. For $a = 1$, $1^p = 1 \equiv 1 \pmod{p}$. The base cases hold.

Inductive Step: Assume the statement holds for k , such that $k^p \equiv k \pmod{p}$. We want to show that $(k + 1)^p \equiv k + 1 \pmod{p}$. Using the Binomial Theorem, we expand $(k + 1)^p$:

$$(k + 1)^p = k^p + \binom{p}{1}k^{p-1} + \binom{p}{2}k^{p-2} + \cdots + \binom{p}{p-1}k + 1$$

For any prime p , the binomial coefficients $\binom{p}{i}$ for $1 \leq i \leq p - 1$ are defined as:

$$\binom{p}{i} = \frac{p!}{i!(p - i)!}$$

Since p is prime and $1 \leq i < p$, the p in the numerator is not cancelled by any factor in the denominator. Thus, p divides $\binom{p}{i}$, meaning $\binom{p}{i} \equiv 0 \pmod{p}$ for all $1 \leq i \leq p - 1$. Returning to the expansion:

$$(k + 1)^p \equiv k^p + 0 + 0 + \cdots + 0 + 1 \pmod{p}$$

$$(k + 1)^p \equiv k^p + 1 \pmod{p}$$

By the inductive hypothesis, $k^p \equiv k \pmod{p}$. Substituting this into our expression:

$$(k + 1)^p \equiv k + 1 \pmod{p}$$

Thus, the statement holds for $k + 1$. By induction, $a^p \equiv a \pmod{p}$ for all non-negative integers a . For negative integers, the result follows from the fact that any negative integer is congruent to a positive integer modulo p . ■

Example 3.4.2 (Computing Large Powers). Find the remainder when 2^{100} is divided by 13.

Solution. Since 13 is prime and $\gcd(2, 13) = 1$, Fermat's Little Theorem states $2^{12} \equiv 1 \pmod{13}$. We divide the exponent 100 by 12: $100 = 12 \times 8 + 4$. Thus:

$$2^{100} = (2^{12})^8 \cdot 2^4 \equiv 1^8 \cdot 16 \pmod{13}$$

$$2^{100} \equiv 16 \equiv 3 \pmod{13}$$

The remainder is 3. ■

Example 3.4.3 (Modular Inverses). Find the modular inverse of 3 modulo 17.

Solution. We seek x such that $3x \equiv 1 \pmod{17}$. By Fermat's Little Theorem, $3^{16} \equiv 1 \pmod{17}$, which can be written as $3 \cdot 3^{15} \equiv 1 \pmod{17}$. Thus, $x \equiv 3^{15} \pmod{17}$. Calculating powers:

$$3^3 = 27 \equiv 10 \equiv -7 \pmod{17}$$

$$3^9 = (3^3)^3 \equiv (-7)^3 = -343$$

Since $-343 = 17 \times (-20) - 3$, we have $3^9 \equiv -3 \equiv 14 \pmod{17}$. Then $3^{15} = 3^9 \cdot 3^3 \cdot 3^3 \equiv 14 \cdot (-7) \cdot (-7) = 14 \cdot 49 \equiv 14 \cdot 15 = 210$. Since $210 = 17 \times 12 + 6$, the inverse is 6. (Check: $3 \times 6 = 18 \equiv 1 \pmod{17}$). ■

Example 3.4.4 (Primality Testing). Show that 117 is not prime using $a = 2$.

Solution. If 117 were prime, then $2^{116} \equiv 1 \pmod{117}$ must hold. $2^7 = 128 \equiv 11 \pmod{117}$. $2^{14} \equiv 121 \equiv 4 \pmod{117}$. $2^{28} \equiv 16 \pmod{117}$. $2^{56} \equiv 256 \equiv 22 \pmod{117}$. $2^{112} \equiv 484 \equiv 16 \pmod{117}$. $2^{116} = 2^{112} \cdot 2^4 \equiv 16 \cdot 16 = 256 \equiv 22 \pmod{117}$. Since $22 \not\equiv 1 \pmod{117}$, 117 cannot be prime. ■

Euler's Totient Theorem

Theorem 3.4.5 (Euler's Totient Theorem). If n is a positive integer and a is an integer such that $\gcd(a, n) = 1$, then:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

where $\phi(n)$ is the Euler's totient function.

Proof. Let the set of integers $\{r_1, r_2, \dots, r_{\phi(n)}\}$ be a reduced residue system modulo n . This set contains all integers $1 \leq r_i \leq n$ such that $\gcd(r_i, n) = 1$. Since $\gcd(a, n) = 1$, the set:

$$\{ar_1, ar_2, \dots, ar_{\phi(n)}\}$$

also forms a reduced residue system modulo n . This is because each ar_i is relatively prime to n (product of two numbers relatively prime to n is relatively prime to n), and no two elements are congruent modulo n (if $ar_i \equiv ar_j \pmod{n}$, then $r_i \equiv r_j \pmod{n}$ because $\gcd(a, n) = 1$).

Therefore, the set $\{ar_1, \dots, ar_{\phi(n)}\}$ is simply a permutation of $\{r_1, \dots, r_{\phi(n)}\}$ modulo n . Taking the product of all elements in both sets:

$$(ar_1)(ar_2) \cdots (ar_{\phi(n)}) \equiv r_1 r_2 \cdots r_{\phi(n)} \pmod{n}$$

$$a^{\phi(n)} (r_1 r_2 \cdots r_{\phi(n)}) \equiv r_1 r_2 \cdots r_{\phi(n)} \pmod{n}$$

Let $P = r_1 r_2 \cdots r_{\phi(n)}$. Since each r_i is relatively prime to n , their product P is also relatively prime to n . Thus, we can divide both sides of the congruence by P :

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

This completes the proof. ■

Alternative Proof of Euler's Theorem. Consider the set of integers $G = \{x \in \mathbb{Z} : 1 \leq x \leq n, \gcd(x, n) = 1\}$. This set forms a group under multiplication modulo n , known as the multiplicative group of integers modulo n , denoted by $(\mathbb{Z}/n\mathbb{Z})^\times$ or $U(n)$.

The order of this group G (the number of elements in it) is exactly $\phi(n)$ by the definition of Euler's totient function.

Let k be the order of the element $a \pmod{n}$ in the group $(\mathbb{Z}/n\mathbb{Z})^\times$. This means k is the smallest positive integer such that $a^k \equiv 1 \pmod{n}$. According to the Lagrange's theorem for finite groups, k must divide $|G| = \phi(n)$. Therefore, there exists an integer m such that $\phi(n) = mk$. Raising the element a to the power of $\phi(n)$:

$$a^{\phi(n)} = a^{mk} = (a^k)^m$$

Since $a^k \equiv 1 \pmod{n}$:

$$a^{\phi(n)} \equiv (1)^m \equiv 1 \pmod{n}$$

This completes the proof. ■

Example 3.4.6 (Units Digit of Large Powers). Find the last digit of 7^{222} .

Solution. The last digit is the remainder when divided by 10. Here $n = 10$ and $a = 7$. Since $\gcd(7, 10) = 1$ and $\phi(10) = 10(1 - 1/2)(1 - 1/5) = 4$, Euler's theorem gives $7^4 \equiv 1 \pmod{10}$. Dividing the exponent by 4: $222 = 4 \times 55 + 2$.

$$7^{222} = (7^4)^{55} \cdot 7^2 \equiv 1^{55} \cdot 49 \pmod{10}$$

$$7^{222} \equiv 9 \pmod{10}$$

The last digit is 9. ■

Example 3.4.7 (RSA Cryptography). In RSA, we choose two large primes p, q and let $n = pq$. The encryption exponent e is chosen such that $\gcd(e, \phi(n)) = 1$. The decryption exponent d is the modular inverse of e modulo $\phi(n)$. Show that $(m^e)^d \equiv m \pmod{n}$ for a message m where $\gcd(m, n) = 1$.

Solution. Since $ed \equiv 1 \pmod{\phi(n)}$, we have $ed = k\phi(n) + 1$ for some integer k . Then:

$$m^{ed} = m^{k\phi(n)+1} = (m^{\phi(n)})^k \cdot m^1$$

By Euler's Totient Theorem, since $\gcd(m, n) = 1$, $m^{\phi(n)} \equiv 1 \pmod{n}$. Substituting this:

$$m^{ed} \equiv (1)^k \cdot m \equiv m \pmod{n}$$

This ensures that the decrypted message is identical to the original message. ■

Example 3.4.8 (Divisibility Properties). Prove that for any integer a relatively prime to 35, $a^{12} - 1$ is divisible by 35.

Solution. Note that $35 = 5 \times 7$. We calculate $\phi(35) = \phi(5)\phi(7) = 4 \times 6 = 24$. However, we can find a smaller exponent using the fact that n is a product of primes. By Fermat's Little Theorem, $a^4 \equiv 1 \pmod{5}$, which implies $(a^4)^3 = a^{12} \equiv 1 \pmod{5}$. By Fermat's Little Theorem, $a^6 \equiv 1 \pmod{7}$, which implies $(a^6)^2 = a^{12} \equiv 1 \pmod{7}$. Since $a^{12} \equiv 1$ for both moduli 5 and 7, and $\gcd(5, 7) = 1$, it follows that $a^{12} \equiv 1 \pmod{35}$. Thus $35 \mid (a^{12} - 1)$. ■

Exercise 3.4.9. Solve the following exercises.

1. Determine whether the set $\{0, 5, 10, 15, 20, 25, 30\}$ forms a Complete Residue System (CRS) modulo 7. Justify your answer.
2. Construct a Reduced Residue System (RRS) modulo 12. How many elements does it contain?
3. Calculate the value of $\phi(360)$ using the formula $\phi(m) = m \prod_{p|m} (1 - \frac{1}{p})$.
4. Use Fermat's Little Theorem to find the remainder when 2^{100} is divided by 13.

5. Use Euler's Theorem to find the last two digits of 7^{803} .
6. Solve the following system of linear congruences using the Chinese Remainder Theorem (CRT):

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

7. Verify the identity $\sum_{d|n} \phi(d) = n$ for $n = 12$ by listing all divisors and their totient values.
8. Prove that if $\{r_1, r_2, \dots, r_m\}$ is a CRS modulo m and $\gcd(a, m) = 1$, then $\{ar_1 + b, ar_2 + b, \dots, ar_m + b\}$ is also a CRS modulo m .
9. Calculate $\phi(n)$ for $n = p^k$ where p is a prime and $k \geq 1$. Use this to evaluate $\phi(2^5 \cdot 3^4)$.
10. If $n = p_1^{a_1} p_2^{a_2} \dots p_k^{a_k}$ is the prime factorization of n , show that $\phi(n) = \phi(p_1^{a_1}) \phi(p_2^{a_2}) \dots \phi(p_k^{a_k})$.
11. Find the smallest positive integer x that satisfies $x \equiv 1 \pmod{5}$, $x \equiv 2 \pmod{6}$, and $x \equiv 3 \pmod{7}$.
12. Prove that for $n > 2$, $\phi(n)$ is always an even integer.
13. Show that $a^{12} - 1$ is divisible by 35 for every integer a such that $\gcd(a, 35) = 1$.
14. Prove that if $\gcd(m, n) = 1$, then $m^{\phi(n)} + n^{\phi(m)} \equiv 1 \pmod{mn}$.
15. Determine all positive integers n such that $\phi(n) = n/2$.
16. A certain number leaves a remainder of 1 when divided by 2, 2 when divided by 3, and 3 when divided by 5. Find the general form of all such numbers.
17. Show that the set $\{a, a^2, a^3, \dots, a^{\phi(m)}\}$ is *not* necessarily an RRS modulo m . Give a counterexample.
18. Use the properties of $\phi(n)$ to prove that if $d|n$, then $\phi(d)|\phi(n)$ is not necessarily true, but $\phi(d) \leq \phi(n)$ is true.
19. Solve the system: $x \equiv 5 \pmod{6}$ and $x \equiv 4 \pmod{11}$. Find the smallest positive solution.
20. Find the number of integers between 1 and 1000 that are relatively prime to 77.

Challenges:

21. Prove that for any $n > 1$, the sum of all integers x such that $1 \leq x < n$ and $\gcd(x, n) = 1$ is equal to $\frac{1}{2}n\phi(n)$.
22. Let n be a composite integer. Prove that $(n-1)! \equiv 0 \pmod{n}$ for all $n > 4$. How does this relate to the RRS of n ?
23. Show that if p and q are distinct primes, then $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$.
24. Prove that there exist infinitely many n such that $10|\phi(n)$.
25. Solve the system of congruences $x \equiv a_i \pmod{m_i}$ where the m_i are *not* pairwise relatively prime:

$$x \equiv 2 \pmod{6}$$

$$x \equiv 8 \pmod{10}$$

Provide the necessary and sufficient condition for a solution to exist in such cases.

26. Let $f(n)$ be a multiplicative function. Prove that the function $F(n) = \sum_{d|n} f(d)$ is also multiplicative. Use this to provide a general proof for $\sum_{d|n} \phi(d) = n$.
27. Prove that if p is a prime of the form $4k + 3$ and $x^2 + 1 \equiv 0 \pmod{p}$, then no such x exists. Use Fermat's Little Theorem.
28. Find all solutions to the equation $\phi(n) = 24$.
29. Let n be an integer such that $n > 1$ and $2^n - 1$ is prime. Prove that n must be prime and that n divides $\phi(2^n - 1)$.
30. Prove Gauss's generalization of Euler's Theorem: For any integer a , $a^n \equiv a^{n-\phi(n)} \pmod{n}$ is false, but find a similar form that holds for all a (even those not coprime to n).
31. Let n be an integer greater than 2. Prove that the sum of all integers in a reduced residue system modulo n is a multiple of n .
32. Let $n = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ be the prime factorization of n . Prove that the number of idempotent elements modulo n (integers x such that $x^2 \equiv x \pmod{n}$) is exactly 2^k .
33. Prove that for any integer $n > 1$, n does not divide $2^n - 1$.
34. Let a and b be any two positive integers. Prove the generalized formula for the Euler totient function:
- $$\phi(ab) = \frac{\phi(a)\phi(b) \cdot \gcd(a, b)}{\phi(\gcd(a, b))}$$
35. Characterize all positive integers n for which the condition $\phi(n) \mid n$ is satisfied.
36. Prove that there exist infinitely many positive integers n such that $\phi(n)$ is a perfect square.
37. Let p be a prime number such that $p > 3$. Prove that the sum of the squares of the members of a reduced residue system modulo p is congruent to 0 modulo p .
38. Consider the system of congruences $x \equiv a \pmod{m}$ and $x \equiv b \pmod{n}$. Prove that a solution exists if and only if $\gcd(m, n) \mid (a - b)$, and if a solution exists, show it is unique modulo $\text{lcm}(m, n)$.
39. Provide a combinatorial proof for the identity $n = \sum_{d|n} \phi(d)$ by considering the set of fractions $\left\{ \frac{1}{n}, \frac{2}{n}, \dots, \frac{n}{n} \right\}$ and their reduction to lowest terms.

Chapter 4

4.1 Order of an Integer Modulo n

Definition 4.1.1. Let $n > 1$ and a be an integer such that $\gcd(a, n) = 1$. The **order** of a modulo n , often denoted by $\text{ord}_n(a)$, is the smallest positive integer k such that:

$$a^k \equiv 1 \pmod{n}$$

Theorem 4.1.2. Let $n > 1$ and $\gcd(a, n) = 1$. Let $k = \text{ord}_n(a)$ be the smallest positive integer such that $a^k \equiv 1 \pmod{n}$. Then the following properties hold:

1. $a^m \equiv 1 \pmod{n}$ if and only if $k \mid m$.
2. $k \mid \phi(n)$.
3. $a^i \equiv a^j \pmod{n}$ if and only if $i \equiv j \pmod{k}$.

Proof. We provide the proofs for each property as follows:

Proof of Property 1 First, suppose that $k \mid m$. Then $m = qk$ for some integer q . We have:

$$a^m = a^{qk} = (a^k)^q \equiv 1^q \equiv 1 \pmod{n}$$

Conversely, suppose that $a^m \equiv 1 \pmod{n}$. By the Division Algorithm, there exist integers q and r such that $m = qk + r$, where $0 \leq r < k$. Substituting this into the congruence:

$$a^m = a^{qk+r} = (a^k)^q \cdot a^r \equiv 1^q \cdot a^r \equiv a^r \pmod{n}$$

Since $a^m \equiv 1 \pmod{n}$, we have $a^r \equiv 1 \pmod{n}$. If $r > 0$, this contradicts the definition of k as the *smallest* positive integer such that $a^k \equiv 1 \pmod{n}$ (since $r < k$). Therefore, we must have $r = 0$, which implies $m = qk$, or $k \mid m$.

Proof of Property 2 By Euler's Totient Theorem, since $\gcd(a, n) = 1$, we have:

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

From Property 1 established above, any integer m such that $a^m \equiv 1 \pmod{n}$ must be a multiple of the order k . Setting $m = \phi(n)$, it follows immediately that $k \mid \phi(n)$.

Proof of Property 3 Without loss of generality, assume $i \geq j$. The congruence $a^i \equiv a^j \pmod{n}$ can be rewritten as:

$$a^i - a^j \equiv 0 \pmod{n} \implies a^j(a^{i-j} - 1) \equiv 0 \pmod{n}$$

Since $\gcd(a, n) = 1$, we also have $\gcd(a^j, n) = 1$. This allows us to divide the congruence by a^j :

$$a^{i-j} \equiv 1 \pmod{n}$$

From Property 1, $a^{i-j} \equiv 1 \pmod{n}$ if and only if $k \mid (i - j)$. By definition of modular arithmetic, $k \mid (i - j)$ is equivalent to:

$$i \equiv j \pmod{k}$$

This completes the proof of all properties. ■

Lemma 4.1.3. Let $\text{ord}_n(a) = k$. Then $\text{ord}_n(a^m) = \frac{k}{\gcd(m, k)}$.

Proof. Let $d = \gcd(m, k)$ and let $h = \text{ord}_n(a^m)$. We wish to show $h = k/d$. First, note that:

$$(a^m)^{k/d} = (a^k)^{m/d} \equiv 1^{m/d} \equiv 1 \pmod{n}$$

Since the power of a^m is congruent to 1, the order h must divide the exponent: $h \mid (k/d)$.

Conversely, since $(a^m)^h \equiv 1 \pmod{n}$, we have $a^{mh} \equiv 1 \pmod{n}$. By the properties of the order, the order of a (which is k) must divide the exponent mh . Thus, $k \mid mh$. Dividing both sides of this divisibility relation by d :

$$\frac{k}{d} \mid h \left(\frac{m}{d} \right)$$

Since $\gcd(k/d, m/d) = 1$, by Euclid's Lemma, we must have $\frac{k}{d} \mid h$. Because $h \mid (k/d)$ and $(k/d) \mid h$, and both are positive, it follows that $h = k/d$. ■

Theorem 4.1.4. If the integer n has a primitive root, then it has exactly $\phi(\phi(n))$ distinct primitive roots modulo n .

Proof. Suppose n has a primitive root g . By definition, $\text{ord}_n(g) = \phi(n)$. Recall that the set of integers $\{g^1, g^2, g^3, \dots, g^{\phi(n)}\}$ forms a reduced residue system modulo n . Therefore, any other primitive root of n must be congruent to g^k for some $1 \leq k \leq \phi(n)$.

Using the Lemma established above, the order of g^k modulo n is:

$$\text{ord}_n(g^k) = \frac{\text{ord}_n(g)}{\gcd(k, \text{ord}_n(g))} = \frac{\phi(n)}{\gcd(k, \phi(n))}$$

For g^k to be a primitive root, its order must be exactly $\phi(n)$. Setting the expression equal to $\phi(n)$:

$$\phi(n) = \frac{\phi(n)}{\gcd(k, \phi(n))}$$

This equality holds if and only if:

$$\gcd(k, \phi(n)) = 1$$

Thus, g^k is a primitive root modulo n if and only if k is relatively prime to $\phi(n)$.

The number of such integers k in the range $1 \leq k \leq \phi(n)$ is, by the definition of Euler's totient function, exactly $\phi(\phi(n))$. Thus, there are $\phi(\phi(n))$ primitive roots modulo n . ■

Example 4.1.5. Find the order of 2 modulo 7.

Solution. We compute the powers of 2 modulo 7:

$$2^1 = 2 \equiv 2 \pmod{7}$$

$$2^2 = 4 \equiv 4 \pmod{7}$$

$$2^3 = 8 \equiv 1 \pmod{7}$$

The smallest positive integer k such that $2^k \equiv 1 \pmod{7}$ is 3. Therefore, $\text{ord}_7(2) = 3$. Note that 3 divides $\phi(7) = 6$. ■

Example 4.1.6. Find the order of 3 modulo 11.

Solution. We compute the powers of 3 modulo 11:

- $3^1 \equiv 3 \pmod{11}$
- $3^2 \equiv 9 \pmod{11}$
- $3^3 \equiv 27 \equiv 5 \pmod{11}$
- $3^4 \equiv 15 \equiv 4 \pmod{11}$

- $3^5 \equiv 12 \equiv 1 \pmod{11}$

The smallest power resulting in $1 \pmod{11}$ is 5. Thus, $\text{ord}_{11}(3) = 5$. Note that 5 divides $\phi(11) = 10$. ■

Example 4.1.7. Find the order of 10 modulo 13.

Solution. We calculate powers of 10 (mod 13):

- $10^1 \equiv 10 \pmod{13}$
- $10^2 = 100 = 13 \times 7 + 9 \equiv 9 \pmod{13}$
- $10^3 \equiv 10 \times 9 = 90 = 13 \times 6 + 12 \equiv 12 \pmod{13}$
- $10^4 \equiv 10 \times 12 = 120 = 13 \times 9 + 3 \equiv 3 \pmod{13}$
- $10^5 \equiv 10 \times 3 = 30 \equiv 4 \pmod{13}$
- $10^6 \equiv 10 \times 4 = 40 \equiv 1 \pmod{13}$

Thus, $\text{ord}_{13}(10) = 6$. Note that 6 divides $\phi(13) = 12$. ■

4.2 Primitive Roots for Primes

Definition 4.2.1. If $\text{ord}_n(g) = \phi(n)$, then g is called a **primitive root** modulo n .

Existence and Properties For a prime p , a primitive root always exists. If g is a primitive root modulo p , then the set of its powers $\{g^1, g^2, g^3, \dots, g^{p-1}\}$ reduced modulo p is a permutation of the set $\{1, 2, 3, \dots, p-1\}$. In algebraic terms, this means the multiplicative group $(\mathbb{Z}/p\mathbb{Z})^\times$ is cyclic, and g is a generator of the group.

The number of primitive roots modulo n (if any exist) is exactly $\phi(\phi(n))$. For a prime p , there are exactly $\phi(p-1)$ primitive roots.

Example 4.2.2. Find all primitive roots modulo 5.

Solution. We have $\phi(5) = 4$. An integer g is a primitive root if $\text{ord}_5(g) = 4$.

- For $g = 2$: $2^1 \equiv 2, 2^2 \equiv 4, 2^3 \equiv 8 \equiv 3, 2^4 \equiv 16 \equiv 1 \pmod{5}$. Order is 4.
- For $g = 3$: $3^1 \equiv 3, 3^2 \equiv 9 \equiv 4, 3^3 \equiv 12 \equiv 2, 3^4 \equiv 6 \equiv 1 \pmod{5}$. Order is 4.
- For $g = 4$: $4^1 \equiv 4, 4^2 \equiv 16 \equiv 1 \pmod{5}$. Order is 2.

The primitive roots modulo 5 are 2 and 3. The number of roots is $\phi(4) = 2$, which matches our findings. ■

Example 4.2.3. Determine if 2 is a primitive root modulo 7 and find the total number of primitive roots.

Solution. For $p = 7$, $\phi(7) = 6$. The number of primitive roots is $\phi(6) = \phi(2)\phi(3) = 1 \times 2 = 2$. From our previous section, we calculated $\text{ord}_7(2) = 3$. Since $3 \neq 6$, 2 is **not** a primitive root modulo 7. Checking $g = 3$:

- $3^1 \equiv 3, 3^2 \equiv 2, 3^3 \equiv 6, 3^4 \equiv 4, 3^5 \equiv 5, 3^6 \equiv 1 \pmod{7}$.

Since $\text{ord}_7(3) = 6$, 3 is a primitive root. The other primitive root can be found by 3^k where $\gcd(k, 6) = 1$, which is $3^5 \equiv 5 \pmod{7}$. ■

Example 4.2.4. Find the number of primitive roots for $p = 11$ and test $g = 2$.

Solution. For $p = 11$, we have $\phi(11) = 10$. The number of primitive roots is $\phi(10) = 4$. We test $g = 2$:

- $2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16 \equiv 5, 2^5 = 10 \equiv -1 \pmod{11}$.

Since $2^5 \equiv -1 \pmod{11}$, the order must be 10 (because the order must divide 10 and $2^5 \not\equiv 1$, but $(2^5)^2 = 2^{10} \equiv 1$). Since $\text{ord}_{11}(2) = 10 = \phi(11)$, 2 is a primitive root modulo 11. The other three primitive roots are $2^3 \equiv 8, 2^7 \equiv 128 \equiv 7$, and $2^9 \equiv 512 \equiv 6 \pmod{11}$. ■

4.3 Composite Numbers Having Primitive Roots

While every prime number p possesses a primitive root, this is not true for all composite numbers. The classification of integers possessing primitive roots was completed by Gauss.

Theorem 4.3.1. *For $k \geq 3$, the integer 2^k has no primitive roots.*

Proof. For reasons that will become clear later, we start by showing that if a is an odd integer, then for $k \geq 3$

$$a^{2^{k-2}} \equiv 1 \pmod{2^k}$$

If $k = 3$, this congruence becomes $a^2 \equiv 1 \pmod{8}$, which is certainly true (indeed, $1^2 \equiv 3^2 \equiv 5^2 \equiv 7^2 \equiv 1 \pmod{8}$). For $k > 3$, we proceed by induction on k . Assume that the asserted congruence holds for the integer k ; that is, $a^{2^{k-2}} \equiv 1 \pmod{2^k}$. This is equivalent to the equation

$$a^{2^{k-2}} = 1 + b2^k$$

where b is an integer. Squaring both sides, we obtain

$$\begin{aligned} a^{2^{k-1}} &= (a^{2^{k-2}})^2 = 1 + 2(b2^k) + (b2^k)^2 \\ &= 1 + 2^{k+1}(b + b^2 2^{k-1}) \\ &\equiv 1 \pmod{2^{k+1}} \end{aligned}$$

so that the asserted congruence holds for $k + 1$ and, hence, for all $k \geq 3$.

Now the integers that are relatively prime to 2^k are precisely the odd integers, so that $\phi(2^k) = 2^{k-1}$. By what was just proved, if a is an odd integer and $k \geq 3$,

$$a^{\phi(2^k)/2} \equiv 1 \pmod{2^k}$$

and, consequently, there are no primitive roots of 2^k . ■

Theorem 4.3.2. *If $\gcd(m, n) = 1$, where $m > 2$ and $n > 2$, then the integer mn has no primitive roots.*

Proof. Consider any integer a for which $\gcd(a, mn) = 1$; then $\gcd(a, m) = 1$ and $\gcd(a, n) = 1$. Put $h = \text{lcm}(\phi(m), \phi(n))$ and $d = \gcd(\phi(m), \phi(n))$.

Because $\phi(m)$ and $\phi(n)$ are both even (Theorem 7.4), surely $d \geq 2$. In consequence,

$$h = \frac{\phi(m)\phi(n)}{d} \leq \frac{\phi(mn)}{2}$$

Now Euler's theorem asserts that $a^{\phi(m)} \equiv 1 \pmod{m}$. Raising this congruence to the $\phi(n)/d$ power, we get

$$a^h = (a^{\phi(m)})^{\phi(n)/d} \equiv 1^{\phi(n)/d} \equiv 1 \pmod{m}$$

Similar reasoning leads to $a^h \equiv 1 \pmod{n}$. Together with the hypothesis $\gcd(m, n) = 1$, these congruences force the conclusion that

$$a^h \equiv 1 \pmod{mn}$$

The point we wish to make is that the order of any integer relatively prime to mn does not exceed $\phi(mn)/2$, whence there can be no primitive roots for mn . ■

Some special cases of the theorem are of particular interest, and we list these below.

Corollary 4.3.3. *The integer n fails to have a primitive root if either*

- (a) n is divisible by two odd primes, or
- (b) n is of the form $n = 2^m p^k$, where p is an odd prime and $m \geq 2$.

The significant feature of this last series of results is that it restricts our search for primitive roots to the integers $2, 4, p^k$, and $2p^k$, where p is an odd prime. In this section, we prove that each of the numbers just mentioned has a primitive root, the major task being the establishment of the existence of primitive roots for powers of an odd prime. The argument is somewhat long-winded, but otherwise routine; for the sake of clarity, it is broken down into several steps.

Lemma 4.3.4. *If p is an odd prime, then there exists a primitive root r of p such that $r^{p-1} \not\equiv 1 \pmod{p^2}$.*

Proof. From Theorem 8.6, it is known that p has primitive roots. Choose any one, call it r . If $r^{p-1} \not\equiv 1 \pmod{p^2}$, then we are finished. In the contrary case, replace r by $r' = r + p$, which is also a primitive root of p . Then employing the binomial theorem,

$$(r')^{p-1} \equiv (r + p)^{p-1} \equiv r^{p-1} + (p-1)pr^{p-2} \pmod{p^2}$$

But we have assumed that $r^{p-1} \equiv 1 \pmod{p^2}$; hence,

$$(r')^{p-1} \equiv 1 - pr^{p-2} \pmod{p^2}$$

Because r is a primitive root of p , $\gcd(r, p) = 1$, and therefore $p \nmid r^{p-2}$. The outcome of all this is that $(r')^{p-1} \not\equiv 1 \pmod{p^2}$, which proves the lemma. ■

Corollary 4.3.5. *If p is an odd prime, then p^2 has a primitive root; in fact, for a primitive root r of p , either r or $r + p$ (or both) is a primitive root of p^2 .*

Proof. The assertion is almost obvious: If r is a primitive root of p , then the order of r modulo p^2 is either $p-1$ or $p(p-1) = \phi(p^2)$. The foregoing proof shows that if r has order $p-1$ modulo p^2 , then $r + p$ is a primitive root of p^2 . ■

As an illustration of this corollary, we observe that 3 is a primitive root of 7; and that both 3 and 10 are primitive roots of 7^2 . Also, 14 is a primitive root of 29, but not of 29^2 .

To reach our goal, another somewhat technical lemma is needed.

Lemma 4.3.6. *Let p be an odd prime and let r be a primitive root of p with the property that $r^{p-1} \not\equiv 1 \pmod{p^2}$. Then for each positive integer $k \geq 2$,*

$$r^{p^{k-2}(p-1)} \not\equiv 1 \pmod{p^k}$$

Proof. The proof proceeds by induction on k . By hypothesis, the assertion holds for $k = 2$. Let us assume that it is true for some $k \geq 2$ and show that it is true for $k + 1$. Because $\gcd(r, p^{k-1}) = \gcd(r, p^k) = 1$, Euler's theorem indicates that

$$r^{p^{k-2}(p-1)} = r^{\phi(p^{k-1})} \equiv 1 \pmod{p^{k-1}}$$

Hence, there exists an integer a satisfying

$$r^{p^{k-2}(p-1)} = 1 + ap^{k-1}$$

where $p \nmid a$ by our induction hypothesis. Raise both sides of this last equation to the p th power and expand to obtain

$$r^{p^{k-1}(p-1)} = (1 + ap^{k-1})^p \equiv 1 + ap^k \pmod{p^{k+1}}$$

Because the integer a is not divisible by p , we have

$$r^{p^{k-1}(p-1)} \not\equiv 1 \pmod{p^{k+1}}$$

This completes the induction step, thereby proving the lemma. ■

The hard work, for the moment, is over. We now stitch the pieces together to prove that the powers of any odd prime have a primitive root.

Theorem 4.3.7. *If p is an odd prime number and $k \geq 1$, then there exists a primitive root for p^k .*

Proof. The two lemmas allow us to choose a primitive root r of p for which $r^{p^{k-2}(p-1)} \not\equiv 1 \pmod{p^k}$; in fact, any integer r satisfying the condition $r^{p-1} \not\equiv 1 \pmod{p^2}$ will do. We argue that such an r serves as a primitive root for all powers of p .

Let n be the order of r modulo p^k . In compliance with Theorem 8.1, n must divide $\phi(p^k) = p^{k-1}(p-1)$. Because $r^n \equiv 1 \pmod{p^k}$ yields $r^n \equiv 1 \pmod{p}$, we also have $p-1 \mid n$ (Theorem 8.1 serves again). Consequently, n assumes the form $n = p^m(p-1)$, where $0 \leq m \leq k-1$. If it happened that $n \neq p^{k-1}(p-1)$, then $p^{k-2}(p-1)$ would be divisible by n and we would arrive at

$$r^{p^{k-2}(p-1)} \equiv 1 \pmod{p^k}$$

contradicting the way in which r was initially chosen. Therefore, $n = p^{k-1}(p-1)$ and r is a primitive root for p^k . ■

This leaves only the case $2p^k$ for our consideration.

Corollary 4.3.8. *There are primitive roots for $2p^k$, where p is an odd prime and $k \geq 1$.*

Proof. Let r be a primitive root for p^k . There is no harm in assuming that r is an odd integer; for, if it is even, then $r + p^k$ is odd and is still a primitive root for p^k . Then $\gcd(r, 2p^k) = 1$. The order n of r modulo $2p^k$ must divide

$$\phi(2p^k) = \phi(2)\phi(p^k) = \phi(p^k)$$

But $r^n \equiv 1 \pmod{2p^k}$ implies that $r^n \equiv 1 \pmod{p^k}$, and therefore $\phi(p^k) \mid n$. Together these divisibility conditions force $n = \phi(2p^k)$, making r a primitive root of $2p^k$. ■

The prime 5 has $\phi(4) = 2$ primitive roots, namely, the integers 2 and 3. Because

$$2^{5-1} \equiv 16 \not\equiv 1 \pmod{25} \quad \text{and} \quad 3^{5-1} \equiv 6 \not\equiv 1 \pmod{25}$$

these also serve as primitive roots for 5^2 and, hence, for all higher powers of 5. The proof of the last corollary guarantees that 3 is a primitive root for all numbers of the form $2 \cdot 5^k$.

We summarize what has been accomplished in the following theorem.

Theorem 4.3.9 (Gauss's Classification). *A positive integer $n > 1$ has a primitive root if and only if n is one of the following:*

$$2, 4, p^k, 2p^k$$

where p is an odd prime and $k \geq 1$.

Proof. By virtue of Theorems 4.3.1 and 4.3.2, the only positive integers with primitive roots are those mentioned in the statement of our theorem. It may be checked that 1 is a primitive root for 2, and 3 is a primitive root of 4. We have just finished proving that primitive roots exist for any power of an odd prime and for twice such a power. ■

Theorem 4.3.10. *If p is an odd prime and r is a primitive root of p such that $r^{p-1} \not\equiv 1 \pmod{p^2}$, then r is a primitive root modulo p^k for all integers $k \geq 1$.*

Proof. To establish that r is a primitive root modulo p^k , we first show by induction that for every integer $k \geq 2$, the congruence $r^{p^{k-2}(p-1)} \not\equiv 1 \pmod{p^k}$ holds. For the base case $k = 2$, the condition $r^{p-1} \not\equiv 1 \pmod{p^2}$ is given by the hypothesis. Assume that the assertion holds for some $k \geq 2$, so that $r^{p^{k-2}(p-1)} \not\equiv 1 \pmod{p^k}$. By Euler's Totient Theorem, since $\gcd(r, p) = 1$, we know that $r^{\phi(p^{k-1})} = r^{p^{k-2}(p-1)} \equiv 1 \pmod{p^{k-1}}$. Combining this with our inductive hypothesis, we can write $r^{p^{k-2}(p-1)} = 1 + ap^{k-1}$ for some integer a where $p \nmid a$. Raising both sides to the p -th power and expanding via the binomial theorem, we obtain:

$$r^{p^{k-1}(p-1)} = (1 + ap^{k-1})^p = 1 + p(ap^{k-1}) + \binom{p}{2}(ap^{k-1})^2 + \cdots + (ap^{k-1})^p$$

Simplifying the terms, we see that

$$r^{p^{k-1}(p-1)} = 1 + ap^k + \frac{p-1}{2}a^2p^{2k-1} + \cdots + a^p p^{p(k-1)}$$

Since $k \geq 2$, the exponents $2k-1$ and $p(k-1)$ are both greater than or equal to $k+1$. Consequently, all terms in the expansion except for the first two are divisible by p^{k+1} . This allows us to write the congruence $r^{p^{k-1}(p-1)} \equiv 1 + ap^k \pmod{p^{k+1}}$. Given that $p \nmid a$, it follows that $ap^k \not\equiv 0 \pmod{p^{k+1}}$, and thus $r^{p^{k-1}(p-1)} \not\equiv 1 \pmod{p^{k+1}}$, which completes the induction.

Now, let n denote the order of r modulo p^k . By the properties of the order of an integer, n must divide $\phi(p^k) = p^{k-1}(p-1)$. Furthermore, the congruence $r^n \equiv 1 \pmod{p^k}$ implies that $r^n \equiv 1 \pmod{p}$. Because r is a primitive root modulo p , its order is $p-1$, which forces $(p-1) \mid n$. These conditions taken together imply that n must be of the form $p^m(p-1)$ for some integer $0 \leq m \leq k-1$. If m were strictly less than $k-1$, then n would necessarily divide $p^{k-2}(p-1)$, implying $r^{p^{k-2}(p-1)} \equiv 1 \pmod{p^k}$. This, however, contradicts the result obtained through induction. Therefore, we must have $m = k-1$, which means $n = p^{k-1}(p-1) = \phi(p^k)$. It follows that r is a primitive root modulo p^k . ■

Example 4.3.11. Find the four primitive roots of 26 and the eight primitive roots of 25.

Solution. To find the primitive roots of 26, we observe that $\phi(26) = \phi(2)\phi(13) = 1 \cdot 12 = 12$. The number of primitive roots is $\phi(\phi(26)) = \phi(12) = 4$. Since 26 is of the form $2p$, we first find the primitive roots of $p = 13$. The primitive roots of 13 are 2, 6, 7, and 11. A primitive root g of p is a primitive root of $2p$ if g is odd. If g is even, then $g + p$ is a primitive root of $2p$. Applying this:

- $r = 2$ is even, so $2 + 13 = 15$.
- $r = 6$ is even, so $6 + 13 = 19$.
- $r = 7$ is odd, so it remains 7.
- $r = 11$ is odd, so it remains 11.

Thus, the four primitive roots of 26 are 7, 11, 15, and 19.

To find the primitive roots of 25, we note that $\phi(25) = 20$. The number of primitive roots is $\phi(20) = 8$. We first find a primitive root modulo 5, which is $g = 2$. Since $2^{5-1} = 16 \not\equiv 1 \pmod{25}$, 2 is a primitive root of 25. All other primitive roots are of the form $2^k \pmod{25}$ where $\gcd(k, 20) = 1$. The values for k are $\{1, 3, 7, 9, 11, 13, 17, 19\}$. Calculating these powers:

- $2^1 = 2$
- $2^3 = 8$
- $2^7 = 128 \equiv 3 \pmod{25}$
- $2^9 = 12$
- $2^{11} = 23$
- $2^{13} = 17$
- $2^{17} = 22$
- $2^{19} = 13$

The eight primitive roots of 25 are 2, 3, 8, 12, 13, 17, 22, and 23. ■

Example 4.3.12. Determine all primitive roots of 3^2 , 3^3 , and 3^4 .

Solution. The integer $g = 2$ is a primitive root of $p = 3$. Since $2^{3-1} = 4 \not\equiv 1 \pmod{3^2}$, $g = 2$ is a primitive root for all powers 3^k .

For $3^2 = 9$, we have $\phi(9) = 6$. The number of primitive roots is $\phi(6) = 2$. The roots are of the form $2^k \pmod{9}$ where $\gcd(k, 6) = 1$, specifically $k \in \{1, 5\}$.

- $2^1 = 2$
- $2^5 = 32 \equiv 5 \pmod{9}$

The primitive roots of 9 are 2 and 5.

For $3^3 = 27$, we have $\phi(27) = 18$. The number of primitive roots is $\phi(18) = 6$. The roots are $2^k \pmod{27}$ where $\gcd(k, 18) = 1$, specifically $k \in \{1, 5, 7, 11, 13, 17\}$.

- $2^1 = 2$
- $2^5 = 32 \equiv 5 \pmod{27}$
- $2^7 = 128 \equiv 20 \pmod{27}$
- $2^{11} \equiv 23 \pmod{27}$
- $2^{13} \equiv 11 \pmod{27}$
- $2^{17} \equiv 14 \pmod{27}$

The primitive roots of 27 are 2, 5, 11, 14, 20, and 23.

For $3^4 = 81$, we have $\phi(81) = 54$. The number of primitive roots is $\phi(54) = 18$. The roots are $2^k \pmod{81}$ where $\gcd(k, 54) = 1$, specifically $k \in \{1, 5, 7, 11, 13, 17, 19, 23, 25, 29, 31, 35, 37, 41, 43, 47, 49, 53\}$. Calculating these powers modulo 81 yields the following set of primitive roots: $\{2, 32, 47, 23, 11, 14, 56, 5, 20,$ ■

Theorem 4.3.13 (Euler's Criterion for Primitive Roots). *Let p be an odd prime. An integer g such that $\gcd(g, p) = 1$ is a primitive root modulo p if and only if for every prime divisor q of $p - 1$, the following condition holds:*

$$g^{\frac{p-1}{q}} \not\equiv 1 \pmod{p}$$

Proof. Let $k = \text{ord}_p(g)$ denote the multiplicative order of g modulo p . Recall that k is the smallest positive integer such that $g^k \equiv 1 \pmod{p}$. By Fermat's Little Theorem, we know that $g^{p-1} \equiv 1 \pmod{p}$, which implies that k must divide $p - 1$.

($\Rightarrow$) **Necessary Condition:** Assume g is a primitive root modulo p . By definition, the order of g is $p - 1$. Let q be any prime divisor of $p - 1$. Since $q > 1$, the exponent $\frac{p-1}{q}$ satisfies:

$$0 < \frac{p-1}{q} < p-1$$

Since $p - 1$ is the *smallest* positive exponent such that $g^{p-1} \equiv 1 \pmod{p}$, raising g to any smaller positive power cannot yield 1. Therefore:

$$g^{\frac{p-1}{q}} \not\equiv 1 \pmod{p}$$

($\Leftarrow$) **Sufficient Condition:** Assume that for every prime divisor q of $p - 1$, $g^{\frac{p-1}{q}} \not\equiv 1 \pmod{p}$. We wish to show that $k = p - 1$.

Suppose, for the sake of contradiction, that g is not a primitive root. This means $k < p - 1$. Since $k \mid (p - 1)$, the integer $m = \frac{p-1}{k}$ must be greater than 1. Let q be a prime divisor of m . Because m divides $p - 1$, q is also a prime divisor of $p - 1$.

Since $q \mid m$, we can write $m = q \cdot n$ for some integer n . We substitute $m = \frac{p-1}{k}$ into this equation:

$$\frac{p-1}{k} = q \cdot n \implies \frac{p-1}{q} = k \cdot n$$

This equation shows that the order k divides $\frac{p-1}{q}$. Consequently, we can express $g^{\frac{p-1}{q}}$ as a power of g^k :

$$g^{\frac{p-1}{q}} = g^{k \cdot n} = (g^k)^n$$

Since k is the order of g , $g^k \equiv 1 \pmod{p}$. Thus:

$$(g^k)^n \equiv 1^n \equiv 1 \pmod{p}$$

This implies $g^{\frac{p-1}{q}} \equiv 1 \pmod{p}$, which contradicts the hypothesis that $g^{\frac{p-1}{q}} \not\equiv 1 \pmod{p}$ for all prime factors of $p-1$.

Therefore, the assumption that $k < p-1$ is false. We conclude that $k = p-1$, meaning g is a primitive root modulo p . ■

Example 4.3.14. Determine if $g = 3$ is a primitive root modulo $p = 7$.

Solution. Here $p = 7$, so $p-1 = 6$. The prime factors of 6 are $q \in \{2, 3\}$. According to Euler's Criterion, $g = 3$ is a primitive root if and only if $3^{\frac{6}{q}} \not\equiv 1 \pmod{7}$ for all prime factors q .

We verify the condition for each prime factor q :

1. For $q = 2$: The exponent is $\frac{6}{2} = 3$.

$$3^3 = 27 = 3(7) + 6 \equiv 6 \pmod{7}$$

Since $6 \not\equiv 1 \pmod{7}$, the condition is met for $q = 2$.

2. For $q = 3$: The exponent is $\frac{6}{3} = 2$.

$$3^2 = 9 \equiv 2 \pmod{7}$$

Since $2 \not\equiv 1 \pmod{7}$, the condition is met for $q = 3$.

Since the condition holds for all prime factors of $p-1$, we conclude that 3 is a primitive root modulo 7. ■

Example 4.3.15. Verify that $g = 2$ is a primitive root modulo $p = 13$.

Solution. Here $p = 13$, so $p-1 = 12$. The prime factorization of 12 is $2^2 \cdot 3$. Thus, the distinct prime factors are $q \in \{2, 3\}$. We must check that $2^{\frac{12}{q}} \not\equiv 1 \pmod{13}$ for each q .

1. For $q = 2$: The exponent is $\frac{12}{2} = 6$.

$$2^6 = 64$$

We verify $64 \pmod{13}$. Since $13 \times 5 = 65$, we have:

$$64 \equiv -1 \equiv 12 \pmod{13}$$

Since $12 \not\equiv 1 \pmod{13}$, the condition is met.

2. For $q = 3$: The exponent is $\frac{12}{3} = 4$.

$$2^4 = 16 \equiv 3 \pmod{13}$$

Since $3 \not\equiv 1 \pmod{13}$, the condition is met.

As the condition holds for all prime factors, 2 is a primitive root modulo 13. ■

Example 4.3.16. Find a primitive root modulo $p = 17$ by testing $g = 2$ and $g = 3$.

Solution. Here $p = 17$, so $p-1 = 16$. The only prime factor of 16 is $q = 2$. To test if a number g is a primitive root, we only need to check one condition:

$$g^{\frac{16}{2}} = g^8 \not\equiv 1 \pmod{17}$$

Test $g = 2$: We calculate $2^8 \pmod{17}$. First, note that $2^4 = 16 \equiv -1 \pmod{17}$. Squaring both sides:

$$2^8 = (2^4)^2 \equiv (-1)^2 \equiv 1 \pmod{17}$$

Since the result is 1, Euler's Criterion fails. Therefore, 2 is **not** a primitive root modulo 17.

Test $g = 3$: We calculate $3^8 \pmod{17}$. First, calculate powers of 3:

$$3^1 = 3, \quad 3^2 = 9, \quad 3^3 = 27 \equiv 10, \quad 3^4 = 81$$

Reducing 3^4 : $81 = 17(4) + 13$, so $3^4 \equiv 13 \equiv -4 \pmod{17}$. Now, square 3^4 to get 3^8 :

$$3^8 = (3^4)^2 \equiv (-4)^2 = 16 \equiv -1 \pmod{17}$$

Since $-1 \not\equiv 1 \pmod{17}$, the condition is satisfied. Therefore, 3 is a primitive root modulo 17. ■

4.4 The Legendre Symbol and Law of Reciprocity

4.4.1 The Legendre Symbol

Definition 4.4.1. Let p be an odd prime and a be an integer. The **Legendre symbol** $\left(\frac{a}{p}\right)$ is defined as:

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } \gcd(a, p) = 1 \text{ and } a \text{ is a quadratic residue modulo } p, \\ -1 & \text{if } \gcd(a, p) = 1 \text{ and } a \text{ is a quadratic non-residue modulo } p, \\ 0 & \text{if } p \mid a. \end{cases}$$

Theorem 4.4.2 (Euler's Criterion). Let p be an odd prime and a be an integer coprime to p . Then:

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}$$

Proof. Since p is prime, there exists a primitive root g modulo p . Any integer a coprime to p can be written as $a \equiv g^k \pmod{p}$ for some $1 \leq k \leq p-1$. Note that a is a quadratic residue if and only if k is even.

We evaluate $a^{\frac{p-1}{2}}$:

$$a^{\frac{p-1}{2}} \equiv (g^k)^{\frac{p-1}{2}} \equiv g^{\frac{k(p-1)}{2}} \pmod{p}$$

Case 1: a is a quadratic residue. Then k is even, let $k = 2m$.

$$g^{\frac{2m(p-1)}{2}} = (g^{p-1})^m \equiv 1^m \equiv 1 \pmod{p}$$

Thus, $a^{\frac{p-1}{2}} \equiv 1 \equiv \left(\frac{a}{p}\right) \pmod{p}$.

Case 2: a is a non-residue. Then k is odd, let $k = 2m + 1$.

$$g^{\frac{(2m+1)(p-1)}{2}} = g^{m(p-1)} \cdot g^{\frac{p-1}{2}} \equiv 1 \cdot g^{\frac{p-1}{2}} \pmod{p}$$

Since g is a primitive root, $g^{\frac{p-1}{2}} \equiv -1 \pmod{p}$ (it is the unique element of order 2). Thus, $a^{\frac{p-1}{2}} \equiv -1 \equiv \left(\frac{a}{p}\right) \pmod{p}$. ■

Example 4.4.3. Determine the value of $\left(\frac{8}{11}\right)$ using Euler's Criterion.

Solution. Here $a = 8$ and $p = 11$. By Euler's Criterion:

$$\left(\frac{8}{11}\right) \equiv 8^{\frac{11-1}{2}} \equiv 8^5 \pmod{11}$$

We compute $8^5 \pmod{11}$: First, note that $8 \equiv -3 \pmod{11}$.

$$(-3)^5 = -243$$

Now reduce -243 modulo 11. $243 = 11 \times 22 + 1$, so $243 \equiv 1 \pmod{11}$. Therefore, $-243 \equiv -1 \pmod{11}$.

$$\left(\frac{8}{11}\right) = -1$$

Thus, 8 is a quadratic non-residue modulo 11. ■

Example 4.4.4. Compute $\left(\frac{12}{13}\right)$.

Solution. Using Euler's Criterion for $a = 12 \equiv -1 \pmod{13}$:

$$\left(\frac{12}{13}\right) \equiv 12^{\frac{13-1}{2}} \equiv (-1)^6 \pmod{13}$$

Since the exponent 6 is even, $(-1)^6 = 1$. Thus, $\left(\frac{12}{13}\right) = 1$. ■

Theorem 4.4.5. Let p be an odd prime and let a and b be integers that are relatively prime to p . Then the Legendre symbol has the following properties:

- (a) If $a \equiv b \pmod{p}$, then $(a/p) = (b/p)$.
- (b) $(a^2/p) = 1$.
- (c) $(a/p) \equiv a^{(p-1)/2} \pmod{p}$.
- (d) $(ab/p) = (a/p)(b/p)$.
- (e) $(1/p) = 1$ and $(-1/p) = (-1)^{(p-1)/2}$.

Proof. If $a \equiv b \pmod{p}$, then the two congruences $x^2 \equiv a \pmod{p}$ and $x^2 \equiv b \pmod{p}$ have exactly the same solutions, if any at all. Thus, $x^2 \equiv a \pmod{p}$ and $x^2 \equiv b \pmod{p}$ are both solvable, or neither one has a solution. This is reflected in the statement $(a/p) = (b/p)$.

Regarding property (b), observe that the integer a trivially satisfies the congruence $x^2 \equiv a^2 \pmod{p}$; hence, $(a^2/p) = 1$. Property (c) is just the corollary to Theorem 9.1 rephrased in terms of the Legendre symbol. We use (c) to establish property (d):

$$(ab/p) \equiv (ab)^{(p-1)/2} \equiv a^{(p-1)/2} b^{(p-1)/2} \equiv (a/p)(b/p) \pmod{p}$$

Now the Legendre symbol assumes only the values 1 or -1 . If $(ab/p) \neq (a/p)(b/p)$, we would have $1 \equiv -1 \pmod{p}$ or $2 \equiv 0 \pmod{p}$; this cannot occur, because $p > 2$. It follows that

$$(ab/p) = (a/p)(b/p)$$

Finally, we observe that the first equality in property (e) is a special case of property (b), whereas the second one is obtained from property (c) upon setting $a = -1$. Because the quantities $(-1/p)$ and $(-1)^{(p-1)/2}$ are either 1 or -1 , the resulting congruence

$$(-1/p) \equiv (-1)^{(p-1)/2} \pmod{p}$$

implies that $(-1/p) = (-1)^{(p-1)/2}$. ■

Corollary 4.4.6. If p is an odd prime, then

$$(-1/p) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv 3 \pmod{4} \end{cases}$$

Proof. By Euler's Criterion, $(-1/p) = (-1)^{(p-1)/2}$. If $p \equiv 1 \pmod{4}$, then $p = 4k + 1$ for some integer k , so $\frac{p-1}{2} = 2k$ is even and $(-1/p) = (-1)^{2k} = 1$. If $p \equiv 3 \pmod{4}$, then $p = 4k + 3$, so $\frac{p-1}{2} = 2k + 1$ is odd and $(-1/p) = (-1)^{2k+1} = -1$. ■

Theorem 4.4.7. If p is an odd prime, then

$$\sum_{a=1}^{p-1} (a/p) = 0$$

Hence, there are precisely $(p-1)/2$ quadratic residues and $(p-1)/2$ quadratic non-residues of p .

Proof. Let r be a primitive root of p . We know that, modulo p , the powers $r, r^2, \dots, r^{p-1}$ are just a permutation of the integers $1, 2, \dots, p-1$. Thus, for any a lying between 1 and $p-1$, inclusive, there exists a unique positive integer k ($1 \leq k \leq p-1$), such that $a \equiv r^k \pmod{p}$. By appropriate use of Euler's criterion, we have

$$(a/p) = (r^k/p) \equiv (r^k)^{(p-1)/2} = (r^{(p-1)/2})^k \equiv (-1)^k \pmod{p} \quad (1)$$

where, because r is a primitive root of p , $r^{(p-1)/2} \equiv -1 \pmod{p}$. But (a/p) and $(-1)^k$ are equal to either 1 or -1 , so that equality holds in Eq. (1). Now add up the Legendre symbols in question to obtain

$$\sum_{a=1}^{p-1} (a/p) = \sum_{k=1}^{p-1} (-1)^k = 0$$

which is the desired conclusion. ■

Corollary 4.4.8. *The quadratic residues of an odd prime p are congruent modulo p to the even powers of a primitive root r of p ; the quadratic nonresidues are congruent to the odd powers of r .*

Proof. Let r be a primitive root of p . Any integer a such that $1 \leq a \leq p-1$ can be expressed as $a \equiv r^k \pmod{p}$ for a unique k in the range $1 \leq k \leq p-1$. Using Euler's criterion and the property that $r^{(p-1)/2} \equiv -1 \pmod{p}$, we have:

$$(a/p) \equiv (r^k)^{(p-1)/2} = (r^{(p-1)/2})^k \equiv (-1)^k \pmod{p}$$

Since both sides of the congruence are either 1 or -1 , it follows that $(a/p) = (-1)^k$.

By definition, a is a quadratic residue of p if and only if $(a/p) = 1$. From the identity above, $(a/p) = 1$ if and only if $(-1)^k = 1$, which occurs precisely when k is an even integer. Conversely, a is a quadratic nonresidue if and only if $(a/p) = -1$, which occurs if and only if $(-1)^k = -1$, meaning k is an odd integer. ■

Example 4.4.9. Identify the quadratic residues and nonresidues of $p = 7$ by using a primitive root.

Solution. First, we find a primitive root for $p = 7$. Testing $r = 3$:

$$\begin{aligned} 3^1 &\equiv 3 \pmod{7} \\ 3^2 &\equiv 2 \pmod{7} \\ 3^3 &\equiv 6 \pmod{7} \\ 3^4 &\equiv 4 \pmod{7} \\ 3^5 &\equiv 5 \pmod{7} \\ 3^6 &\equiv 1 \pmod{7} \end{aligned}$$

Since the order of 3 modulo 7 is $6 = \phi(7)$, $r = 3$ is a primitive root. According to the corollary, the quadratic residues are the even powers of r , and the nonresidues are the odd powers:

- **Even powers (Residues):** $3^2 \equiv 2$, $3^4 \equiv 4$, and $3^6 \equiv 1$.
- **Odd powers (Nonresidues):** $3^1 \equiv 3$, $3^3 \equiv 6$, and $3^5 \equiv 5$.

Thus, the quadratic residues of 7 are $\{1, 2, 4\}$ and the quadratic nonresidues are $\{3, 5, 6\}$. ■

Example 4.4.10. Identify the quadratic residues and nonresidues of $p = 11$ by using the primitive root $r = 2$.

Solution. We verify that $r = 2$ is a primitive root of 11 by calculating its powers: $2^1 = 2$, $2^2 = 4$, $2^3 = 8$, $2^4 = 5$, $2^5 = 10$, $2^6 = 9$, $2^7 = 7$, $2^8 = 3$, $2^9 = 6$, $2^{10} = 1$. Using the corollary, we classify the residues based on the parity of the exponent k in $2^k \pmod{11}$:

- **Quadratic Residues (Even k):**

- $2^2 \equiv 4$
- $2^4 \equiv 5$
- $2^6 \equiv 9$
- $2^8 \equiv 3$
- $2^{10} \equiv 1$

• **Quadratic Nonresidues (Odd k):**

- $2^1 \equiv 2$
- $2^3 \equiv 8$
- $2^5 \equiv 10$
- $2^7 \equiv 7$
- $2^9 \equiv 6$

The set of quadratic residues modulo 11 is $\{1, 3, 4, 5, 9\}$, and the set of quadratic nonresidues is $\{2, 6, 7, 8, 10\}$. ■

Lemma 4.4.11 (Gauss Lemma). *Let p be an odd prime and let $\gcd(a, p) = 1$. If n denotes the number of integers in the set*

$$S = \left\{ a, 2a, 3a, \dots, \left(\frac{p-1}{2} \right) a \right\}$$

whose remainders upon division by p exceed $p/2$, then

$$(a/p) = (-1)^n$$

Proof. Because $\gcd(a, p) = 1$, none of the $(p-1)/2$ integers in S is congruent to zero and no two are congruent to each other modulo p . Let $r_1, \dots, r_m$ be those remainders upon division by p such that $0 < r_i < p/2$, and let $s_1, \dots, s_n$ be those remainders such that $p > s_i > p/2$. Then $m + n = (p-1)/2$, and the integers

$$r_1, \dots, r_m \quad \text{and} \quad p - s_1, \dots, p - s_n$$

are all positive and less than $p/2$.

To prove that these integers are all distinct, it suffices to show that no $p - s_i$ is equal to any r_j . Assume to the contrary that $p - s_i = r_j$ for some choice of i and j . Then there exist integers u and v , with $1 \leq u, v \leq (p-1)/2$, satisfying $s_i \equiv ua \pmod{p}$ and $r_j \equiv va \pmod{p}$. Hence,

$$(u+v)a \equiv s_i + r_j = p \equiv 0 \pmod{p}$$

which says that $u+v \equiv 0 \pmod{p}$. But the latter congruence cannot take place, because $1 < u+v \leq p-1$.

The point we wish to bring out is that the $(p-1)/2$ numbers

$$r_1, \dots, r_m, \quad p - s_1, \dots, p - s_n$$

are simply the integers $1, 2, \dots, (p-1)/2$, not necessarily in order of appearance. Thus, their product is $[(p-1)/2]!$:

$$\begin{aligned} \left(\frac{p-1}{2} \right)! &= r_1 \cdots r_m (p - s_1) \cdots (p - s_n) \\ &\equiv r_1 \cdots r_m (-s_1) \cdots (-s_n) \pmod{p} \\ &\equiv (-1)^n r_1 \cdots r_m s_1 \cdots s_n \pmod{p} \end{aligned}$$

But we know that $r_1, \dots, r_m, s_1, \dots, s_n$ are congruent modulo p to $a, 2a, \dots, [(p-1)/2]a$ in some order, so that

$$\begin{aligned} \left(\frac{p-1}{2} \right)! &\equiv (-1)^n a \cdot 2a \cdots \left(\frac{p-1}{2} \right) a \pmod{p} \\ &\equiv (-1)^n a^{(p-1)/2} \left(\frac{p-1}{2} \right)! \pmod{p} \end{aligned}$$

Because $[(p-1)/2]!$ is relatively prime to p , it may be canceled from both sides of this congruence to give

$$1 \equiv (-1)^n a^{(p-1)/2} \pmod{p}$$

or, upon multiplying by $(-1)^n$,

$$a^{(p-1)/2} \equiv (-1)^n \pmod{p}$$

Use of Euler's criterion now completes the argument:

$$(a/p) \equiv a^{(p-1)/2} \equiv (-1)^n \pmod{p}$$

which implies that $(a/p) = (-1)^n$. ■

Illustration: Let $p = 13$ and $a = 5$. Then $(p-1)/2 = 6$, so that

$$S = \{5, 10, 15, 20, 25, 30\}$$

Modulo 13, the members of S are the same as the integers 5, 10, 2, 7, 12, 4. Three of these (10, 7, 12) are greater than $13/2$; hence, $n = 3$, and Gauss' Lemma says that

$$(5/13) = (-1)^3 = -1$$

Theorem 4.4.12. *If p is an odd prime, then*

$$(2/p) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{8} \text{ or } p \equiv 7 \pmod{8} \\ -1 & \text{if } p \equiv 3 \pmod{8} \text{ or } p \equiv 5 \pmod{8} \end{cases}$$

Proof. According to Gauss' Lemma, $(2/p) = (-1)^n$, where n is the number of integers in the set

$$S = \left\{ 1 \cdot 2, 2 \cdot 2, 3 \cdot 2, \dots, \left(\frac{p-1}{2} \right) \cdot 2 \right\}$$

which, upon division by p , have remainders greater than $p/2$. The members of S are all less than p , so that it suffices to count the number that exceed $p/2$. For $1 \leq k \leq (p-1)/2$, we have $2k > p/2$ if and only if $k > p/4$. If $[\cdot]$ denotes the greatest integer function, then there are $[p/4]$ integers in S less than $p/2$; hence,

$$n = \frac{p-1}{2} - \left[\frac{p}{4} \right]$$

is the number of integers that are greater than $p/2$.

Now we have four possibilities; for, any odd prime has one of the forms $8k+1$, $8k+3$, $8k+5$, or $8k+7$. A simple calculation shows that

- if $p = 8k+1$, then $n = 4k - [2k + 1/4] = 4k - 2k = 2k$
- if $p = 8k+3$, then $n = 4k + 1 - [2k + 3/4] = 4k + 1 - 2k = 2k + 1$
- if $p = 8k+5$, then $n = 4k + 2 - [2k + 1 + 1/4] = 4k + 2 - (2k + 1) = 2k + 1$
- if $p = 8k+7$, then $n = 4k + 3 - [2k + 1 + 3/4] = 4k + 3 - (2k + 1) = 2k + 2$

Thus, when p is of the form $8k+1$ or $8k+7$, n is even and $(2/p) = 1$; on the other hand, when p assumes the form $8k+3$ or $8k+5$, n is odd and $(2/p) = -1$. ■

Corollary 4.4.13. *If p is an odd prime, then*

$$(2/p) = (-1)^{(p^2-1)/8}$$

Proof. An odd prime p must be of the form $8k \pm 1$ or $8k \pm 3$. If $p = 8k \pm 1$, then $p^2 = 64k^2 \pm 16k + 1$, so $(p^2-1)/8 = 8k^2 \pm 2k$, which is an even integer. In this case, $(-1)^{(p^2-1)/8} = 1$, which matches the condition $(2/p) = 1$ when $p \equiv 1, 7 \pmod{8}$. If $p = 8k \pm 3$, then $p^2 = 64k^2 \pm 48k + 9$, so $(p^2-1)/8 = 8k^2 \pm 6k + 1$, which is an odd integer. In this case, $(-1)^{(p^2-1)/8} = -1$, which matches the condition $(2/p) = -1$ when $p \equiv 3, 5 \pmod{8}$. ■

Example 4.4.14. Use Gauss' Lemma to evaluate the Legendre symbol $(3/11)$.

Solution. Here $p = 11$ and $a = 3$. We calculate $(p-1)/2 = 5$. The set S to consider is:

$$S = \{1 \cdot 3, 2 \cdot 3, 3 \cdot 3, 4 \cdot 3, 5 \cdot 3\} = \{3, 6, 9, 12, 15\}$$

Now we reduce these members modulo 11:

$$3 \equiv 3 \pmod{11}$$

$$6 \equiv 6 \pmod{11}$$

$$9 \equiv 9 \pmod{11}$$

$$12 \equiv 1 \pmod{11}$$

$$15 \equiv 4 \pmod{11}$$

The residues modulo 11 are $\{3, 6, 9, 1, 4\}$. We compare these to $p/2 = 5.5$. The residues that exceed 5.5 are 6 and 9. Thus, $n = 2$. According to Gauss' Lemma:

$$(3/11) = (-1)^n = (-1)^2 = 1$$

This indicates that 3 is a quadratic residue modulo 11. (Indeed, $5^2 = 25 \equiv 3 \pmod{11}$). ■

Example 4.4.15. Determine the value of $(2/17)$ and $(2/19)$.

Solution. The theorem states that $(2/p) = 1$ if $p \equiv 1, 7 \pmod{8}$ and $(2/p) = -1$ if $p \equiv 3, 5 \pmod{8}$.

- For $p = 17$: Since $17 = 2 \times 8 + 1$, we have $17 \equiv 1 \pmod{8}$. Therefore, $(2/17) = 1$.
- For $p = 19$: Since $19 = 2 \times 8 + 3$, we have $19 \equiv 3 \pmod{8}$. Therefore, $(2/19) = -1$. ■

Example 4.4.16. Evaluate $(2/31)$.

Solution. For $p = 31$, we calculate the exponent:

$$\frac{p^2 - 1}{8} = \frac{31^2 - 1}{8} = \frac{961 - 1}{8} = \frac{960}{8} = 120$$

Since 120 is an even integer, the result is:

$$(2/31) = (-1)^{120} = 1$$

This matches the fact that $31 \equiv 7 \pmod{8}$, which also yields $(2/31) = 1$. ■

Example 4.4.17. Evaluate $(6/13)$ using multiplicativity and the quadratic character of 2.

Solution. Using the property $(ab/p) = (a/p)(b/p)$, we have:

$$(6/13) = (2/13)(3/13)$$

First, evaluate $(2/13)$. Since $13 \equiv 5 \pmod{8}$, the theorem gives $(2/13) = -1$. Second, evaluate $(3/13)$ via Gauss' Lemma. $(p-1)/2 = 6$.

$$S = \{3, 6, 9, 12, 15, 18\} \equiv \{3, 6, 9, 12, 2, 5\} \pmod{13}$$

Comparing to $p/2 = 6.5$, the residues greater than 6.5 are $\{9, 12\}$, so $n = 2$. Thus $(3/13) = (-1)^2 = 1$. Combining the results:

$$(6/13) = (-1) \cdot (1) = -1$$

Example 4.4.18. Determine if the quadratic congruence $x^2 \equiv 2 \pmod{23}$ is solvable.

Solution. The congruence is solvable if and only if $(2/23) = 1$. We check the value of 23 $\pmod{8}$:

$$23 = 2 \times 8 + 7 \implies 23 \equiv 7 \pmod{8}$$

Since $p \equiv 7 \pmod{8}$, the theorem confirms that $(2/23) = 1$. Therefore, the congruence is solvable. (Specifically, $5^2 = 25 \equiv 2 \pmod{23}$, so $x \equiv 5$ is a solution). ■

4.4.2 The Law of Quadratic Reciprocity

Theorem 4.4.19 (Quadratic Reciprocity Law). *Let p and q be distinct odd primes. Then:*

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$$

Equivalently:

$$\left(\frac{p}{q}\right) = \begin{cases} -\left(\frac{q}{p}\right) & \text{if } p \equiv 1 \pmod{4} \text{ and } q \equiv 3 \pmod{4} \\ \left(\frac{q}{p}\right) & \text{otherwise} \end{cases}$$

Note: We also use the supplementary laws: 1. $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$ 2. $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}$

Proof. We use a geometric proof involving lattice points (Eisenstein's Proof). Consider a rectangle in the xy -plane with vertices $(0, 0)$, $(p/2, 0)$, $(p/2, q/2)$, $(0, q/2)$. The total number of interior lattice points (points with integer coordinates) is:

$$N = \frac{p-1}{2} \cdot \frac{q-1}{2}$$

Draw the diagonal $y = \frac{q}{p}x$. No lattice points lie on this line since p and q are distinct primes. According to Gauss's Lemma arguments, the value of $\left(\frac{q}{p}\right)$ is determined by the parity of the number of lattice points below the diagonal, denoted μ_1 :

$$\mu_1 = \sum_{x=1}^{(p-1)/2} \left\lfloor \frac{qx}{p} \right\rfloor \implies \left(\frac{q}{p}\right) = (-1)^{\mu_1}$$

Similarly, the number of lattice points above the diagonal, μ_2 , determines $\left(\frac{p}{q}\right)$:

$$\mu_2 = \sum_{y=1}^{(q-1)/2} \left\lfloor \frac{py}{q} \right\rfloor \implies \left(\frac{p}{q}\right) = (-1)^{\mu_2}$$

The total number of points is $N = \mu_1 + \mu_2$. Therefore:

$$\left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{\mu_2} (-1)^{\mu_1} = (-1)^{\mu_1 + \mu_2} = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$$

■

Example 4.4.20. Calculate the Legendre symbol $\left(\frac{29}{53}\right)$.

Solution. Both 29 and 53 are odd primes. Check modulo 4: $29 \equiv 1 \pmod{4}$. Since at least one prime is $1 \pmod{4}$, the reciprocity is positive (no sign flip).

$$\left(\frac{29}{53}\right) = \left(\frac{53}{29}\right)$$

Reduce 53 modulo 29: $53 = 1(29) + 24$.

$$\left(\frac{53}{29}\right) = \left(\frac{24}{29}\right)$$

Factor $24 = 4 \times 6 = 2^2 \times 2 \times 3$.

$$\left(\frac{24}{29}\right) = \left(\frac{2^2}{29}\right) \left(\frac{2}{29}\right) \left(\frac{3}{29}\right) = 1 \cdot \left(\frac{2}{29}\right) \left(\frac{3}{29}\right)$$

1. For $\left(\frac{2}{29}\right)$: Since $29 \equiv 5 \pmod{8}$, $\left(\frac{2}{29}\right) = -1$.

2. For $\left(\frac{3}{29}\right)$: $29 \equiv 2 \pmod{3}$ and $3 \equiv 3 \pmod{4}$. By reciprocity:

$$\left(\frac{3}{29}\right) = \left(\frac{29}{3}\right) = \left(\frac{2}{3}\right) = -1$$

Substituting back:

$$\left(\frac{29}{53}\right) = 1 \cdot (-1) \cdot (-1) = 1$$

■

Example 4.4.21. Determine if 219 is a quadratic residue modulo 383. (Note: 383 is prime).

Solution. We need to evaluate $\left(\frac{219}{383}\right)$. First, factor $219 = 3 \times 73$.

$$\left(\frac{219}{383}\right) = \left(\frac{3}{383}\right) \left(\frac{73}{383}\right)$$

Part 1: Calculate $\left(\frac{3}{383}\right)$. Since $383 \equiv -1 \equiv 3 \pmod{4}$, and $3 \equiv 3 \pmod{4}$, we must flip the sign.

$$\left(\frac{3}{383}\right) = -\left(\frac{383}{3}\right)$$

$$383 \equiv 2 \pmod{3}.$$

$$-\left(\frac{2}{3}\right) = -(-1) = 1$$

Part 2: Calculate $\left(\frac{73}{383}\right)$. $73 \equiv 1 \pmod{4}$, so no sign flip.

$$\left(\frac{73}{383}\right) = \left(\frac{383}{73}\right)$$

$$383 = 5(73) + 18.$$

$$\left(\frac{18}{73}\right) = \left(\frac{2 \cdot 3^2}{73}\right) = \left(\frac{2}{73}\right) \cdot 1$$

Since $73 \equiv 1 \pmod{8}$, $\left(\frac{2}{73}\right) = 1$.

Conclusion:

$$\left(\frac{219}{383}\right) = 1 \cdot 1 = 1$$

Therefore, 219 is a quadratic residue modulo 383. ■

Corollary 4.4.22. If p and q are distinct odd primes, then

$$(p/q)(q/p) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \text{ or } q \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv q \equiv 3 \pmod{4} \end{cases}$$

Proof. The Law of Quadratic Reciprocity states that for distinct odd primes p and q , the product of their Legendre symbols is given by:

$$(p/q)(q/p) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$$

The value on the right-hand side is 1 if the exponent $\frac{p-1}{2} \cdot \frac{q-1}{2}$ is even, and -1 if the exponent is odd.

The product $\frac{p-1}{2} \cdot \frac{q-1}{2}$ is even if and only if at least one of the factors is even. Since an odd prime p is of the form $4k+1$ or $4k+3$, we see that:

- If $p \equiv 1 \pmod{4}$, then $p = 4k+1$, so $\frac{p-1}{2} = 2k$ (even).
- If $p \equiv 3 \pmod{4}$, then $p = 4k+3$, so $\frac{p-1}{2} = 2k+1$ (odd).

Thus, if $p \equiv 1 \pmod{4}$ or $q \equiv 1 \pmod{4}$, at least one factor in the exponent is even, making the product even and the result 1. If both $p \equiv 3 \pmod{4}$ and $q \equiv 3 \pmod{4}$, then both factors are odd, making the product odd and the result -1 . ■

Multiplying each side of the equation of the Quadratic Reciprocity Law by (q/p) and using the fact that $(q/p)^2 = 1$, we could also formulate this as follows:

Corollary 4.4.23. *If p and q are distinct odd primes, then*

$$(p/q) = \begin{cases} (q/p) & \text{if } p \equiv 1 \pmod{4} \text{ or } q \equiv 1 \pmod{4} \\ -(q/p) & \text{if } p \equiv q \equiv 3 \pmod{4} \end{cases}$$

Proof. Multiplying the previous result by (q/p) and noting that $(q/p)^2 = 1$, we see that (p/q) equals (q/p) when the exponent is even and $-(q/p)$ when the exponent is odd. ■

Theorem 4.4.24. *There are no positive integers x, y , and z that satisfy the equation*

$$x^n + y^n = z^n$$

for any integer value of n greater than 2.

Historical Note

The story of Fermat's Last Theorem (FLT) is one of the most famous in the history of mathematics, spanning over 350 years.

The Origin (1637) The theorem was first conjectured by the French lawyer and amateur mathematician **Pierre de Fermat**. While studying a copy of Diophantus's *Arithmetica*, Fermat came across the section discussing Pythagorean triples (solutions to $x^2 + y^2 = z^2$). He wrote a note in the margin of the book:

"It is impossible to separate a cube into two cubes, or a fourth power into two fourth powers, or in general, any power higher than the second, into two like powers. I have discovered a truly marvelous proof of this, which this margin is too narrow to contain."

Early Progress For centuries, mathematicians struggled to rediscover Fermat's "marvelous proof." Fermat himself left a proof for the case $n = 4$ using his method of *infinite descent*. Other milestones included:

- **Leonhard Euler** (1770) proved the case $n = 3$.
- **Adrien-Marie Legendre** and **Peter Gustav Lejeune Dirichlet** (1825) proved the case $n = 5$.
- **Sophie Germain** (early 19th century) made a significant breakthrough by proving a condition that applied to a whole class of primes (Sophie Germain primes), moving away from solving n one-by-one.
- **Ernst Kummer** (1847) proved the theorem for all "regular primes" using complex numbers and ideal theory, though "irregular primes" remained an obstacle.

The Bridge: Modularity and Elliptic Curves In the 20th century, the problem moved into the realm of modern algebraic geometry. In 1955, Japanese mathematicians **Yutaka Taniyama** and **Goro Shimura** proposed a conjecture (the Taniyama–Shimura–Weil conjecture) suggesting a deep link between elliptic curves and modular forms.

In the 1980s, **Gerhard Frey** suggested that if a counterexample to Fermat's Last Theorem existed, it could be turned into a very strange elliptic curve (the Frey curve). **Ken Ribet** eventually proved the *epsilon conjecture*, showing that such a curve could not be modular. This meant that proving the Taniyama–Shimura conjecture would automatically prove Fermat's Last Theorem.

The Proof (1994) **Andrew Wiles**, an English mathematician who had been obsessed with the theorem since childhood, worked in near-total secrecy for seven years to prove the Modularity Theorem for semistable elliptic curves. In 1993, he announced his proof, but a significant gap was found during peer review.

Working with his former student **Richard Taylor**, Wiles spent another year resolving the error. The final, corrected proof was published in 1995 in the journal *Annals of Mathematics*, officially closing one of the longest chapters in mathematical history.

Significance

While Fermat's Last Theorem itself has few direct applications, the *effort* to solve it revolutionized mathematics. It led to the birth of algebraic number theory, the development of ring theory, and the unification of disparate fields through the Langlands program.

Exercise 4.4.25. Solve the following exercises:

1. Find the order of 2 and 3 modulo 13.
2. Determine $\text{ord}_{17}(5)$. Use this to decide if 5 is a primitive root of 17.
3. If a has order hk modulo n , prove that a^h has order k modulo n .
4. Find all integers n such that $\text{ord}_n(a) = \phi(n)$ for $n = 10, 12, 14$.
5. Prove that if $\gcd(a, n) = 1$, then $\text{ord}_n(a)$ must divide $\phi(n)$.
6. Find all primitive roots modulo 11.
7. How many primitive roots exist modulo 23? List the three smallest ones.
8. Show that 2 is a primitive root modulo 19, but not modulo 17.
9. Find a primitive root for the prime $p = 31$.
10. For the prime $p = 7$, verify that the set $\{g, g^2, \dots, g^6\}$ forms a reduced residue system modulo 7, where g is a primitive root.
11. Identify which of the following composite integers possess primitive roots: 8, 9, 10, 12, 15, 18, 25, 26, 30.
12. Find a primitive root modulo 25 and use it to find a primitive root modulo 50.
13. Prove that 2^k has no primitive roots for $k \geq 3$ by showing $a^{2^{k-2}} \equiv 1 \pmod{2^k}$ for any odd a .
14. Find a primitive root modulo 49.
15. Show that if g is a primitive root modulo p^2 , then g is a primitive root modulo p^k for all $k \geq 2$.
16. Use Euler's Criterion to evaluate the Legendre symbols $\left(\frac{3}{7}\right)$ and $\left(\frac{2}{11}\right)$.
17. Determine whether $x^2 \equiv 5 \pmod{13}$ is solvable using Euler's Criterion.
18. Prove that $a^{(p-1)/2} \equiv \left(\frac{a}{p}\right) \pmod{p}$ holds for $p = 5$ and all $a \in \{1, 2, 3, 4\}$.
19. Prove that the number of quadratic residues modulo p is exactly $(p-1)/2$.
20. Use the properties of the Legendre symbol to evaluate $\left(\frac{14}{23}\right)$.
21. Use the Law of Quadratic Reciprocity to evaluate $\left(\frac{11}{31}\right)$.
22. Determine the solvability of $x^2 \equiv 21 \pmod{71}$.
23. Evaluate $\left(\frac{31}{101}\right)$ using the Law of Quadratic Reciprocity.

24. Find all primes p for which 3 is a quadratic residue (i.e., $\left(\frac{3}{p}\right) = 1$).
25. Use Quadratic Reciprocity to evaluate $\left(\frac{7}{p}\right)$ for $p = 29$ and $p = 37$.

Challenges:

26. Prove that if p is a prime of the form $p = 2^n + 1$, then every quadratic non-residue modulo p is a primitive root of p .
27. Show that the product of all primitive roots modulo p is congruent to 1 (mod p) if $\phi(p-1) > 1$.
28. Let p be an odd prime. Prove that the sum of the quadratic residues modulo p is $\frac{p(p-1)}{4}$ if $p > 3$.
29. Use Quadratic Reciprocity to prove that there are infinitely many primes of the form $4n + 1$.
30. Let p be a prime. Prove that $\sum_{a=1}^{p-1} \left(\frac{a}{p}\right) = 0$ and use this to find $\sum_{a=1}^{p-1} \left(\frac{a^2+a}{p}\right)$.
31. Show that if p is an odd prime, then $\sum_{x=0}^{p-1} \left(\frac{x^2+ax+b}{p}\right) = \begin{cases} p-1 & \text{if } a^2 - 4b \equiv 0 \pmod{p} \\ -1 & \text{if } a^2 - 4b \not\equiv 0 \pmod{p} \end{cases}$.
32. Prove that 3 is a primitive root modulo every Fermat prime $F_n = 2^{2^n} + 1$ with $n \geq 1$.
33. Let $p = 2q + 1$ be a prime, where q is also an odd prime. Show that the integer a is a primitive root modulo p if and only if $a \not\equiv -1 \pmod{p}$ and $\left(\frac{a}{p}\right) = -1$.
34. Regarding Fermat's Last Theorem (FLT): Prove the case $n = 4$ is impossible in positive integers by showing that $x^4 + y^4 = z^2$ has no solutions in positive integers using the method of infinite descent.
35. If p and $q = 2p + 1$ are both odd primes, prove that q divides the Mersenne number $2^p - 1$ if and only if $q \equiv 7 \pmod{8}$.
36. Let g be a primitive root of p . Prove that $\text{ind}_g(-1) = (p-1)/2$, where ind denotes the discrete logarithm (index).
37. Discuss the implications of Fermat's Last Theorem statement: Show that if $x^n + y^n = z^n$ had a solution for $n = k$, it would imply a solution for all n being multiples of k . Use this to explain why proving FLT for $n = 4$ and all odd primes p is sufficient.

Chapter 5

Theory of Polynomials Equations

A polynomial $P(x)$ of degree n is an expression of the form:

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where $a_n, a_{n-1}, \dots, a_0$ are constants (coefficients) and $a_n \neq 0$. The theory of polynomial equations studies the properties, roots, and behaviors of such expressions.

5.1 The Division Algorithm

Fundamental to the study of polynomials is the Division Algorithm, which provides the basis for the Remainder and Factor Theorems.

Theorem 5.1.1 (Division Algorithm). *Given polynomials $f(x)$ and $g(x)$ with $g(x) \neq 0$, there exist unique polynomials $q(x)$ (the quotient) and $r(x)$ (the remainder) such that:*

$$f(x) = g(x)q(x) + r(x)$$

where either $r(x) = 0$ or the degree of $r(x)$ is strictly less than the degree of $g(x)$.

Proof. Existence: Let $S = \{f(x) - g(x)k(x) \mid k(x) \in F[x]\}$. If $0 \in S$, let $r(x) = 0$. Otherwise, let $r(x)$ be an element of S with minimal degree. Suppose $\deg r \geq \deg g$. Let the leading terms of $r(x)$ and $g(x)$ be ax^m and bx^n respectively ($m \geq n$). Then $r_1(x) = r(x) - \frac{a}{b}x^{m-n}g(x)$ is in S and has $\deg r_1 < \deg r$, contradicting the minimality of $\deg r$. Thus $\deg r < \deg g$.

Uniqueness: Suppose $f = gq_1 + r_1 = gq_2 + r_2$. Then $g(q_1 - q_2) = r_2 - r_1$. If $q_1 \neq q_2$, the degree of the left side is at least $\deg g$, while the degree of the right side is strictly less than $\deg g$. This is a contradiction, so $q_1 = q_2$, which implies $r_1 = r_2$. ■

Example 5.1.2. Divide $f(x) = 3x^3 - 2x^2 + 4x - 5$ by $g(x) = x - 2$.

Solution. We set up the division in the long division format:

$$\begin{array}{r}
 3x^2 + 4x + 12 \\
 x - 2 \overline{) 3x^3 - 2x^2 + 4x - 5} \\
 \underline{-(3x^3 - 6x^2)} \\
 4x^2 + 4x \\
 \underline{-(4x^2 - 8x)} \\
 12x - 5 \\
 \underline{-(12x - 24)} \\
 19
 \end{array}$$

Final result: The quotient is $q(x) = 3x^2 + 4x + 12$ and the remainder is $r(x) = 19$. We can express this as:

$$3x^3 - 2x^2 + 4x - 5 = (x - 2)(3x^2 + 4x + 12) + 19$$

■

Example 5.1.3. Divide $f(x) = x^4 - 3x^2 + x + 1$ by $g(x) = x^2 + 1$.

Solution. Note that we include a $0x^3$ term in the dividend to act as a placeholder for alignment:

$$\begin{array}{r}
 x^2 - 4 \\
 \hline
 x^2 + 0x + 1 \overline{) x^4 + 0x^3 - 3x^2 + x + 1} \\
 \underline{-(x^4 + 0x^3 + x^2)} \\
 -4x^2 + x + 1 \\
 \underline{-(-4x^2 + 0x - 4)} \\
 x + 5
 \end{array}$$

Final result: The quotient is $q(x) = x^2 - 4$ and the remainder is $r(x) = x + 5$. We can express this as:

$$x^4 - 3x^2 + x + 1 = (x^2 + 1)(x^2 - 4) + (x + 5)$$

■

5.2 The Remainder and Factor Theorems

Theorem 5.2.1 (The Remainder Theorem). *If a polynomial $f(x)$ is divided by $(x - c)$, then the remainder is $f(c)$.*

Proof. By the Division Algorithm, when $f(x)$ is divided by a linear polynomial $(x - c)$, the remainder r must be a constant (since its degree must be less than 1). We write:

$$f(x) = (x - c)q(x) + r$$

Evaluating this expression at $x = c$:

$$f(c) = (c - c)q(c) + r = 0 \cdot q(c) + r = r$$

Thus, $r = f(c)$.

■

Theorem 5.2.2 (The Factor Theorem). *A polynomial $f(x)$ has a factor $(x - c)$ if and only if $f(c) = 0$.*

Proof. ($\Rightarrow$) Suppose $(x - c)$ is a factor of $f(x)$. Then $f(x) = (x - c)q(x)$ for some polynomial $q(x)$. Evaluating at $x = c$ gives $f(c) = (c - c)q(c) = 0$.

($\Leftarrow$) Suppose $f(c) = 0$. By the Remainder Theorem, the remainder of dividing $f(x)$ by $(x - c)$ is $r = f(c) = 0$. Therefore, by the Division Algorithm, $f(x) = (x - c)q(x) + 0$, which implies that $(x - c)$ is a factor of $f(x)$.

■

5.3 The Fundamental Theorem of Algebra

The Fundamental Theorem of Algebra (FTA) is a cornerstone of mathematical analysis and complex numbers.

Theorem 5.3.1 (Fundamental Theorem of Algebra). *Every non-constant polynomial with complex coefficients has at least one complex root.*

Corollary 5.3.2. A polynomial $f(x)$ of degree $n \geq 1$ can be factored into exactly n linear factors:

$$f(x) = a_n(x - c_1)(x - c_2) \dots (x - c_n)$$

where $c_1, c_2, \dots, c_n$ are the complex roots (not necessarily distinct) of $f(x)$.

Proof. By FTA, $f(x)$ has a root c_1 . By the Factor Theorem, $f(x) = (x - c_1)q_1(x)$, where $q_1(x)$ has degree $n - 1$. Applying FTA repeatedly to the resulting quotients $q_i(x)$ until the quotient is a constant a_n yields exactly n factors. ■

Vieta's Formulas (Roots and Coefficients)

Vieta's formulas describe the relationship between the coefficients of a polynomial and the sums and products of its roots.

Theorem 5.3.3 (Vieta's Formulas). If $x_1, x_2, \dots, x_n$ are the roots of $a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 = 0$, then:

$$\sum_{i=1}^n x_i = -\frac{a_{n-1}}{a_n}, \quad \sum_{1 \leq i < j \leq n} x_i x_j = \frac{a_{n-2}}{a_n} \text{ and } x_1 x_2 \dots x_n = (-1)^n \frac{a_0}{a_n}$$

Proof. By the Fundamental Theorem of Algebra, a polynomial $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$ with roots $x_1, x_2, \dots, x_n$ can be factored as:

$$P(x) = a_n(x - x_1)(x - x_2) \dots (x - x_n)$$

Expanding the right-hand side using the properties of binomial multiplication, we obtain:

$$P(x) = a_n \left[x^n - \left(\sum_{i=1}^n x_i \right) x^{n-1} + \left(\sum_{1 \leq i < j \leq n} x_i x_j \right) x^{n-2} - \dots + (-1)^n (x_1 x_2 \dots x_n) \right]$$

Comparing the coefficients of the expanded form with the standard form $a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$, we equate the corresponding terms:

$$\begin{aligned} a_{n-1} &= a_n \left(- \sum x_i \right) \implies \sum x_i = -\frac{a_{n-1}}{a_n} \\ a_{n-2} &= a_n \left(\sum x_i x_j \right) \implies \sum x_i x_j = \frac{a_{n-2}}{a_n} \\ a_0 &= a_n \left((-1)^n \prod x_i \right) \implies \prod x_i = (-1)^n \frac{a_0}{a_n} \end{aligned}$$

This completes the proof for the general case. ■

Example 5.3.4. Consider the cubic equation $2x^3 - 6x^2 + 4x - 8 = 0$. Find the sum and the product of its roots.

Solution. Here, $a_3 = 2$, $a_2 = -6$, $a_1 = 4$, and $a_0 = -8$. Using Vieta's formulas:

- The sum of the roots is: $\sum x_i = -\frac{a_2}{a_3} = -\frac{-6}{2} = 3$.
- The product of the roots (for $n = 3$) is: $x_1 x_2 x_3 = (-1)^3 \frac{a_0}{a_3} = -1 \left(\frac{-8}{2} \right) = 4$.

Example 5.3.5. Find a quadratic polynomial $P(x) = x^2 + bx + c$ whose roots are $3 + \sqrt{2}$ and $3 - \sqrt{2}$.

Solution. Let $x_1 = 3 + \sqrt{2}$ and $x_2 = 3 - \sqrt{2}$. For a monic quadratic ($a_2 = 1$):

- The coefficient $b = -(x_1 + x_2) = -(3 + \sqrt{2} + 3 - \sqrt{2}) = -6$.
- The constant term $c = x_1 x_2 = (3 + \sqrt{2})(3 - \sqrt{2}) = 3^2 - (\sqrt{2})^2 = 9 - 2 = 7$.

Thus, the polynomial is $P(x) = x^2 - 6x + 7$. ■

The Conjugate Root Theorems

Theorem 5.3.6 (Complex Conjugate Root Theorem). *If $f(x)$ is a polynomial with **real coefficients** and $a + bi$ is a root of $f(x)$, then its conjugate $a - bi$ is also a root of $f(x)$.*

Proof. Let $f(x) = \sum a_k x^k$ where $a_k \in \mathbb{R}$. We know that for any complex numbers u, v , $\overline{u + v} = \bar{u} + \bar{v}$ and $\overline{uv} = \bar{u}\bar{v}$. Since a_k are real, $\bar{a}_k = a_k$. Thus:

$$f(\bar{z}) = \sum a_k (\bar{z})^k = \sum \bar{a}_k \bar{z}^k = \overline{\sum a_k z^k} = \overline{f(z)}$$

If z is a root, $f(z) = 0$. Therefore, $f(\bar{z}) = \bar{0} = 0$, so $\bar{z}$ is also a root. ■

Theorem 5.3.7 (Irrational Conjugate Root Theorem). *If $f(x)$ is a polynomial with **rational coefficients** and $a + \sqrt{b}$ is a root (where $\sqrt{b}$ is irrational), then $a - \sqrt{b}$ is also a root of $f(x)$.*

Proof. Define a mapping $\sigma(u + v\sqrt{b}) = u - v\sqrt{b}$ for $u, v \in \mathbb{Q}$. This mapping is an automorphism of the field $\mathbb{Q}(\sqrt{b})$, meaning $\sigma(x + y) = \sigma(x) + \sigma(y)$ and $\sigma(xy) = \sigma(x)\sigma(y)$. For any rational coefficient a_k , $\sigma(a_k) = a_k$. Following the same logic as the complex case:

$$f(z^*) = f(\sigma(z)) = \sum a_k (\sigma(z))^k = \sigma\left(\sum a_k z^k\right) = \sigma(f(z))$$

Since $f(z) = 0$ and $\sigma(0) = 0$, it follows that $f(z^*) = 0$. ■

The Rational Root Theorem

Theorem 5.3.8 (Rational Root Theorem). *If the polynomial $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ has integer coefficients, then every rational root $x = p/q$ written in lowest terms ($\gcd(p, q) = 1$) must satisfy:*

- p is a factor of the constant term a_0 .
- q is a factor of the leading coefficient a_n .

Proof. Since p/q is a root of $P(x)$, we have:

$$a_n \left(\frac{p}{q}\right)^n + a_{n-1} \left(\frac{p}{q}\right)^{n-1} + \cdots + a_1 \left(\frac{p}{q}\right) + a_0 = 0$$

Multiply both sides by q^n to clear the denominators:

$$a_n p^n + a_{n-1} p^{n-1} q + \cdots + a_1 p q^{n-1} + a_0 q^n = 0$$

To prove $p \mid a_0$, isolate the term $a_0 q^n$:

$$a_0 q^n = -p(a_n p^{n-1} + a_{n-1} p^{n-2} q + \cdots + a_1 q^{n-1})$$

This shows that p divides $a_0 q^n$. Since $\gcd(p, q) = 1$, it follows that $\gcd(p, q^n) = 1$. By Euclid's Lemma, p must divide a_0 .

To prove $q \mid a_n$, isolate the term $a_n p^n$:

$$a_n p^n = -q(a_{n-1} p^{n-1} + a_{n-2} p^{n-2} q + \cdots + a_0 q^{n-1})$$

This shows that q divides $a_n p^n$. Since $\gcd(p, q) = 1$, it follows that $\gcd(q, p^n) = 1$. By Euclid's Lemma, q must divide a_n . ■

Example 5.3.9. Find all possible rational roots of $P(x) = 2x^3 + x^2 - 13x + 6$ and identify the actual roots.

Solution. According to the Rational Root Theorem:

- p must be a factor of 6: $\pm 1, \pm 2, \pm 3, \pm 6$.

- q must be a factor of 2: 1, 2.

The possible rational roots p/q are: $\pm 1, \pm 2, \pm 3, \pm 6, \pm 1/2, \pm 3/2$. Testing values:

- $P(2) = 2(8) + 4 - 26 + 6 = 0$. Thus, 2 is a root.
- $P(1/2) = 2(1/8) + 1/4 - 13/2 + 6 = 1/4 + 1/4 - 13/2 + 6 = 1/2 - 13/2 + 6 = 0$. Thus, $1/2$ is a root.
- $P(-3) = 2(-27) + 9 + 39 + 6 = -54 + 54 = 0$. Thus, -3 is a root.

The rational roots are 2, $1/2$, and -3 . ■

Example 5.3.10. Use the Rational Root Theorem to prove that $\sqrt{2}$ is irrational.

Solution. Consider the polynomial $P(x) = x^2 - 2$. The roots of this polynomial are $\pm\sqrt{2}$. If $P(x)$ has a rational root p/q :

- p must be a factor of -2 : $\pm 1, \pm 2$.
- q must be a factor of 1: 1.

The only possible rational roots are $\pm 1, \pm 2$. Testing these: $P(1) = -1$, $P(-1) = -1$, $P(2) = 2$, $P(-2) = 2$. Since none of the possible rational roots are actual roots, $x^2 - 2 = 0$ has no rational solutions. Therefore, $\sqrt{2}$ must be irrational. ■

5.4 Synthetic Division and its Applications

Synthetic division is a shorthand method of polynomial division, specifically used for dividing a polynomial $f(x)$ by a linear factor of the form $(x - c)$. It simplifies the long division process by focusing solely on coefficients.

1. The Division Algorithm via Synthetic Division

To divide $f(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$ by $(x - c)$, we arrange the coefficients in a row and use the following iterative process:

1. Bring down the leading coefficient a_n .
2. Multiply the brought-down value by c and place it under the next coefficient.
3. Add the values in the column.
4. Repeat until the last column is reached.

Example 5.4.1. Divide $f(x) = 2x^3 - 7x^2 + 0x + 5$ by $x - 3$.

Solution. Here, $c = 3$. The coefficients are 2, -7 , 0, 5.

$$\begin{array}{r|rrrr} 3 & 2 & -7 & 0 & 5 \\ \downarrow & & 6 & -3 & -9 \\ \hline & 2 & -1 & -3 & -4 \end{array}$$

The bottom row provides the coefficients of the quotient and the remainder. The quotient is $q(x) = 2x^2 - x - 3$ and the remainder is $r = -4$. ■

Example 5.4.2 (Division with Missing Terms). Divide $f(x) = 4x^4 - 2x^3 - 3x + 1$ by $x - 2$.

Solution. We must include a placeholder coefficient of 0 for the missing x^2 term. The coefficients are 4, -2 , 0, -3 , 1 and $c = 2$.

$$\begin{array}{r|rrrrr} 2 & 4 & -2 & 0 & -3 & 1 \\ \downarrow & & 8 & 12 & 24 & 42 \\ \hline & 4 & 6 & 12 & 21 & 43 \end{array}$$

The quotient is $q(x) = 4x^3 + 6x^2 + 12x + 21$ and the remainder is $r = 43$. ■

Example 5.4.3 (Polynomial Evaluation / Remainder Theorem). If $f(x) = x^4 + 2x^3 - x^2 + 5$, evaluate $f(-3)$ using synthetic division.

Solution. By the Remainder Theorem, $f(-3)$ is the remainder when $f(x)$ is divided by $x + 3$. The coefficients (with $0x$ as a placeholder) are 1, 2, -1, 0, 5 and $c = -3$.

$$\begin{array}{r|rrrrr} -3 & 1 & 2 & -1 & 0 & 5 \\ \downarrow & & -3 & 3 & -6 & 18 \\ \hline & 1 & -1 & 2 & -6 & 23 \end{array}$$

Since the remainder is 23, we have $f(-3) = 23$. ■

2. Polynomial Evaluation (The Remainder Theorem)

Synthetic division is often used to evaluate a polynomial $f(x)$ at $x = c$. According to the Remainder Theorem, $f(c)$ is exactly the remainder obtained when $f(x)$ is divided by $(x - c)$. This is also known as **Horner's Method**, which is computationally more efficient than direct substitution.

3. Factoring and Finding Roots

If synthetic division of $f(x)$ by $(x - c)$ results in a remainder of 0, then by the Factor Theorem, $(x - c)$ is a factor of $f(x)$, and c is a root. The resulting row of coefficients represents the *depressed polynomial*, which has a degree exactly one less than the original. This allows us to find remaining roots by solving a simpler equation.

Example 5.4.4. Given that $x = -2$ is a root of $f(x) = x^3 - 3x^2 - 4x + 12$, find the remaining roots.

Solution. Perform synthetic division with $c = -2$:

$$\begin{array}{r|rrrr} -2 & 1 & -3 & -4 & 12 \\ \downarrow & & -2 & 10 & -12 \\ \hline & 1 & -5 & 6 & 0 \end{array}$$

The remainder is 0, confirming $x + 2$ is a factor. The depressed polynomial is $x^2 - 5x + 6$. Factoring the quadratic: $x^2 - 5x + 6 = (x - 2)(x - 3)$. Thus, the roots are $x = -2, 2, 3$. ■

Example 5.4.5 (Factoring a Cubic Polynomial). Show that $x + 1$ is a factor of $f(x) = x^3 - 7x - 6$ and find the remaining factors.

Solution. $x + 1$ is a factor if the remainder when dividing by $c = -1$ is zero. The coefficients are 1, 0, -7, -6.

$$\begin{array}{r|rrrr} -1 & 1 & 0 & -7 & -6 \\ \downarrow & & -1 & 1 & 6 \\ \hline & 1 & -1 & -6 & 0 \end{array}$$

The remainder is 0, so $(x + 1)$ is a factor. The depressed polynomial is $x^2 - x - 6$. Factoring the quadratic: $x^2 - x - 6 = (x - 3)(x + 2)$. Thus, $f(x) = (x + 1)(x - 3)(x + 2)$. ■

Example 5.4.6 (Finding a Missing Constant). Find the value of k such that $x - 2$ is a factor of $f(x) = x^3 - kx^2 + 3x - 2$.

Solution. For $x - 2$ to be a factor, synthetic division by $c = 2$ must result in a remainder of 0.

$$\begin{array}{r|rrrr} 2 & 1 & -k & 3 & -2 \\ \downarrow & & 2 & 4 - 2k & 14 - 4k \\ \hline & 1 & 2 - k & 7 - 2k & 12 - 4k \end{array}$$

Set the remainder to zero: $12 - 4k = 0 \implies 4k = 12 \implies k = 3$. ■

4. Constructing a Polynomial from Roots

While synthetic division is a tool for division, the **Reverse Synthetic Division** (or synthetic multiplication) can be used to construct a polynomial given its roots $r_1, r_2, \dots, r_n$.

Starting with the first root r_1 , the polynomial is $(x - r_1)$. To multiply by the next factor $(x - r_2)$, we take the current coefficients and perform a step-wise accumulation:

- Start with coefficients of the current polynomial.
- To multiply by $(x - r_i)$, shift the coefficients one place to the left (representing multiplication by x) and add the current coefficients multiplied by $-r_i$.

Example 5.4.7. Construct a monic polynomial with roots 1, -2 , and 4.

Solution. Start with root $r_1 = 1$: coefficients are $(1, -1)$ for $x - 1$. Multiply by $(x - (-2)) = (x + 2)$:

$$\begin{array}{r} (1, -1, 0) \quad (\text{shifted for } x) \\ + (0, 2, -2) \quad (\text{multiplied by } 2) \\ \hline (1, 1, -2) \quad \text{yielding } x^2 + x - 2 \end{array}$$

Multiply by $(x - 4)$:

$$\begin{array}{r} (1, 1, -2, 0) \quad (\text{shifted for } x) \\ + (0, -4, -4, 8) \quad (\text{multiplied by } -4) \\ \hline (1, -3, -6, 8) \end{array}$$

The resulting polynomial is $f(x) = x^3 - 3x^2 - 6x + 8$. ■

Example 5.4.8 (Division by $(ax - b)$). Divide $f(x) = 2x^3 - 5x^2 + 3x + 7$ by $2x - 4$.

Solution. Synthetic division is designed for $(x - c)$. We rewrite $2x - 4$ as $2(x - 2)$. We first divide by $(x - 2)$ (using $c = 2$):

$$\begin{array}{r|rrrr} 2 & 2 & -5 & 3 & 7 \\ \downarrow & & 4 & -2 & 2 \\ \hline & 2 & -1 & 1 & 9 \end{array}$$

This gives $f(x) = (x - 2)(2x^2 - x + 1) + 9$. To divide by $2(x - 2)$, we divide the quotient coefficients by 2: $q(x) = \frac{1}{2}(2x^2 - x + 1) = x^2 - 0.5x + 0.5$. The remainder remains 9.

$$f(x) = (2x - 4)(x^2 - 0.5x + 0.5) + 9$$
■

Exercise 5.4.9. Solve the following exercises:

1. Use the Remainder Theorem to find the remainder when $f(x) = 2x^4 - 3x^3 + x^2 - 5x + 7$ is divided by $x - 2$.
2. Determine if $x + 3$ is a factor of the polynomial $P(x) = x^4 + 3x^3 - x^2 - 2x + 3$ using the Factor Theorem.
3. Perform synthetic division to find the quotient and remainder for $(3x^3 - 5x^2 + 10x - 4) \div (x - 1)$.
4. According to the Fundamental Theorem of Algebra, how many roots (including multiplicities) does the equation $x^7 - 4x^5 + 2x^2 - i = 0$ have?
5. If the roots of $x^2 - 7x + 12 = 0$ are α and β , use the relation between roots and coefficients to find the value of $\alpha^2 + \beta^2$.
6. Find a polynomial of degree 3 with real coefficients that has roots $x = 1$ and $x = 2 + i$.

7. A polynomial $f(x)$ leaves a remainder of 5 when divided by $x - 1$ and a remainder of 7 when divided by $x - 2$. Find the remainder when $f(x)$ is divided by $(x - 1)(x - 2)$.
8. Given the cubic equation $x^3 - 6x^2 + 11x - 6 = 0$, find the sum of the roots and the product of the roots using Vieta's formulas.
9. Use synthetic division to evaluate $f(-2)$ for $f(x) = x^5 + 4x^4 - 7x^2 + 10x - 12$.
10. If $2 + \sqrt{3}$ is a root of a polynomial with rational coefficients, what other value must also be a root? Construct the simplest quadratic equation for these roots.
11. Find the value of k such that $x - 3$ is a factor of $x^3 - kx^2 + 2x - 12$.
12. Solve the equation $x^3 - 7x^2 + 14x - 8 = 0$ given that its roots are in a geometric progression.
13. If α, β, γ are the roots of $2x^3 - 3x^2 + 5x - 1 = 0$, calculate the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.
14. Show that the polynomial $x^4 + 2x^2 + 1$ has no real roots and find its complex roots in the form $a + bi$.
15. Use synthetic division to divide $x^4 - 16$ by $x + 2$. Write the result in the form $Q(x) + \frac{R}{D(x)}$.
16. Construct a polynomial of the lowest possible degree with integer coefficients having $1 + 3i$ and $\sqrt{2}$ as roots.
17. If the sum of two roots of the equation $x^3 - 5x^2 - 2x + 24 = 0$ is zero, find all the roots of the equation.
18. Prove that $x - a$ is a factor of $x^n - a^n$ for all positive integers n .
19. Find the quotient and remainder when $x^6 - 1$ is divided by $x - 1$ using synthetic division.
20. If α, β are the roots of $ax^2 + bx + c = 0$, construct an equation whose roots are $\frac{1}{\alpha^2}$ and $\frac{1}{\beta^2}$.

Challenges:

21. Given that α, β, γ are roots of $x^3 + px + q = 0$, show that the sum of the fifth powers of the roots satisfies $\sum \alpha^5 = 5qr$, where r is derived from the coefficients. (More specifically, evaluate $\alpha^5 + \beta^5 + \gamma^5$ in terms of p and q).
22. Solve the equation $x^4 - 10x^3 + 26x^2 - 10x + 1 = 0$ by treating it as a reciprocal equation.
23. If the roots of $x^3 - px^2 + qx - r = 0$ are in arithmetic progression, prove that $2p^3 - 9pq + 27r = 0$.
24. Find the condition that the roots of $x^3 + 3px^2 + 3qx + r = 0$ may be in geometric progression, and solve the equation $x^3 - 7x^2 + 14x - 8 = 0$ using this condition.
25. Let $f(x)$ be a polynomial of degree n . If $f(k) = \frac{1}{k}$ for $k = 1, 2, \dots, n+1$, find the value of $f(n+2)$.
26. Determine the values of a and b such that $x^4 + ax^3 + bx^2 + 8x + 1$ is a perfect square of a quadratic polynomial.
27. If $\alpha, \beta, \gamma, \delta$ are roots of $x^4 + px^3 + qx^2 + rx + s = 0$, find the value of $\sum \alpha^2\beta$ in terms of the coefficients.
28. Prove that for $n > 1$, the polynomial $f(x) = nx^{n+1} - (n+1)x^n + 1$ is divisible by $(x-1)^2$.
29. Let $P(x)$ be a polynomial with integer coefficients. If $P(a) = P(b) = P(c) = -1$ where a, b, c are distinct integers, prove that $P(x) = 0$ has no integer roots.
30. Use the Fundamental Theorem of Algebra and the Conjugate Root Theorem to prove that every polynomial of odd degree with real coefficients must have at least one real root.

Chapter 6

Theory of Cubic and Biquadratic Equations

The solution of higher-degree polynomial equations involves reducing the given equation into simpler forms through algebraic transformations. This section covers the foundational methods and rules governing cubic and quartic (biquadratic) polynomials.

6.1 Theory of Symmetric Functions of Roots

A symmetric function of the roots of a polynomial is an algebraic expression involving the roots such that the expression remains identical regardless of how the roots are permuted (interchanged). In the context of the theory of equations, these functions allow us to extract properties of the roots directly from the coefficients of the polynomial.

6.1.1 Elementary Symmetric Functions

Given a polynomial $f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0$ with roots $\alpha_1, \alpha_2, \dots, \alpha_n$, the **Elementary Symmetric Functions** (e_k) are defined as:

$$\begin{aligned} e_1 &= \sum \alpha_i = \alpha_1 + \alpha_2 + \cdots + \alpha_n \\ e_2 &= \sum_{i < j} \alpha_i \alpha_j = \alpha_1 \alpha_2 + \alpha_1 \alpha_3 + \cdots + \alpha_{n-1} \alpha_n \\ e_3 &= \sum_{i < j < k} \alpha_i \alpha_j \alpha_k \\ &\vdots \\ e_n &= \alpha_1 \alpha_2 \cdots \alpha_n \end{aligned}$$

By Vieta's Formulas, we have the identity: $e_k = (-1)^k \frac{a_{n-k}}{a_n}$.

6.1.2 Fundamental Theorem of Symmetric Functions

Theorem 6.1.1. *Every symmetric polynomial with coefficients in a field F can be expressed as a polynomial in the elementary symmetric functions $e_1, e_2, \dots, e_n$ with coefficients in F .*

Proof. The proof typically proceeds by induction on the number of variables n and the degree of the polynomial. One defines a lexicographical ordering on the terms of the symmetric polynomial. By identifying the leading term $c \alpha_1^{k_1} \alpha_2^{k_2} \cdots \alpha_n^{k_n}$, it can be shown that $k_1 \geq k_2 \geq \cdots \geq k_n$. One then subtracts a term of the form $c e_1^{k_1 - k_2} e_2^{k_2 - k_3} \cdots e_n^{k_n}$ from the original polynomial. This subtraction results in a symmetric polynomial with a "lower" leading term. Successive applications of this process eventually reduce the polynomial to zero, effectively expressing it as a sum of products of e_i . ■

6.1.3 Newton's Sums (Newton-Girard Formulas)

To calculate sums of higher powers of roots, we use **Newton's Sums**. Let $s_k = \sum \alpha_i^k$. For the polynomial $x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0$:

$$\begin{aligned}s_1 + a_{n-1} &= 0 \\s_2 + a_{n-1}s_1 + 2a_{n-2} &= 0 \\s_3 + a_{n-1}s_2 + a_{n-2}s_1 + 3a_{n-3} &= 0\end{aligned}$$

Proof. Consider $f(x) = \prod(x - \alpha_i)$. Then $f'(x) = \sum_{i=1}^n \frac{f(x)}{x - \alpha_i}$. Using the geometric series expansion for $\frac{1}{x - \alpha_i} = \frac{1}{x} \left(1 + \frac{\alpha_i}{x} + \frac{\alpha_i^2}{x^2} + \dots\right)$, we find that $f'(x) = \frac{nf(x)}{x} + \frac{s_1 f(x)}{x^2} + \frac{s_2 f(x)}{x^3} + \dots$. Equating the coefficients of x^{-k} after multiplying by the denominator yields the recursive formulas. ■

For the following examples, assume α, β, γ are the roots of the cubic equation $ax^3 + bx^2 + cx + d = 0$. Let $e_1 = -b/a, e_2 = c/a, e_3 = -d/a$.

Example 6.1.2. If α, β, γ are roots of $x^3 + px + q = 0$, find the value of $\sum \alpha^2$.

Solution. From Vieta's formulas:

- $\alpha + \beta + \gamma = 0$ (coefficient of x^2 is 0)
- $\alpha\beta + \beta\gamma + \gamma\alpha = p$

We use the identity:

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

Substituting the known values:

$$\sum \alpha^2 = (0)^2 - 2(p) = -2p$$

Example 6.1.3. Find the value of $\sum \alpha^2$ in terms of the coefficients.

Solution. We use the square of the first elementary symmetric function:

$$(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \gamma\alpha)$$

Rearranging for the sum of squares:

$$\sum \alpha^2 = e_1^2 - 2e_2 = \left(-\frac{b}{a}\right)^2 - 2\left(\frac{c}{a}\right) = \frac{b^2 - 2ac}{a^2}$$

Example 6.1.4. Evaluate $\sum \alpha^3$ for the depressed cubic $x^3 + px + q = 0$.

Solution. Here $e_1 = 0, e_2 = p, e_3 = -q$. Using Newton's Sums:

1. $s_1 + 0 = 0 \implies s_1 = 0$
2. $s_2 + 0(s_1) + 2p = 0 \implies s_2 = -2p$
3. $s_3 + 0(s_2) + p(s_1) + 3q = 0 \implies s_3 = -3q$

Thus, $\sum \alpha^3 = -3q$. ■

Example 6.1.5. Find the value of $\sum \frac{1}{\alpha}$ (where no root is zero).

Solution. This is a symmetric function. Finding a common denominator:

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} = \frac{e_2}{e_3}$$

Substituting coefficients: $\frac{c/a}{-d/a} = -\frac{c}{d}$. ■

Example 6.1.6. Evaluate the cross-product sum $\sum \alpha^2\beta = \alpha^2\beta + \alpha^2\gamma + \beta^2\alpha + \beta^2\gamma + \gamma^2\alpha + \gamma^2\beta$.

Solution. Consider the product $e_1e_2 = (\alpha + \beta + \gamma)(\alpha\beta + \beta\gamma + \gamma\alpha)$. Expanding this gives:

$$\begin{aligned} e_1e_2 &= (\alpha^2\beta + \alpha\beta\gamma + \alpha^2\gamma) + (\alpha\beta^2 + \beta^2\gamma + \alpha\beta\gamma) + (\alpha\beta\gamma + \beta\gamma^2 + \gamma^2\alpha) \\ e_1e_2 &= \sum \alpha^2\beta + 3\alpha\beta\gamma \end{aligned}$$

Thus, $\sum \alpha^2\beta = e_1e_2 - 3e_3$. ■

Example 6.1.7. Express $\sum \alpha^2\beta^2$ in terms of coefficients.

Solution. We square e_2 :

$$e_2^2 = (\alpha\beta + \beta\gamma + \gamma\alpha)^2 = (\alpha\beta)^2 + (\beta\gamma)^2 + (\gamma\alpha)^2 + 2(\alpha\beta^2\gamma + \beta\gamma^2\alpha + \gamma\alpha^2\beta)$$

Factoring out $\alpha\beta\gamma$ from the last three terms:

$$e_2^2 = \sum \alpha^2\beta^2 + 2\alpha\beta\gamma(\beta + \gamma + \alpha) = \sum \alpha^2\beta^2 + 2e_3e_1$$

Therefore, $\sum \alpha^2\beta^2 = e_2^2 - 2e_1e_3$. ■

Example 6.1.8. Find the discriminant of a quadratic $ax^2 + bx + c = 0$ in terms of symmetric functions.

Solution. The discriminant D is $a^2(\alpha - \beta)^2$. Expanding $(\alpha - \beta)^2 = \alpha^2 + \beta^2 - 2\alpha\beta$. We know $\alpha^2 + \beta^2 = e_1^2 - 2e_2$. So, $(\alpha - \beta)^2 = e_1^2 - 2e_2 - 2e_2 = e_1^2 - 4e_2$. Substituting $e_1 = -b/a$ and $e_2 = c/a$:

$$D = a^2 \left(\frac{b^2}{a^2} - \frac{4c}{a} \right) = a^2 \left(\frac{b^2 - 4ac}{a^2} \right) = b^2 - 4ac$$

Example 6.1.9. Find the sum of the squares of the reciprocals of the roots, $\sum \frac{1}{\alpha^2}$, for a cubic equation with roots α, β, γ .

Solution. We seek the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$. Finding a common denominator, we have:

$$\sum \frac{1}{\alpha^2} = \frac{\beta^2\gamma^2 + \alpha^2\gamma^2 + \alpha^2\beta^2}{(\alpha\beta\gamma)^2} = \frac{\sum \alpha^2\beta^2}{e_3^2}$$

From a previous identity, we know that $\sum \alpha^2\beta^2 = e_2^2 - 2e_1e_3$. Substituting this into our expression:

$$\sum \frac{1}{\alpha^2} = \frac{e_2^2 - 2e_1e_3}{e_3^2}$$

In terms of the general cubic $ax^3 + bx^2 + cx + d = 0$:

$$\sum \frac{1}{\alpha^2} = \frac{(c/a)^2 - 2(-b/a)(-d/a)}{(-d/a)^2} = \frac{c^2 - 2bd}{d^2}$$

Example 6.1.10. For the cubic equation $x^3 + px^2 + qx + r = 0$, evaluate the symmetric product $(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha)$.

Solution. We use the fact that $e_1 = \alpha + \beta + \gamma = -p$. This allows us to simplify the factors:

$$\alpha + \beta = e_1 - \gamma, \quad \beta + \gamma = e_1 - \alpha, \quad \gamma + \alpha = e_1 - \beta$$

The product becomes $(e_1 - \alpha)(e_1 - \beta)(e_1 - \gamma)$. Note that this is the same as the expansion of the polynomial $f(x) = (x - \alpha)(x - \beta)(x - \gamma)$ evaluated at $x = e_1$. Since $f(x) = x^3 - e_1x^2 + e_2x - e_3$, we substitute $x = e_1$:

$$f(e_1) = e_1^3 - e_1(e_1^2) + e_2e_1 - e_3 = e_1e_2 - e_3$$

Substituting the coefficients $e_1 = -p, e_2 = q, e_3 = -r$:

$$(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha) = (-p)(q) - (-r) = r - pq$$

Example 6.1.11. Find the sum of the fourth powers of the roots, $s_4 = \sum \alpha^4$, for the cubic $x^3 + px + q = 0$.

Solution. Here $e_1 = 0, e_2 = p, e_3 = -q$. Using Newton's Sums ($s_k + a_1 s_{k-1} + a_2 s_{k-2} + \dots = 0$):

1. $s_1 = -a_1 = 0$.
2. $s_2 = -a_1 s_1 - 2a_2 = 0 - 2p = -2p$.
3. $s_3 = -a_1 s_2 - a_2 s_1 - 3a_3 = 0 - 0 - 3q = -3q$.
4. $s_4 = -a_1 s_3 - a_2 s_2 - a_3 s_1$. (For $k > n$, the constant multiplier is just the coefficient).

$$s_4 = -(0)(-3q) - (p)(-2p) - (q)(0) = 2p^2$$

Thus, $\sum \alpha^4 = 2p^2$. ■

Example 6.1.12. Evaluate $\sum \frac{\alpha^2 + \beta^2}{\alpha\beta}$ in terms of e_1, e_2, e_3 for a cubic equation.

Solution. Expanding the summation:

$$\sum \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{\alpha^2 + \beta^2}{\alpha\beta} + \frac{\beta^2 + \gamma^2}{\beta\gamma} + \frac{\gamma^2 + \alpha^2}{\gamma\alpha}$$

Find a common denominator $e_3 = \alpha\beta\gamma$:

$$\frac{\gamma(\alpha^2 + \beta^2) + \alpha(\beta^2 + \gamma^2) + \beta(\gamma^2 + \alpha^2)}{e_3} = \frac{\alpha^2\gamma + \beta^2\gamma + \alpha\beta^2 + \alpha\gamma^2 + \beta\gamma^2 + \alpha^2\beta}{e_3}$$

The numerator is exactly the symmetric function $\sum \alpha^2\beta$. From a previous example, we know $\sum \alpha^2\beta = e_1 e_2 - 3e_3$. Therefore:

$$\sum \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{e_1 e_2 - 3e_3}{e_3} = \frac{e_1 e_2}{e_3} - 3$$
■

Example 6.1.13. Calculate the sum of the squares of the differences of the roots, $\sum(\alpha - \beta)^2$, for a cubic equation.

Solution. Expanding the terms:

$$\begin{aligned} \sum(\alpha - \beta)^2 &= (\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2 \\ &= (\alpha^2 - 2\alpha\beta + \beta^2) + (\beta^2 - 2\beta\gamma + \gamma^2) + (\gamma^2 - 2\gamma\alpha + \alpha^2) \\ &= 2(\alpha^2 + \beta^2 + \gamma^2) - 2(\alpha\beta + \beta\gamma + \gamma\alpha) \end{aligned}$$

We know that $\sum \alpha^2 = e_1^2 - 2e_2$. Substituting this in:

$$\sum(\alpha - \beta)^2 = 2(e_1^2 - 2e_2) - 2e_2 = 2e_1^2 - 6e_2$$

In terms of $ax^3 + bx^2 + cx + d = 0$:

$$\sum(\alpha - \beta)^2 = 2\left(-\frac{b}{a}\right)^2 - 6\left(\frac{c}{a}\right) = \frac{2b^2 - 6ac}{a^2}$$
■

Example 6.1.14 (Cubic Reciprocals). Let α, β, γ be the roots of the equation $x^3 - 4x^2 + x + 6 = 0$. Calculate the value of $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2}$.

Solution. From Vieta's formulas, we have: $e_1 = \alpha + \beta + \gamma = 4$, $e_2 = \alpha\beta + \beta\gamma + \gamma\alpha = 1$, $e_3 = \alpha\beta\gamma = -6$. We seek the sum of the squares of the reciprocals:

$$\sum \frac{1}{\alpha^2} = \frac{\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2}{(\alpha\beta\gamma)^2} = \frac{e_2^2 - 2e_1e_3}{e_3^2}$$

Substituting the known values:

$$\sum \frac{1}{\alpha^2} = \frac{(1)^2 - 2(4)(-6)}{(-6)^2} = \frac{1 + 48}{36} = \frac{49}{36}$$

■

Example 6.1.15 (Quartic Power Sums). Consider the quartic equation $x^4 + x^3 - 2x^2 + 4 = 0$. Find the sum of the cubes of its roots, $s_3 = \alpha^3 + \beta^3 + \gamma^3 + \delta^3$.

Solution. We use Newton's Sums for the polynomial $x^4 + a_1x^3 + a_2x^2 + a_3x + a_4 = 0$, where: $a_1 = 1$, $a_2 = -2$, $a_3 = 0$, $a_4 = 4$. The Newton formulas are:

$$1. \quad s_1 + a_1 = 0 \implies s_1 + 1 = 0 \implies s_1 = -1.$$

$$2. \quad s_2 + a_1s_1 + 2a_2 = 0 \implies s_2 + (1)(-1) + 2(-2) = 0 \implies s_2 - 1 - 4 = 0 \implies s_2 = 5.$$

$$3. \quad s_3 + a_1s_2 + a_2s_1 + 3a_3 = 0 \implies s_3 + (1)(5) + (-2)(-1) + 3(0) = 0.$$

$$s_3 + 5 + 2 = 0 \implies s_3 = -7$$

Thus, the sum of the cubes of the roots is -7 .

■

Example 6.1.16 (Shifted Products). If α, β, γ are the roots of $x^3 - x - 1 = 0$, evaluate the product $(1 + \alpha)(1 + \beta)(1 + \gamma)$.

Solution. Let $f(x) = x^3 - x - 1$. We know that $f(x) = (x - \alpha)(x - \beta)(x - \gamma)$. We wish to find $(1 + \alpha)(1 + \beta)(1 + \gamma)$. Notice that:

$$(1 + \alpha)(1 + \beta)(1 + \gamma) = (-1)(-1)(-1)(-1 - \alpha)(-1 - \beta)(-1 - \gamma) = -(-1 - \alpha)(-1 - \beta)(-1 - \gamma)$$

This is equivalent to $-f(-1)$. Let us evaluate $f(-1)$:

$$f(-1) = (-1)^3 - (-1) - 1 = -1 + 1 - 1 = -1$$

Therefore:

$$(1 + \alpha)(1 + \beta)(1 + \gamma) = -(-1) = 1$$

■

Example 6.1.17 (Symmetric Grouping). For the equation $x^3 - 2x^2 + 5x - 4 = 0$, calculate the sum $\sum \alpha^2\beta = \alpha^2\beta + \alpha^2\gamma + \beta^2\alpha + \beta^2\gamma + \gamma^2\alpha + \gamma^2\beta$.

Solution. From Vieta's formulas: $e_1 = 2$, $e_2 = 5$, $e_3 = 4$. We use the symmetric identity: $e_1e_2 = \sum \alpha^2\beta + 3e_3$. Rearranging for our target sum:

$$\sum \alpha^2\beta = e_1e_2 - 3e_3$$

Substituting the values:

$$\sum \alpha^2\beta = (2)(5) - 3(4) = 10 - 12 = -2$$

■

Example 6.1.18 (High-Degree Power Sum). If $x^3 - 3x + 1 = 0$, find the sum of the fourth powers of its roots, $\sum \alpha^4$.

Solution. Coefficients: $a_1 = 0, a_2 = -3, a_3 = 1$. Using Newton's Sums:

1. $s_1 = -a_1 = 0$.
2. $s_2 = -a_1s_1 - 2a_2 = 0 - 2(-3) = 6$.
3. $s_3 = -a_1s_2 - a_2s_1 - 3a_3 = 0 - (-3)(0) - 3(1) = -3$.
4. For s_4 , since $k > n = 3$, the formula is $s_4 + a_1s_3 + a_2s_2 + a_3s_1 = 0$:

$$s_4 + (0)(-3) + (-3)(6) + (1)(0) = 0$$

$$s_4 - 18 = 0 \implies s_4 = 18$$

The sum of the fourth powers is 18. ■

Example 6.1.19 (Complex Combinations). Find the value of $\sum \frac{\alpha}{\beta+\gamma}$ for the roots of $x^3 + 2x^2 - 3x + 1 = 0$.

Solution. Let $e_1 = -2$. We know $\beta + \gamma = e_1 - \alpha = -2 - \alpha$. The sum becomes:

$$\sum \frac{\alpha}{-2-\alpha} = \frac{\alpha}{-2-\alpha} + \frac{\beta}{-2-\beta} + \frac{\gamma}{-2-\gamma}$$

Let $y = \frac{x}{-2-x}$. We want to find the sum of the roots of the transformed equation. Solving for x :
 $y(-2-x) = x \implies -2y - xy = x \implies -2y = x(1+y) \implies x = \frac{-2y}{1+y}$. Substitute x into $x^3 + 2x^2 - 3x + 1 = 0$:

$$\left(\frac{-2y}{1+y}\right)^3 + 2\left(\frac{-2y}{1+y}\right)^2 - 3\left(\frac{-2y}{1+y}\right) + 1 = 0$$

Multiply by $(1+y)^3$:

$$\begin{aligned} -8y^3 + 2(4y^2)(1+y) - 3(-2y)(1+y)^2 + (1+y)^3 &= 0 \\ -8y^3 + 8y^2 + 8y^3 + 6y(1+2y+y^2) + (1+3y+3y^2+y^3) &= 0 \\ 8y^2 + 6y + 12y^2 + 6y^3 + 1 + 3y + 3y^2 + y^3 &= 0 \\ 7y^3 + 23y^2 + 9y + 1 &= 0 \end{aligned}$$

The sum of the roots of this equation in y is $\sum y = -23/7$. ■

6.1.4 Descartes' Rule of Signs

Descartes' Rule of Signs provides an upper bound on the number of positive and negative real roots of a polynomial with real coefficients.

Theorem 6.1.20 (Descartes' Rule of Signs). Let $f(x)$ be a polynomial with real coefficients ordered by descending variable exponent.

- The number of positive real roots is either equal to the number of sign changes in the coefficients of $f(x)$, or less than it by an even number.
- The number of negative real roots is either equal to the number of sign changes in the coefficients of $f(-x)$, or less than it by an even number.

Example 6.1.21. Analyze the roots of $f(x) = x^4 + 3x^2 + 2x - 5$.

Solution. The coefficients are $(+1, +3, +2, -5)$. There is 1 sign change. Thus, there is exactly 1 positive real root. For $f(-x) = (-x)^4 + 3(-x)^2 + 2(-x) - 5 = x^4 + 3x^2 - 2x - 5$. The coefficients are $(+1, +3, -2, -5)$. There is 1 sign change. Thus, there is exactly 1 negative real root. Since the degree is 4 and there are 2 real roots, the remaining 2 roots must be complex. ■

Example 6.1.22. Examine the nature of the roots of the polynomial:

$$f(x) = x^5 - x^4 + 3x^3 + 9x^2 - x + 5$$

Solution. First, we analyze the positive real roots by counting the sign changes in $f(x)$: The signs of the coefficients are: $(+1, -1, +3, +9, -1, +5)$.

- From $+1$ to -1 : 1st change.
- From -1 to $+3$: 2nd change.
- From $+3$ to $+9$: no change.
- From $+9$ to -1 : 3rd change.
- From -1 to $+5$: 4th change.

The number of sign changes is 4. Thus, there are **4, 2, or 0 positive real roots**.

Next, we analyze the negative real roots by examining $f(-x)$:

$$f(-x) = (-x)^5 - (-x)^4 + 3(-x)^3 + 9(-x)^2 - (-x) + 5 = -x^5 - x^4 - 3x^3 + 9x^2 + x + 5$$

The signs are: $(-1, -1, -3, +9, +1, +5)$. There is only **1 sign change** (from -3 to $+9$). Thus, there is exactly **1 negative real root**. ■

Example 6.1.23. Determine the number of real and complex roots for:

$$f(x) = 2x^4 + 5x^2 + 7$$

Solution. The coefficients of $f(x)$ are all positive: $(+2, +5, +7)$. There are **0 sign changes**, meaning there are **no positive real roots**.

Now consider $f(-x)$:

$$f(-x) = 2(-x)^4 + 5(-x)^2 + 7 = 2x^4 + 5x^2 + 7$$

Again, there are **0 sign changes**, meaning there are **no negative real roots**.

Since the degree of the polynomial is 4 and there are 0 real roots, we conclude that there are **4 complex roots** (occurring in two conjugate pairs). ■

Example 6.1.24. Apply Descartes' Rule of Signs to $f(x) = x^3 + x - 1$.

Solution. For $f(x)$, the signs are $(+1, +1, -1)$. There is **1 sign change**, so there is exactly **1 positive real root**.

For $f(-x)$:

$$f(-x) = (-x)^3 + (-x) - 1 = -x^3 - x - 1$$

The signs are $(-1, -1, -1)$. There are **0 sign changes**, so there are **no negative real roots**.

Since there is only 1 real root for a cubic equation, the remaining **2 roots must be complex**. ■

Example 6.1.25. Find the maximum possible number of real roots for:

$$f(x) = 3x^6 - 2x^4 - x^2 + x - 1$$

Solution. For $f(x)$, the signs of the non-zero coefficients are: $(+3, -2, -1, +1, -1)$.

- $+3$ to -2 : 1st change.
- -1 to $+1$: 2nd change.
- $+1$ to -1 : 3rd change.

Total changes = 3. Possible positive real roots: **3 or 1**.

For $f(-x)$:

$$f(-x) = 3(-x)^6 - 2(-x)^4 - (-x)^2 + (-x) - 1 = 3x^6 - 2x^4 - x^2 - x - 1$$

The signs are: $(+3, -2, -1, -1, -1)$. There is only **1 sign change** (from $+3$ to -2). Possible negative real roots: **1**.

Total maximum real roots = $3(\text{pos}) + 1(\text{neg}) = 4$. Total minimum complex roots = $6(\text{degree}) - 4 = 2$. ■

6.2 Transformation of Equations: Removal of Terms

In the theory of equations, a common requirement is to transform a given polynomial $f(x)$ into a new polynomial $g(y)$ such that a specific term is missing. This process is essentially a shift of the origin. The most frequent application is the removal of the second term (x^{n-1}) to simplify the solution process for cubic and quartic equations.

6.2.1 1. General Principle of Transformation

Let the given equation of degree n be:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \cdots + a_0 = 0$$

To transform this equation into one in y where a specific term is absent, we substitute $x = y + h$. By the Binomial Theorem, $x^n = (y + h)^n = y^n + n h y^{n-1} + \frac{n(n-1)}{2} h^2 y^{n-2} + \cdots$.

If we wish to remove the **second term** (y^{n-1}), we collect the coefficients of y^{n-1} after substitution:

$$a_n(nh) + a_{n-1} = 0 \implies h = -\frac{a_{n-1}}{na_n}$$

If we wish to remove the **third term** (y^{n-2}), we would set the coefficient of y^{n-2} to zero, which usually results in a quadratic equation for h .

6.2.2 2. Method of Successive Synthetic Division

Once h is determined, the coefficients of the transformed equation:

$$g(y) = A_n y^n + A_{n-1} y^{n-1} + \cdots + A_0 = 0$$

are found by evaluating $f(x)$ and its derivatives at h . Mathematically, this is equivalent to Taylor's expansion: $A_k = \frac{f^{(k)}(h)}{k!}$.

Practically, this is achieved through **Repeated Synthetic Division** (Horner's Method):

1. Divide $f(x)$ by $(x - h)$ using synthetic division. The remainder is A_0 .
2. Divide the resulting quotient by $(x - h)$. The new remainder is A_1 .
3. Repeat this n times. The successive remainders (from last to first) are the coefficients $A_n, A_{n-1}, \dots, A_0$.

6.2.3 3. Method of Long Division

While synthetic division is more compact, polynomial long division can also be used. To find the coefficients A_i , we repeatedly divide by the linear factor $(x - h)$.

$$f(x) = (x - h)Q_1(x) + A_0$$

$$Q_1(x) = (x - h)Q_2(x) + A_1$$

This continues until the quotient is a constant. The remainders $A_0, A_1, \dots$ are the coefficients of the polynomial in powers of $(x - h)$.

Example 6.2.1. Remove the x^2 term from the equation $x^3 - 6x^2 + 10x - 3 = 0$.

Solution. Here $n = 3$, $a_3 = 1$, $a_2 = -6$. To remove the x^2 term, we find h :

$$h = -\frac{a_2}{na_3} = -\frac{-6}{3(1)} = 2$$

Substitute $x = y + 2$. We perform repeated synthetic division by 2:

$$\begin{array}{r|rrrr}
 2 & 1 & -6 & 10 & -3 \\
 & & 2 & -8 & 4 \\
 \hline
 2 & 1 & -4 & 2 & 1 \rightarrow A_0 \\
 & & 2 & -4 & \\
 \hline
 2 & 1 & -2 & -2 & \rightarrow A_1 \\
 & & 2 & & \\
 \hline
 & 1 & 0 & & \rightarrow A_2
 \end{array}$$

The transformed equation is $y^3 - 2y + 1 = 0$. ■

Another Method.

Solution. Let the given equation be:

$$x^3 - 6x^2 + 10x - 3 = 0 \quad \dots (1)$$

We want to remove the second term (the x^2 term). Let us apply the transformation:

$$x = y + h$$

Substituting this value of x into equation (1):

$$(y + h)^3 - 6(y + h)^2 + 10(y + h) - 3 = 0$$

Expanding the terms using the binomial theorem:

$$(y^3 + 3y^2h + 3yh^2 + h^3) - 6(y^2 + 2yh + h^2) + 10(y + h) - 3 = 0$$

Now, we arrange the terms in descending powers of y :

$$y^3 + (3h - 6)y^2 + (3h^2 - 12h + 10)y + (h^3 - 6h^2 + 10h - 3) = 0 \quad \dots (2)$$

To remove the second term, the coefficient of y^2 must be equal to zero. Therefore:

$$3h - 6 = 0 \implies 3h = 6 \implies h = 2$$

Now we substitute $h = 2$ into the coefficients of the remaining terms in equation (2).

Coefficient of y :

$$3h^2 - 12h + 10 = 3(2)^2 - 12(2) + 10 = 3(4) - 24 + 10 = 12 - 24 + 10 = -2$$

Constant term:

$$h^3 - 6h^2 + 10h - 3 = (2)^3 - 6(2)^2 + 10(2) - 3 = 8 - 24 + 20 - 3 = 1$$

Substituting these values back into equation (2), the transformed equation is:

$$y^3 - 2y + 1 = 0$$
■

Example 6.2.2. Remove the x^3 term from $x^4 + 4x^3 + 2x^2 - 4x - 2 = 0$.

Solution. $n = 4$, $a_4 = 1$, $a_3 = 4$. Calculate h :

$$h = -\frac{4}{4(1)} = -1$$

Substitute $x = y - 1$. Perform repeated synthetic division by -1 :

$$\begin{array}{r|rrrrrr}
 -1 & 1 & 4 & 2 & -4 & -2 \\
 & & -1 & -3 & 1 & 3 \\
 \hline
 -1 & 1 & 3 & -1 & -3 & 1 & \rightarrow A_0 \\
 & & -1 & -2 & 3 & \\
 \hline
 -1 & 1 & 2 & -3 & 0 & & \rightarrow A_1 \\
 & & -1 & -1 & \\
 \hline
 -1 & 1 & 1 & -4 & & & \rightarrow A_2 \\
 & & -1 & \\
 \hline
 & 1 & 0 & & & & \rightarrow A_3
 \end{array}$$

The transformed equation is $y^4 - 4y^2 + 1 = 0$. ■

Method of increasing roots

Solution. Let the given equation be:

$$x^4 + 4x^3 + 2x^2 - 4x - 2 = 0 \quad \dots (1)$$

We apply the transformation $x = y + h$. Substituting this into equation (1):

$$(y + h)^4 + 4(y + h)^3 + 2(y + h)^2 - 4(y + h) - 2 = 0$$

We expand the terms using the Binomial Theorem. We are primarily interested in the coefficients of the highest powers to determine h .

$$\begin{aligned}
 (y + h)^4 &= y^4 + 4y^3h + 6y^2h^2 + \dots \\
 4(y + h)^3 &= 4(y^3 + 3y^2h + \dots) = 4y^3 + 12y^2h + \dots
 \end{aligned}$$

Arranging the expanded equation in descending powers of y , the first two terms are:

$$y^4 + (4h + 4)y^3 + \dots = 0$$

To remove the second term (the x^3 term), the coefficient of y^3 must be zero:

$$4h + 4 = 0 \implies 4h = -4 \implies h = -1$$

Now we substitute $h = -1$ into the transformed equation $x = y - 1$. We can calculate the new coefficients by substituting $x = y - 1$ directly into (1):

$$(y - 1)^4 + 4(y - 1)^3 + 2(y - 1)^2 - 4(y - 1) - 2 = 0$$

Expanding the terms:

$$\begin{aligned}
 (y - 1)^4 &= y^4 - 4y^3 + 6y^2 - 4y + 1 \\
 4(y - 1)^3 &= \quad + 4y^3 - 12y^2 + 12y - 4 \\
 2(y - 1)^2 &= \quad \quad + 2y^2 - 4y + 2 \\
 -4(y - 1) &= \quad \quad \quad - 4y + 4 \\
 \text{Constants} &= -2
 \end{aligned}$$

Summing the coefficients vertically:

$$\begin{aligned}
 y^4 : & 1 \\
 y^3 : & -4 + 4 = 0 \\
 y^2 : & 6 - 12 + 2 = -4 \\
 y^1 : & -4 + 12 - 4 - 4 = 0 \\
 y^0 : & 1 - 4 + 2 + 4 - 2 = 1
 \end{aligned}$$

Thus, the transformed equation is:

$$y^4 - 4y^2 + 1 = 0$$

Example 6.2.3. Remove the x^2 term from $x^3 - 3x^2 + 12x + 16 = 0$ using long division logic.

Solution. $h = -(-3)/(3 \cdot 1) = 1$. We divide $f(x)$ by $(x - 1)$ repeatedly.

1. $(x^3 - 3x^2 + 12x + 16) \div (x - 1) = (x^2 - 2x + 10)$ with Remainder **26** (A_0).
2. $(x^2 - 2x + 10) \div (x - 1) = (x - 1)$ with Remainder **9** (A_1).
3. $(x - 1) \div (x - 1) = 1$ with Remainder **0** (A_2).

The leading coefficient $A_3 = 1$. The transformed equation is $y^3 + 9y + 26 = 0$.

Method of Long Division

Solution. Let the given equation be:

$$f(x) = x^3 - 3x^2 + 12x + 16 = 0$$

We diminish the roots by a quantity h . Let $x = y + h$. To remove the second term (x^2), the value of h is given by the formula $h = -\frac{a_1}{na_0}$, where n is the degree of the equation, a_0 is the coefficient of x^n , and a_1 is the coefficient of x^{n-1} .

Here, $n = 3$, $a_0 = 1$, and $a_1 = -3$.

$$h = -\frac{-3}{3(1)} = \frac{3}{3} = 1$$

Thus, we must diminish the roots by $h = 1$. We apply the method of synthetic division (repeated division) by 1 to find the coefficients of the transformed equation in y .

The scheme is as follows:

1	1	-3	12	16
↓	1	-2	10	
1	1	-2	10	26
↓	1	-1		
1	1	-1	9	
↓	1			
1	1	0		

Explanation of steps:

1. We divide the original coefficients (1, -3, 12, 16) by 1. The remainder is **26**.
2. We divide the resulting quotient (1, -2, 10) by 1. The remainder is **9**.
3. We divide the resulting quotient (1, -2) by 1. The remainder is **0**.
4. The final leading coefficient remains **1**.

The coefficients of the transformed equation are the final quotient and the remainders read in reverse order: **1, 0, 9, 26**. Therefore, the transformed equation is:

$$1 \cdot y^3 + 0 \cdot y^2 + 9 \cdot y + 26 = 0$$

$$y^3 + 9y + 26 = 0$$

Example 6.2.4. Remove the x^2 term from $x^3 + 3x^2 + 6x + 4 = 0$.

Solution. $h = -3/(3 \cdot 1) = -1$. Perform synthetic division by -1 :

$$\begin{array}{r|rrrr}
 -1 & 1 & 3 & 6 & 4 \\
 & & -1 & -2 & -4 \\
 \hline
 -1 & 1 & 2 & 4 & 0 \rightarrow A_0 \\
 & & -1 & -1 & \\
 \hline
 -1 & 1 & 1 & 3 & \rightarrow A_1 \\
 & & -1 & & \\
 \hline
 & 1 & 0 & & \rightarrow A_2
 \end{array}$$

The transformed equation is $y^3 + 3y = 0$. ■

Example 6.2.5. Transform $x^4 - 8x^3 + 7x^2 + 2x - 15 = 0$ into an equation without the x^3 term.

Solution. $h = -(-8)/(4 \cdot 1) = 2$. Perform repeated synthetic division by 2:

$$\begin{array}{r|rrrrr}
 2 & 1 & -8 & 7 & 2 & -15 \\
 & & 2 & -12 & -10 & -16 \\
 \hline
 2 & 1 & -6 & -5 & -8 & -31 \rightarrow A_0 \\
 & & 2 & -8 & -26 & \\
 \hline
 2 & 1 & -4 & -13 & -34 & \rightarrow A_1 \\
 & & 2 & -4 & & \\
 \hline
 2 & 1 & -2 & -17 & & \rightarrow A_2 \\
 & & 2 & & & \\
 \hline
 & 1 & 0 & & & \rightarrow A_3
 \end{array}$$

The transformed equation is $y^4 - 17y^2 - 34y - 31 = 0$. ■

Example 6.2.6. Remove the x^2 term from $2x^3 - 6x^2 + 3x + 5 = 0$.

Solution. $h = -(-6)/(3 \cdot 2) = 1$. Perform repeated synthetic division by 1:

$$\begin{array}{r|rrrr}
 1 & 2 & -6 & 3 & 5 \\
 & & 2 & -4 & -1 \\
 \hline
 1 & 2 & -4 & -1 & 4 \rightarrow A_0 \\
 & & 2 & -2 & \\
 \hline
 1 & 2 & -2 & -3 & \rightarrow A_1 \\
 & & 2 & & \\
 \hline
 & 2 & 0 & & \rightarrow A_2
 \end{array}$$

The transformed equation is $2y^3 - 3y + 4 = 0$. ■

6.2.4 Solution of Cubic Equations: Cardano's Method

To solve the general cubic $ax^3 + bx^2 + cx + d = 0$, we first perform a transformation to remove the x^2 term (the depressed cubic).

Transformation to Depressed Cubic

Substituting $x = z - \frac{b}{3a}$ reduces the equation to the form:

$$z^3 + 3Hz + G = 0$$

where H and G are constants derived from a, b, c, d .

Cardano's Formula

To solve $z^3 + 3Hz + G = 0$, we assume $z = u + v$. Cubing both sides and comparing terms, we find:

$$u^3 + v^3 = -G \quad \text{and} \quad u^3v^3 = -H^3$$

This implies u^3 and v^3 are the roots of the quadratic $t^2 + Gt - H^3 = 0$. Using the quadratic formula:

$$u^3, v^3 = \frac{-G \pm \sqrt{G^2 + 4H^3}}{2}$$

The roots are then $z_1 = u + v$, $z_2 = u\omega + v\omega^2$, and $z_3 = u\omega^2 + v\omega$, where ω is the complex cube root of unity.

Example 6.2.7. Solve $x^3 - 15x - 126 = 0$ using Cardano's Method.

Solution. This is already in depressed form $z^3 + 3Hz + G = 0$, with $3H = -15 \implies H = -5$ and $G = -126$. The quadratic for u^3, v^3 is:

$$t^2 - 126t - (-5)^3 = t^2 - 126t + 125 = 0$$

Factorizing gives $(t - 125)(t - 1) = 0$. Thus $u^3 = 125$ and $v^3 = 1$. Taking the real cube roots: $u = 5, v = 1$. The real root is $x = u + v = 5 + 1 = 6$. We established that $x = 6$ is a real root of the polynomial $P(x) = x^3 - 15x - 126$. Therefore, $(x - 6)$ is a factor of $P(x)$. To find the remaining roots, we divide $P(x)$ by $(x - 6)$ using synthetic division:

$$\begin{array}{r|rrrr} 6 & 1 & 0 & -15 & -126 \\ & \downarrow & 6 & 36 & 126 \\ \hline & 1 & 6 & 21 & 0 \end{array}$$

This yields the quadratic quotient $x^2 + 6x + 21 = 0$. We solve for the remaining roots using the quadratic formula:

$$\begin{aligned} x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\ &= \frac{-6 \pm \sqrt{6^2 - 4(1)(21)}}{2} \\ &= \frac{-6 \pm \sqrt{36 - 84}}{2} \\ &= \frac{-6 \pm \sqrt{-48}}{2}. \end{aligned}$$

Thus, the three roots are $6, -3 + 2i\sqrt{3}$ and $-3 - 2i\sqrt{3}$. ■

Example 6.2.8. Solve the depressed cubic equation $x^3 - 9x - 28 = 0$.

Solution. The equation is in the form $x^3 + px + q = 0$ with $p = -9$ and $q = -28$. Let $x = u + v$. By Cardano's identity, we set:

$$3uv = -p \implies uv = 3 \implies u^3v^3 = 27$$

$$u^3 + v^3 = -q \implies u^3 + v^3 = 28$$

We form the quadratic resolvent equation for $t = u^3$ and $t' = v^3$:

$$t^2 - (u^3 + v^3)t + u^3v^3 = 0$$

$$t^2 - 28t + 27 = 0$$

Factorizing the quadratic:

$$(t - 27)(t - 1) = 0$$

Thus, we can assign $u^3 = 27$ and $v^3 = 1$. Taking real cube roots:

$$u = 3, \quad v = 1$$

The real root is $x = u + v = 3 + 1 = 4$.

To find the remaining roots, we divide the original polynomial by $(x - 4)$:

$$(x^3 - 9x - 28) \div (x - 4) = x^2 + 4x + 7$$

Solving $x^2 + 4x + 7 = 0$ using the quadratic formula:

$$x = \frac{-4 \pm \sqrt{16 - 28}}{2} = \frac{-4 \pm \sqrt{-12}}{2} = -2 \pm i\sqrt{3}$$

The roots are $x = 4$, $x = -2 + i\sqrt{3}$, and $x = -2 - i\sqrt{3}$. ■

Example 6.2.9. Solve the depressed cubic equation $x^3 + 6x - 7 = 0$.

Solution. Here $p = 6$ and $q = -7$. Let $x = u + v$.

$$uv = -\frac{p}{3} = -2 \implies u^3v^3 = -8$$

$$u^3 + v^3 = -q = 7$$

The quadratic resolvent is:

$$t^2 - 7t - 8 = 0$$

$$(t - 8)(t + 1) = 0$$

This gives $u^3 = 8$ and $v^3 = -1$. Taking real cube roots implies $u = 2$ and $v = -1$. The real root is $x = u + v = 2 + (-1) = 1$.

Dividing $(x^3 + 6x - 7)$ by $(x - 1)$ yields the quotient $x^2 + x + 7$. Solving $x^2 + x + 7 = 0$:

$$x = \frac{-1 \pm \sqrt{1 - 28}}{2} = \frac{-1 \pm i\sqrt{27}}{2}$$

The roots are $x = 1$, $x = \frac{-1 + 3i\sqrt{3}}{2}$, and $x = \frac{-1 - 3i\sqrt{3}}{2}$. ■

Example 6.2.10. Solve the general cubic equation $x^3 + 3x^2 + 9x - 13 = 0$.

Solution. First, we depress the cubic. Given $ax^3 + bx^2 + cx + d = 0$, we substitute $x = z - \frac{b}{3a}$. Here $a = 1, b = 3$, so we substitute $x = z - 1$.

$$\begin{aligned} (z - 1)^3 + 3(z - 1)^2 + 9(z - 1) - 13 &= 0 \\ (z^3 - 3z^2 + 3z - 1) + 3(z^2 - 2z + 1) + (9z - 9) - 13 &= 0 \\ z^3 + (-3 + 3)z^2 + (3 - 6 + 9)z + (-1 + 3 - 9 - 13) &= 0 \\ z^3 + 6z - 20 &= 0 \end{aligned}$$

Now we solve for z with $p = 6, q = -20$. Let $z = u + v$.

$$uv = -2 \implies u^3v^3 = -8$$

$$u^3 + v^3 = 20$$

The quadratic resolvent is $t^2 - 20t - 8 = 0$. Using the quadratic formula:

$$t = \frac{20 \pm \sqrt{400 - 4(-8)}}{2} = \frac{20 \pm \sqrt{432}}{2} = 10 \pm 6\sqrt{3}$$

Let $u^3 = 10 + 6\sqrt{3}$ and $v^3 = 10 - 6\sqrt{3}$. We observe that $(1 + \sqrt{3})^3 = 1 + 3\sqrt{3} + 3(3) + 3\sqrt{3} = 10 + 6\sqrt{3}$. Similarly, $(1 - \sqrt{3})^3 = 10 - 6\sqrt{3}$. Thus, $u = 1 + \sqrt{3}$ and $v = 1 - \sqrt{3}$.

$$z = u + v = (1 + \sqrt{3}) + (1 - \sqrt{3}) = 2$$

Substituting back to x :

$$x = z - 1 = 2 - 1 = 1$$

Synthetic division of the original polynomial by $(x - 1)$ gives $x^2 + 4x + 13 = 0$. The remaining roots are:

$$x = \frac{-4 \pm \sqrt{16 - 52}}{2} = -2 \pm 3i$$

The roots are $x = 1$, $x = -2 + 3i$, and $x = -2 - 3i$. ■

6.3 Methods of solving Biquadratic Polynomials

6.3.1 Descartes' Method for Solving Biquadratic Equations

Descartes' method reduces a monic quartic equation into two quadratic equations. This method is primarily applied to a **depressed quartic** (a quartic equation with no x^3 term).

Working Rule: Consider the general quartic equation:

$$ax^4 + bx^3 + cx^2 + dx + e = 0$$

Step 1: Removing the second term: Depression.

Divide by a to make it monic. Eliminate the cubic term by substituting $x = z - \frac{b}{4a}$. This yields the depressed form:

$$z^4 + qz^2 + rz + s = 0$$

Step 2: Factorization.

Assume the quartic can be factored into two quadratics with linear coefficients k and $-k$ (since the cubic term is zero):

$$(z^2 + kz + m)(z^2 - kz + n) = 0$$

Step 3: Equating Coefficients.

Expanding the product:

$$z^4 + (n + m - k^2)z^2 + k(n - m)z + mn = 0$$

Comparing this with $z^4 + qz^2 + rz + s = 0$, we obtain the system:

$$1) \quad n + m - k^2 = q \implies n + m = q + k^2 \tag{6.1}$$

$$2) \quad k(n - m) = r \implies n - m = \frac{r}{k} \tag{6.2}$$

$$3) \quad mn = s \tag{6.3}$$

Step 4: The Resolvent Cubic.

Using the identity $(n + m)^2 - (n - m)^2 = 4mn$:

$$(q + k^2)^2 - \left(\frac{r}{k}\right)^2 = 4s$$

Let $t = k^2$. Substituting and rearranging gives the resolvent cubic in t :

$$t^3 + 2qt^2 + (q^2 - 4s)t - r^2 = 0$$

Solve this cubic for a positive real root t . Then $k = \sqrt{t}$.

Step 5: Solve for Roots.

Once k is found, find m and n using the system in Step 3:

$$m = \frac{1}{2} \left((q + k^2) - \frac{r}{k} \right), \quad n = \frac{1}{2} \left((q + k^2) + \frac{r}{k} \right)$$

Finally, solve the two quadratic equations $z^2 + kz + m = 0$ and $z^2 - kz + n = 0$.

Example 6.3.1. Solve $x^4 - 6x^2 - 12x - 8 = 0$.

Solution. Here $q = -6, r = -12, s = -8$. The resolvent cubic for $t = k^2$ is:

$$t^3 + 2(-6)t^2 + ((-6)^2 - 4(-8))t - (-12)^2 = 0$$

$$t^3 - 12t^2 + (36 + 32)t - 144 = 0$$

$$t^3 - 12t^2 + 68t - 144 = 0$$

By trial, $t = 4$ is a root ($64 - 192 + 272 - 144 = 0$). Thus, $k = \sqrt{4} = 2$. We find m and n :

$$n + m = q + k^2 = -6 + 4 = -2$$

$$n - m = r/k = -12/2 = -6$$

Adding the equations: $2n = -8 \implies n = -4$. Subtracting: $2m = 4 \implies m = 2$. The quadratic factors are $(x^2 + 2x + 2)$ and $(x^2 - 2x - 4)$.

$$1. \ x^2 + 2x + 2 = 0 \implies x = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i.$$

$$2. \ x^2 - 2x - 4 = 0 \implies x = \frac{2 \pm \sqrt{4+16}}{2} = 1 \pm \sqrt{5}.$$

The roots are $-1 + i, -1 - i, 1 + \sqrt{5}, 1 - \sqrt{5}$. ■

Example 6.3.2. Solve $x^4 - 10x^2 - 16x + 5 = 0$.

Solution. Here $q = -10, r = -16, s = 5$. The resolvent cubic is:

$$t^3 - 20t^2 + (100 - 20)t - 256 = 0$$

$$t^3 - 20t^2 + 80t - 256 = 0$$

Checking factors of 256, we find $t = 16$:

$$16^3 - 20(16)^2 + 80(16) - 256 = 4096 - 5120 + 1280 - 256 = 0$$

So $k = \sqrt{16} = 4$. We find m and n :

$$n + m = -10 + 16 = 6$$

$$n - m = -16/4 = -4$$

Solving gives $n = 1$ and $m = 5$. The quadratics are $x^2 + 4x + 5 = 0$ and $x^2 - 4x + 1 = 0$.

$$1. \ x^2 + 4x + 5 = 0 \implies x = \frac{-4 \pm \sqrt{16-20}}{2} = -2 \pm i.$$

$$2. \ x^2 - 4x + 1 = 0 \implies x = \frac{4 \pm \sqrt{16-4}}{2} = 2 \pm \sqrt{3}.$$

The roots are $-2 + i, -2 - i, 2 + \sqrt{3}, 2 - \sqrt{3}$. ■

Example 6.3.3. Solve $x^4 + x^2 + 4x - 3 = 0$.

Solution. Here $q = 1, r = 4, s = -3$. The resolvent cubic:

$$t^3 + 2(1)t^2 + (1 - 4(-3))t - 16 = 0$$

$$t^3 + 2t^2 + 13t - 16 = 0$$

The sum of coefficients is 0, so $t = 1$ is a root. Thus $k = 1$. Solving for m, n :

$$n + m = 1 + 1 = 2$$

$$n - m = 4/1 = 4$$

This yields $n = 3, m = -1$. The quadratics are $x^2 + x - 1 = 0$ and $x^2 - x + 3 = 0$.

$$1. x = \frac{-1 \pm \sqrt{1+4}}{2} = \frac{-1 \pm \sqrt{5}}{2}.$$

$$2. x = \frac{1 \pm \sqrt{1-12}}{2} = \frac{1 \pm i\sqrt{11}}{2}.$$

■

Example 6.3.4. Solve $x^4 - 17x^2 - 20x - 6 = 0$.

Solution. Here $q = -17, r = -20, s = -6$. The resolvent cubic:

$$t^3 - 34t^2 + (289 - 4(-6))t - 400 = 0$$

$$t^3 - 34t^2 + 313t - 400 = 0$$

Testing integer values, we find $t = 16$:

$$16^3 - 34(16^2) + 313(16) - 400 = 4096 - 8704 + 5008 - 400 = 0$$

So $t = 16 \implies k = 4$. Find m, n :

$$n + m = -17 + 16 = -1$$

$$n - m = -20/4 = -5$$

Solving gives $n = -3$ and $m = 2$. The quadratics are $x^2 + 4x + 2 = 0$ and $x^2 - 4x - 3 = 0$.

$$1. x = \frac{-4 \pm \sqrt{16-8}}{2} = -2 \pm \sqrt{2}.$$

$$2. x = \frac{4 \pm \sqrt{16+12}}{2} = 2 \pm \sqrt{7}.$$

The roots are $-2 + \sqrt{2}, -2 - \sqrt{2}, 2 + \sqrt{7}, 2 - \sqrt{7}$.

■

6.3.2 Solution of Biquadratic Equations: Ferrari's Method

A biquadratic (quartic) equation $x^4 + ax^3 + bx^2 + cx + d = 0$ can be solved by expressing it as the equality of two perfect squares.

Detailed Explanation of the Method

Step 1: Rearrange and Complete the Square

Move the lower-degree terms to the right side:

$$x^4 + ax^3 = -bx^2 - cx - d$$

Add $\frac{a^2}{4}x^2$ to both sides to complete the square on the left for the terms x^4 and ax^3 :

$$\left(x^2 + \frac{a}{2}x\right)^2 = \left(\frac{a^2}{4} - b\right)x^2 - cx - d$$

Step 2: Introduce the Auxiliary Variable λ

We introduce a parameter λ inside the square on the left side. Expanding $\left(x^2 + \frac{a}{2}x + \lambda\right)^2$ generates extra terms $2\lambda\left(x^2 + \frac{a}{2}x\right) + \lambda^2$. We add these to the right side to maintain equality:

$$\left(x^2 + \frac{a}{2}x + \lambda\right)^2 = \left(\frac{a^2}{4} - b + 2\lambda\right)x^2 + (a\lambda - c)x + (\lambda^2 - d)$$

Let the Right Hand Side (RHS) be $Ax^2 + Bx + C$, where:

$$A = \frac{a^2}{4} - b + 2\lambda, \quad B = a\lambda - c, \quad C = \lambda^2 - d$$

Step 3: Form the Resolvent Cubic

For the RHS $(Ax^2 + Bx + C)$ to be a perfect square $(mx + n)^2$, its discriminant must be zero:

$$B^2 - 4AC = 0$$

Substituting A, B, C creates a cubic equation in λ . This is the **resolvent cubic**.

Step 4: Solve for Roots

Find a real root λ_0 from the resolvent cubic. Substitute λ_0 back into the equation. The RHS becomes a perfect square $(mx + n)^2$. We then solve:

$$x^2 + \frac{a}{2}x + \lambda_0 = \pm(mx + n)$$

This results in two quadratic equations, providing the four roots of the original quartic.

Example 6.3.5. Solve the depressed quartic $x^4 - 10x^2 - 20x - 16 = 0$.

Solution. Here $a = 0, b = -10, c = -20, d = -16$. The identity is $(x^2 + \lambda)^2 = (2\lambda - b)x^2 - cx + (\lambda^2 - d)$. Substituting coefficients:

$$(x^2 + \lambda)^2 = (2\lambda + 10)x^2 + 20x + (\lambda^2 + 16)$$

For the RHS to be a perfect square, the discriminant must be zero:

$$20^2 - 4(2\lambda + 10)(\lambda^2 + 16) = 0$$

Divide by 4:

$$100 - (2\lambda^3 + 32\lambda + 10\lambda^2 + 160) = 0$$

$$2\lambda^3 + 10\lambda^2 + 32\lambda + 60 = 0 \implies \lambda^3 + 5\lambda^2 + 16\lambda + 30 = 0$$

Using the rational root theorem, we test $\lambda = -3$:

$$(-3)^3 + 5(-3)^2 + 16(-3) + 30 = -27 + 45 - 48 + 30 = 0$$

So, $\lambda = -3$. Substitute $\lambda = -3$ back into the RHS equation:

$$A = 2(-3) + 10 = 4, \quad B = 20, \quad C = (-3)^2 + 16 = 25$$

The equation becomes:

$$(x^2 - 3)^2 = 4x^2 + 20x + 25 = (2x + 5)^2$$

Taking square roots:

$$x^2 - 3 = \pm(2x + 5)$$

Case 1: $x^2 - 3 = 2x + 5 \implies x^2 - 2x - 8 = 0 \implies (x - 4)(x + 2) = 0$. Roots: 4, -2.

Case 2: $x^2 - 3 = -(2x + 5) \implies x^2 + 2x + 2 = 0$. Roots: $x = \frac{-2 \pm \sqrt{4-8}}{2} = -1 \pm i$. The roots are 4, -2, $-1 + i$, $-1 - i$. ■

Example 6.3.6. Solve $x^4 + 2x^3 - 7x^2 - 8x + 12 = 0$.

Solution. Here $a = 2, b = -7, c = -8, d = 12$. Rearranging and adding λ :

$$(x^2 + x + \lambda)^2 = \left(\frac{4}{4} - (-7) + 2\lambda\right)x^2 + (2\lambda - (-8))x + (\lambda^2 - 12)$$

$$(x^2 + x + \lambda)^2 = (8 + 2\lambda)x^2 + (2\lambda + 8)x + (\lambda^2 - 12)$$

Set discriminant of RHS to 0 ($B^2 - 4AC = 0$):

$$(2\lambda + 8)^2 - 4(2\lambda + 8)(\lambda^2 - 12) = 0$$

Factor out $4(2\lambda + 8)$ (note that $2\lambda + 8 \neq 0$ is assumed, or checked separately):

$$(2\lambda + 8) - 4(\lambda^2 - 12) = 0$$

This simplification approach assumes $2\lambda + 8 = 0$ is a solution? Indeed, if $A = 0$ and $B = 0$, it is a trivial square. Let's check $\lambda = -4$. If $\lambda = -4$:

$$\text{RHS} = (8 - 8)x^2 + (-8 + 8)x + (16 - 12) = 4 = 2^2$$

So $\lambda = -4$ works perfectly.

$$(x^2 + x - 4)^2 = 2^2$$

$$x^2 + x - 4 = \pm 2$$

Case 1: $x^2 + x - 6 = 0 \implies (x + 3)(x - 2) = 0$. Roots: $-3, 2$.

Case 2: $x^2 + x - 2 = 0 \implies (x + 2)(x - 1) = 0$. Roots: $-2, 1$. The roots are $1, 2, -2, -3$. ■

Example 6.3.7. Solve $x^4 - 2x^3 - 6x^2 + 10x - 3 = 0$.

Solution. Here $a = -2, b = -6, c = 10, d = -3$. Identity: $(x^2 - x + \lambda)^2 = Ax^2 + Bx + C$.

$$A = \frac{4}{4} - (-6) + 2\lambda = 7 + 2\lambda$$

$$B = -2\lambda - 10$$

$$C = \lambda^2 - (-3) = \lambda^2 + 3$$

Discriminant condition:

$$(-2(\lambda + 5))^2 - 4(2\lambda + 7)(\lambda^2 + 3) = 0$$

$$4(\lambda^2 + 10\lambda + 25) - 4(2\lambda^3 + 6\lambda + 7\lambda^2 + 21) = 0$$

$$\lambda^2 + 10\lambda + 25 - 2\lambda^3 - 7\lambda^2 - 6\lambda - 21 = 0$$

$$-2\lambda^3 - 6\lambda^2 + 4\lambda + 4 = 0 \implies \lambda^3 + 3\lambda^2 - 2\lambda - 2 = 0$$

By inspection, $\lambda = 1$ is a root ($1 + 3 - 2 - 2 = 0$). Substitute $\lambda = 1$ back:

$$A = 9, \quad B = -12, \quad C = 4$$

$$(x^2 - x + 1)^2 = 9x^2 - 12x + 4 = (3x - 2)^2$$

Taking roots: Case 1: $x^2 - x + 1 = 3x - 2 \implies x^2 - 4x + 3 = 0 \implies x = 1, 3$.

Case 2: $x^2 - x + 1 = -(3x - 2) \implies x^2 + 2x - 1 = 0$. Using quadratic formula: $x = \frac{-2 \pm \sqrt{4 - 4(1)(-1)}}{2} = -1 \pm \sqrt{2}$. The roots are $1, 3, -1 + \sqrt{2}, -1 - \sqrt{2}$. ■

Example 6.3.8. Solve the equation $x^4 + 1 = 0$ using Ferrari's Method.

Solution. Here $a = 0, b = 0, c = 0, d = 1$.

$$(x^2 + \lambda)^2 = (2\lambda)x^2 + (\lambda^2 - 1)$$

For the RHS to be a perfect square:

$$0^2 - 4(2\lambda)(\lambda^2 - 1) = 0$$

This implies $\lambda = 0$ or $\lambda = \pm 1$. We choose $\lambda = 1$ so that the coefficient of x^2 is positive. Substitute $\lambda = 1$:

$$(x^2 + 1)^2 = 2x^2 + 0 = (\sqrt{2}x)^2$$

$$x^2 + 1 = \pm\sqrt{2}x$$

Case 1: $x^2 - \sqrt{2}x + 1 = 0$.

$$x = \frac{\sqrt{2} \pm \sqrt{2-4}}{2} = \frac{\sqrt{2} \pm i\sqrt{2}}{2}$$

Case 2: $x^2 + \sqrt{2}x + 1 = 0$.

$$x = \frac{-\sqrt{2} \pm \sqrt{2-4}}{2} = \frac{-\sqrt{2} \pm i\sqrt{2}}{2}$$

These are the four complex roots. ■

The Impossibility of Solving Quintic Equations by Radicals

For polynomials of degree 1 through 4 (linear, quadratic, cubic, and quartic), general formulas exist that express the roots using the coefficients and operations of addition, subtraction, multiplication, division, and root extraction (n -th roots). These are known as solutions **by radicals**.

However, for polynomials of degree 5 (quintic) and higher, no such general formula exists. This mathematical impossibility is established by two major results in algebra.

The Abel-Ruffini Theorem

In 1824, Niels Henrik Abel (building on earlier work by Paolo Ruffini) proved that there is no general algebraic formula for the roots of a polynomial of degree $n \geq 5$. This result, known as the **Abel-Ruffini Theorem**, asserts that a generic quintic equation cannot be solved by radicals.

Galois Theory and Solvable Groups

Évariste Galois provided the deeper structural reason for this impossibility by connecting the theory of polynomial equations to Group Theory.

1. **The Galois Group:** Associated with every polynomial is a group of permutations of its roots, known as the Galois Group, denoted as G .
2. **Condition for Solvability:** Galois proved that a polynomial equation is solvable by radicals if and only if its associated Galois group is a **solvable group**. A group is solvable if it can be broken down into a series of abelian (commutative) factor groups.
3. **The Symmetric Group S_n :** For a general polynomial of degree n , the Galois group is the Symmetric Group S_n (the group of all permutations of n elements).
4. **The Failure at $n = 5$:**
 - For $n \leq 4$, the group S_n is solvable. This explains why formulas exist for quadratics, cubics, and quartics.
 - For $n \geq 5$, the group S_n is **not solvable**. Specifically, its normal subgroup, the Alternating Group A_n , is a **simple group** (it cannot be broken down further) for $n \geq 5$.

Conclusion: Because the symmetry group of the general quintic equation (S_5) lacks the necessary structure to be reduced to simple radical operations, it is impossible to construct a general formula for the roots of quintic and higher-degree polynomials.

Exercise 6.3.9. Solve the following exercises:

1. Apply Descartes' Rule of Signs to determine the maximum possible number of positive and negative real roots for the polynomial $f(x) = x^5 - x^4 + 3x^3 + 9x^2 - x + 5$.
2. If α, β, γ are the roots of the cubic equation $x^3 + px + q = 0$, find the value of the symmetric function $\sum \alpha^2$ in terms of p and q .
3. Solve the depressed cubic equation $x^3 + 3x - 4 = 0$ using Cardano's Method.
4. Use Descartes' Rule of Signs to show that the equation $x^6 + 3x^2 + 5x - 7 = 0$ has at least four imaginary roots.
5. If α, β, γ are the roots of $x^3 + qx + r = 0$, calculate the value of $\sum \alpha^2 \beta$ in terms of the coefficients.
6. Solve the biquadratic equation $x^4 - 10x^2 + 9 = 0$ by treating it as a quadratic in x^2 .
7. Find the value of $\sum \frac{1}{\alpha}$ for the roots of the equation $x^3 - 6x^2 + 11x - 6 = 0$ without solving the equation.
8. Use Cardano's Method to find the real root of the equation $x^3 - 15x - 126 = 0$.
9. Determine the nature of the roots of $x^4 + 2x^2 + x - 1 = 0$ using Descartes' Rule of Signs.
10. If α, β, γ are the roots of $x^3 + ax^2 + bx + c = 0$, construct a new cubic equation whose roots are $\frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}, \frac{1}{\alpha\beta}$.
11. Solve the biquadratic equation $x^4 - 2x^3 - 13x^2 + 14x + 24 = 0$, given that the sum of two of its roots is zero.
12. If α, β, γ are roots of $x^3 + px + q = 0$, find the value of the power sum $s_3 = \alpha^3 + \beta^3 + \gamma^3$ using Newton's sums.

Challenges:

13. Transform the cubic equation $x^3 + 6x^2 + 9x + 4 = 0$ into a depressed cubic by removing the x^2 term, and then solve for x using Cardano's Formula.
14. If α, β, γ are the roots of $x^3 + px + q = 0$, express the symmetric function $\sum \alpha^4$ entirely in terms of p and q .
15. Solve the general biquadratic equation $x^4 - 2x^3 - 5x^2 + 10x - 3 = 0$ using Ferrari's Method (reduction to a resolvent cubic).
16. Prove using symmetric functions that if the roots α, β, γ of $x^3 + px + q = 0$ satisfy the relation $\alpha\beta + 1 = 0$, then $q^2 + p + 1 = 0$.
17. In the "Casus Irreducibilis" of Cardano's method, the expression under the square root in the u^3, v^3 calculation is negative. Use this to solve $x^3 - 7x + 6 = 0$ and explain how the three real roots emerge from complex calculations.
18. Find the condition that the roots of the quartic $x^4 + px^3 + qx^2 + rx + s = 0$ should be in Geometric Progression.
19. Use Descartes' Rule of Signs to determine the range of values of k for which the equation $x^4 - 4x^3 + 4x^2 - k = 0$ has four real roots.
20. Let α, β, γ be the roots of $x^3 + px^2 + qx + r = 0$. Find the cubic equation whose roots are $(\alpha - \beta)^2, (\beta - \gamma)^2$, and $(\gamma - \alpha)^2$. (This is the equation of squared differences).